

Unbinned OmniFold measurement of the MINERvA ME FHC CC-inclusive double-differential cross section $d^2\sigma/(dp_T dp_{\parallel})$ and its higher-dimensional extensions MINERvA analysis note — DRAFT

J. R. Bailey

June 10, 2026

Abstract

We apply *unbinned* OmniFold to the MINERvA medium-energy FHC charged-current inclusive cross section, replacing the iterative Bayesian unfolding of the published analysis (arXiv:2106.16210). In two dimensions, $d^2\sigma/(dp_T dp_{\parallel})$, our result reproduces the published spectrum (σ_{tot} ratio 1.011; 94 % of bins within 10 %) with a matching uncertainty budget (median relative σ 6.87 % vs 6.86 %).

Because adding an observable is just adding a feature—with no migration matrix or binning penalty—we then perform the first *simultaneous three- and four-observable* unfolds of MINERvA data, adding available energy (E_{avail}) and three-momentum transfer (q_3). Each higher-dimensional result reproduces the lower-dimensional one under marginalisation (a built-in closure), and four generators we produce at truth level (GENIE, the MINERvA Tune, NuWro, GiBUU) all under-predict the data at low E_{avail} . This is the well-established MINERvA low-recoil/2p2h deficit (Rodrigues 2016; Ascencio 2022): pion final-state interactions move $d\sigma/dE_{\text{avail}}$ by $< 1\%$, whereas adding the empirical Valencia 2p2h fills the quasielastic- Δ dip. Recovering this known feature through an independent method validates the technique; the four-dimensional (E_{avail}, q_3) result additionally resolves the deficit’s momentum-transfer structure, and a bin-identical full-covariance comparison to the published Ascencio low-recoil measurement (arXiv:2110.13372) on their maximal common grid gives $\chi^2/\text{ndf} = 1.68/2$ ($p = 0.43$) — the two measurements are consistent. A fifth axis, the hadronic invariant mass W , localises the remaining high- E_{avail} excess to the deep-inelastic region: with a dedicated 42-bin (E_{avail}, W) covariance, all four generators miss the data in the high- W corner ($E_{\text{avail}} \geq 0.4 \text{ GeV}$, $W \geq 1.8 \text{ GeV}$) at $9\text{--}18\sigma$.

The 2D, 3D and 4D results carry full covariance budgets (universe + bootstrap + ML); in 4D a 160-throw *unified-throw* study shows the block sum underestimates the vertical systematic by a factor ~ 2 , and the published 4D covariance adopts the unified-throw magnitude. The method passes binned and unbinned goodness-of-fit tests plus scalar GBDT/MLP classifier cross-checks. A point-cloud transformer (PET) pipeline operating on the raw reconstructed calorimetric clusters now produces a full-statistics, absolutely-normalised cross section: its closure is unbiased to $\sim 1\%$, it agrees with the production GBDT result at the 9 % (training-configuration) level, and it carries its own frozen-reweighter systematic budget. Finally, removing the muon kinematic cuts from the signal definition yields a first *full-phase-space* CC-inclusive cross section on the tracker fiducial volume, $\sigma_{\text{FPS}} = 4.502 \times 10^{-38} \text{ cm}^2/\text{nucleon}$ (46 % above the restricted phase space, which hides a third of the rate): inside the published region it reproduces the restricted measurement to 0.57 % median per cell, and a three-prior envelope quantifies the extrapolation cost (few-percent where the detector has acceptance, explicit tens-of-percent flags in the dead regions); its systematic campaign is in production. This note documents the experiment and samples, the method, the analysis pipeline, the uncertainty quantification, the validation suite,

the higher-dimensional extensions, the full-phase-space extension, and the statistical methodology (as appendices).

Executive summary

What this note is. This note documents the application of *unbinned machine-learning unfolding* (OmniFold) to MINERvA’s medium-energy neutrino data — first as a faithful reproduction of a published flagship measurement, then as its extension to simultaneous three-, four- and five-observable cross sections that no binned method has produced at MINERvA. It is written so that the first three pages give the complete story; everything afterwards is the supporting technical record.

The method, in plain language. Traditional unfolding corrects detector effects with a migration matrix between histogram bins, so each added observable multiplies the number of bins and the method drowns in dimensions. OmniFold instead trains a classifier to learn, event by event, a weight that morphs the simulation into the data, and pulls that weight back to the true (pre-detector) level through the simulation’s truth–reco pairing. No binning is imposed until the very end, so adding an observable is just adding an input column. This analysis uses gradient-boosted decision trees as the learners and validates the choice extensively (closure tests, coverage, calibrated- classifier checks, neural-network and transformer cross-checks).

Headline results.

- **Reproduction (the demonstrator).** Our unbinned $d^2\sigma/(dp_T dp_{\parallel})$ matches the published measurement (arXiv:2106.16210): total cross-section ratio 1.011, 94 % of bins within 10 %, per-bin pulls sub- 1σ , and a fully independent uncertainty budget whose median per-bin uncertainty (6.87 %) matches the published 6.86 % (§4, §5, Fig. 3).
- **First 3D/4D/5D MINERvA unfolds.** Adding available energy (E_{avail}), momentum transfer (q_3), and hadronic mass (W) produces the first simultaneous three-, four- and five-observable unfolds of MINERvA data; every result reproduces its lower-dimensional predecessor when marginalised — a built-in cross-check (§7).
- **Known physics recovered independently.** All four truth-level generators (GENIE, MINERvA Tune, NuWro, GiBUU) under-predict the data at low E_{avail} ; pion final-state-interaction dials move the spectrum by $< 1\%$ while adding Valencia 2p2h fills the quasielastic- Δ dip — the established MINERvA low-recoil/2p2h deficit, re-found by an independent method (§7.7, Fig. 20).
- **A new localisation of the remaining excess.** The high- E_{avail} excess that 2p2h cannot explain is resolved in the (E_{avail}, W) plane: it concentrates in the deep-inelastic corner ($W \geq 1.8 \text{ GeV}$), where all four generators miss the data at $9\text{--}18\sigma$ on a dedicated covariance — no current model reproduces it (§8, Fig. 28).
- **A methodological finding.** A 160-throw “unified-throw” test of the standard add-the-blocks covariance construction shows it *underestimates* the 4D systematic by a factor ~ 2 in the strongest-migration corner; the published 4D covariance adopts the corrected magnitude (§5.4).

- **Toward raw-data unfolding.** A point-cloud transformer unfolding the actual reconstructed calorimeter clusters now yields an absolutely-normalised, full-statistics cross section, unbiased in closure to $\sim 1\%$ and agreeing with the production result at the 9% (training-limited) level (§9).
- **A first full-phase-space cross section.** Removing the muon kinematic cuts (which hide a third of the signal rate) gives $\sigma_{\text{FPS}} = 4.5 \times 10^{-38} \text{ cm}^2/\text{nucleon}$, anchored to the restricted measurement at 0.6% where they overlap, with a three-prior envelope declaring the extrapolation cost cell-by-cell (§10; systematics in production).

How to read this note. Sections 2–3 define the samples and the method; §4 is the 2D re-production; §5 and App. B are the uncertainty machinery (the appendix is self-contained and tutorial-grade); §6 is the validation suite; the physics is in §7–8; §9 is the point-cloud track; §11 states the planned full-phase-space extension. Appendix A records the resolution of all questions raised in internal review. Every number quoted here is cross-checked in the repository’s validation ledger.

Contents

Executive summary	2
1 Introduction and motivation	5
2 The MINERvA experiment, data and simulation samples	6
2.1 Detector and beam	6
2.2 Data and simulation samples	6
2.3 Signal definition and event selection	7
2.4 Per-event analysis trees	7
2.5 Reco-level control distributions	7
3 The unbinned OmniFold procedure	8
3.1 Algorithm	8
3.2 Estimators and configuration	9
3.3 MINERvA driver and invariants	9
3.4 Cross-section extraction	9
3.5 Binning	10
4 Results	10
4.1 Double-differential cross section	10
4.2 Comparison to the published measurement	12
4.3 Anatomy of the $\chi^2 = 3.66$ tension	14
4.4 Comparison to GENIE MINERvA Tune v1	15
5 Uncertainty quantification	16
5.1 Systematic universes	17
5.2 Statistical uncertainty: Poisson bootstrap	18
5.3 ML/optimisation uncertainty: seedscan	18
5.4 Combination and the unified-throw test	19
6 Validation	19
7 Higher-dimensional extension: available energy	23
7.1 Definitions and pipeline	23
7.2 The available-energy spectrum	24
7.3 Validation against the 2D measurement	24
7.4 Closure: the scatter is physical, not a method bias	25
7.5 Generator comparison in 3D	25
7.6 The 3D systematic-uncertainty campaign	27
7.7 The low-available-energy excess recovers the known low-recoil 2p2h deficit	27
7.8 Fourth axis: three-momentum transfer q_3	29
8 Fifth axis: the (E_{avail}, W) deep-inelastic-tail study	32
8.1 W as a fifth axis	32
8.2 Where the excess lives in the (E_{avail}, W) plane	32
8.3 Generator band: the corner is generator- and tune-robust	33
8.4 The (E_{avail}, W) covariance and significances	33

9	Point-cloud track: PET on the raw reconstructed clusters	34
9.1	From shape check to absolute cross section	34
9.2	PET systematic budget (frozen reweighter)	35
9.3	What the PET track demonstrates	36
10	First full-phase-space results	36
11	Summary and outlook	39
A	Resolution of the open questions from internal review	40
B	Statistical methods	42
B.1	Numerical conventions: inverting a covariance (Cholesky, ridge jitter, pseudo-inverse)	42
B.2	Variance-source procedures	43
B.2.1	ML stochasticity: the seedscan	43
B.2.2	Statistical uncertainty: the Poisson bootstrap	43
B.2.3	Systematic uncertainty: universes	45
B.2.4	How MINERvA builds its published covariance	47
B.2.5	Block-sum combination	48
B.3	Comparing to the paper	49
B.3.1	The 205-bin selection	49
B.3.2	Covariance-aware χ^2/ndf	49
B.3.3	Pull distribution	50
B.3.4	Shape-only χ^2 with Jacobian propagation	50
B.4	Ill-conditioning, rank deficiency, and open questions	51
B.5	Anatomy of the paper-covariance tension	54
B.6	Calibration: coverage toys	55
B.7	Closure tests: detecting bias	56
B.7.1	Truth-reweight closure	56
B.7.2	Hidden-variable closure	57
B.7.3	Alternative-model closure	58
B.8	How the pieces fit together	58
B.9	Multinomial $\rightarrow$ independent Poisson bootstrap limit	59
B.10	The $\pm 1\sigma$ pair covariance estimator: MAT vs legacy	59
C	The MINERvA measurement-dimensionality survey	60

1 Introduction and motivation

The MINERvA medium-energy (ME) FHC charged-current (CC) inclusive double-differential cross section in muon transverse and longitudinal momentum, $d^2\sigma/(dp_T dp_\parallel)$, was published by Ruterbories *et al.* [1] using D’Agostini iterative Bayesian unfolding (IBU) [2] on a binned response matrix. This note reproduces that measurement with *unbinned* OmniFold [3, 4], an unfolding technique based on iterated reweighting with machine-learned likelihood ratios that operates on per-event observables rather than on a fixed histogram binning. First demonstrated experimentally by the H1 collaboration [5, 6], OmniFold is one of a family of machine-learning unfolding methods [7–11] that lift the binned-response limitations of classical IBU, the SVD method [12] and TUnfold [13] (the latter accessed here via RooUnfold [14]).

Why unbinned OmniFold.

- **No binning at unfold time.** OmniFold learns event weights in the full observable space; the result is histogrammed only at the end, so the binning can be chosen (or refined) after unfolding without re-running.
- **Natural path to higher dimensions.** Adding observables is adding input features, not multiplying response-matrix bins — this is the enabling property for the available-energy extension (§7) that has no practical IBU analogue.
- **Different regularisation.** OmniFold’s regulariser is the iteration count and the model capacity of the gradient-boosted decision trees (GBDTs), not the IBU prior-iteration coupling. Comparing the two on the same data quantifies the method-dependence of the result (§4).

Goal and scope. The goal is a faithful, fully uncertainty-quantified reproduction of the published 2D cross section as the demonstrator for unbinned OmniFold on MINERvA data, followed by its higher-dimensional extensions. This note covers the experiment and samples (§2), the method and pipeline (§3), the 2D result and its comparison to the published data and to GENIE MINERvA Tune v1 (§4), the uncertainty budget (§5), the validation suite (§6), the three- and four-observable extensions with available energy and q_3 (§7), the (E_{avail}, W) study that localises the remaining deep-inelastic-tail excess (§8), the point-cloud transformer track (§9), and a summary with the planned full-phase-space extension (§11). Appendix A records the resolution of the questions raised during internal review; Appendix B is the complete statistical methodology.

2 The MINERvA experiment, data and simulation samples

2.1 Detector and beam

MINERvA [15] was a fine-grained scintillator-tracker neutrino experiment on the NuMI beamline at Fermilab. The central tracker is built from hexagonal planes of triangular polystyrene scintillator strips in three stereo orientations, surrounded by electromagnetic and hadronic calorimetry; the magnetized MINOS near detector, 2 m downstream, serves as the muon spectrometer, providing the muon momentum and charge for MINOS-matched tracks. This analysis uses the *medium-energy* (ME) neutrino-mode (FHC) exposure with $\langle E_\nu \rangle \sim 6$ GeV. The neutrino flux is the *a posteriori* constrained NuMI ME prediction of Ref. [16] (PPFX hadron-production weights plus the neutrino–electron-scattering constraint), identical to the flux used by the published measurement [1].

2.2 Data and simulation samples

The data are the twelve ME FHC playlists (ME1A–ME1G, ME1L–ME1P), corresponding to 1.057×10^{21} protons on target — the same exposure as Ref. [1]. After the selection below the data sample contains 4 119 797 reconstructed events.

The reference simulation is GENIE 2.12.6 [17] with the MINERvA Tune v1 applied as per-event weights through the MAT (MINERvA Analysis Toolkit) reweighters: the flux-and-CV weight, the GENIE dial weights, the low-recoil 2p2h enhancement fit, and the non-resonant-pion reduction. Detector response is the full GEANT4-based MINERvA simulation underlying the production AnaTuples. The simulated sample comprises 32 849 103 true signal events (POT-normalised to data exposure event-by-event), processed through the identical reconstruction and selection as the data.

2.3 Signal definition and event selection

The signal definition and phase space follow Ref. [1] exactly. The truth-level signal is a charged-current ν_μ interaction with vertex in the central-tracker fiducial volume, with muon phase space

$$\theta_\mu < 20^\circ, \quad 1.5 < p_\parallel < 60 \text{ GeV}/c, \quad p_T < 4.5 \text{ GeV}/c.$$

The fiducial volume is the tracker region $5980 < z < 8422$ mm within an 850 mm apothem, containing 3.2353×10^{30} nucleons (the fixed tracker-geometry constant; the event loop deliberately does not persist a per-file nucleon count because merging playlists would sum it).

Reconstructed events are selected by requiring: a reconstructed vertex in the same tracker fiducial volume and apothem; a muon track with $\theta_\mu < 20^\circ$ matched to a track in MINOS with a passing MINOS fit; no dead-time overlap; and a neutrino (rather than antineutrino) candidate. The MINOS-match requirement is implemented as `isMinosMatchTrack==1 && minos_trk_is_ok==1`; this replaces an educational stub in the MINERvA-101 tutorial code that accepted essentially every event, and brings the background rate from $\sim 10\%$ to 0.35% , consistent with the published $\sim 0.2\%$. Backgrounds (reconstructed events whose true interaction fails the signal definition) and *fakes* (reconstructed in phase space, true muon outside it) are POT-scaled and subtracted from the reconstructed-data target before unfolding (§3).

2.4 Per-event analysis trees

Unbinned unfolding needs per-event records rather than histograms. A dedicated event loop (MINERvA101/MINERvA-101-Cross-Section/runEventLoopOmniFold.cpp), built on the MAT systematic-universe machinery, writes four flat trees per playlist: `mc_truth_denom` (every true signal event — the efficiency denominator), `mc_signal_reco` (reconstructed, truth-matched simulated events), `mc_background`, and `data`. Each entry carries the truth and reconstructed observables (p_T , $p_\parallel$, and in the extended runs E_{avail} , q_3 , W), the MnvTune central-value weight, and — in the `_universes_full` variants — the full set of systematic-universe weights and laterally-shifted observables (§5). Two invariants make the result truth-tree authoritative:

- **Truth-tree-authoritative reco gate:** a reconstructed signal event is filled only when its (run, subrun, event) key is present in the truth denominator; truth-only events enter as native OmniFold “miss” entries.
- **Completeness** $c = (\text{events in the OmniFold input})/(\text{all truth events})$ is 1.000000 by construction at ME FHC (32 849 103 entries), so the cross-section completeness division is a no-op self-check.

Because the event loop selects one global playlist flux/calibration from the first event, it is always run per playlist and the twelve outputs merged afterwards (with a `TTree::SetMaxTreeSize`-aware merger for the > 100 GB universe dumps). The flux histogram (`pTmu_reweightedflux_integrated`) comes from the corresponding baseline histogram event loop and is POT-weighted across playlists; playlist-to-playlist flux differences are a $\sim 0.2\%$ effect.

2.5 Reco-level control distributions

Figure 1 compares the selected data to the POT-scaled simulation (signal + background) at reconstruction level in all five observables. The data exceed the MnvTune-v1 prediction by a uniform 12% in rate, rising with p_T (from 0.88 in the lowest bin to ~ 1.2 above 1 GeV/c) — the reco-level

image of the normalisation deficit quantified at truth level in §4.4 and §7.5. The simulation describes the *shapes* well, which is what the unfolding relies on; the rate offset is what the data pull corrects.

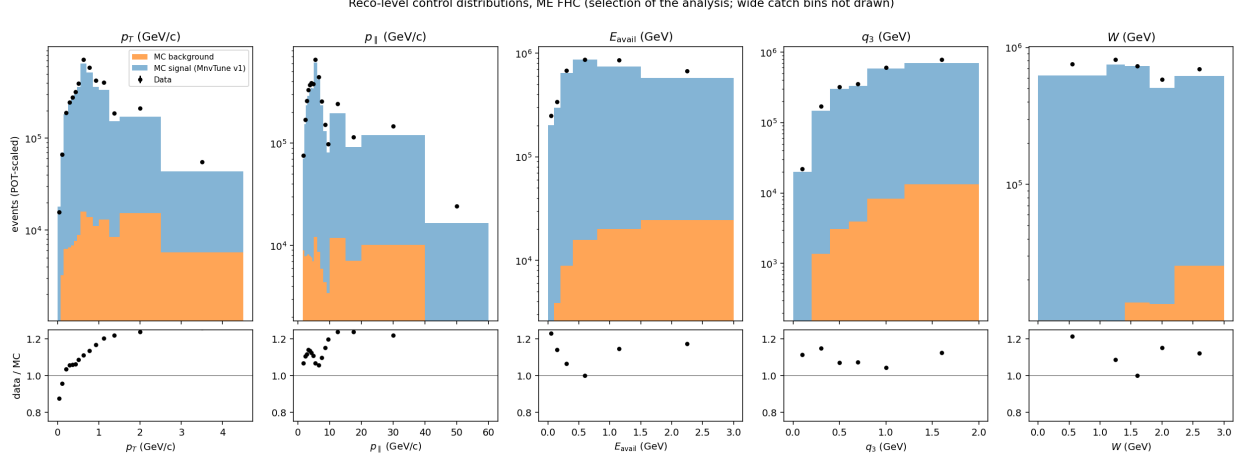


Figure 1: Reco-level control distributions for the analysis selection: data (points) versus POT-scaled simulated signal (blue) plus background (orange), with the data/MC ratio below, for p_T , $p_{\parallel}$, E_{avail} , q_3 and W (wide catch bins not drawn). The uniform $\sim 12\%$ data excess over Mnvtune v1 is the reco-level counterpart of the truth-level generator deficits of §7.5.

3 The unbinned OmniFold procedure

3.1 Algorithm

OmniFold [3, 4] unfolds by iterated reweighting with machine-learned likelihood ratios, operating on per-event observables rather than on a fixed histogram binning. Each iteration has two steps:

1. **Step 1 (detector space).** A classifier is trained to separate the measured data from the reconstructed simulation (the latter weighted by the current push-weights). Its calibrated output p yields the likelihood-ratio reweight $w = p/(1 - p)$ on every reconstructed simulated event. The weight is *pulled back* to truth level through the event pairing: each simulated event is a (truth, reco) pair, so the reco-space weight attaches directly to the truth record. Truth events with no reconstructed partner (the “misses”) receive a step-1 weight from a regressor trained on the reconstructed subsample, $w(\text{truth features})$ — the native OmniFold acceptance treatment, with no separate efficiency correction.
2. **Step 2 (truth space).** A second classifier is trained to separate the original truth-level simulation from the same sample carrying the pulled-back step-1 weights, yielding a smooth truth-space reweight that becomes the next iteration’s push-weight.

After n iterations the push-weights define an efficiency-corrected, truth-level reweighted simulation ensemble; histogramming it (in any binning, chosen after the fact) gives the unfolded spectrum. The regulariser is the iteration count together with the model capacity of the learners.

3.2 Estimators and configuration

We use gradient-boosted decision trees (GBDTs) rather than the neural networks of the original OmniFold papers; §6 validates this choice with calibration tests and MLP/transformer cross-checks. The engine (`unbinned_unfolding/python/omnifold.py`) supports interchangeable backends; the production configuration is **LightGBM** with 100 trees, 8 leaves, learning rate 0.1 (identical settings for the step-1 classifier, step-2 classifier, and miss regressor). Two alternative backends — **exact** (sklearn `GradientBoosting`) and **hist** (sklearn `HistGradientBoosting`, matched 8-leaf configuration) — are retained for method-dependence studies (§4.3). The production iteration count is $n = 5$; doubling to 10 moves the total cross section by 0.026 % (§6). The central value is a single-run unfold with pinned seeds; an ensemble-mean central value (averaging independent reweighting trials) agrees with it to 0.28 %, and ML stochasticity is budgeted explicitly via the seedscan band (§5).

3.3 MINERvA driver and invariants

The MINERvA driver is `2d-unfolding/unfold_2d_omnifold_unbinned.py`; the higher-dimensional driver (`nd-unfolding/unfold_nd_omnifold_unbinned.py`) generalises it to an axis list whose first two entries are always $(p_T, p_{\parallel})$ (carrying the fiducial gate and the per- p_T flux) and **imports** the same 2D helpers. The unfolded observables are the muon $(p_T, p_{\parallel})$, extended by E_{avail} , q_3 , W in the higher-dimensional runs; the phase space is that of §2.3. The pipeline enforces the following contract:

- measured data are masked to the reconstructed phase space before step-1 training;
- the reco-space target is the POT-scaled, non-negative data – background, with signal *fakes* (reconstructed in phase space, truth outside) included in the subtraction so both sides of the step-1 training are fake-free;
- the unfolded yield is filled with $w_{\text{push}} \times w_{\text{truth}}$ (the MnvTune central-value weight rides along), and is *not* divided by an efficiency histogram — OmniFold’s miss treatment already returns an efficiency-corrected truth spectrum;
- closure runs restrict pseudo-data to events passing both reco and truth selections on the same weight footing as production.

3.4 Cross-section extraction

Unfolded truth-level counts are converted to the differential cross section via

$$\frac{d^n \sigma}{dx_1 \cdots dx_n} = \frac{U}{c \Phi N \text{POT} \Delta x_1 \cdots \Delta x_n} \times 10^4,$$

where U is the unfolded yield, c the (unit) completeness, Φ the per- p_T flux integral (broadcast over the other axes), N the fiducial nucleon count (3.2353×10^{30}), and POT the data exposure (1.057×10^{21}). The implementation is `extract_cross_section_2d` and its N-dimensional generalisation `xsec_nd.py` (self-tested to reproduce the frozen lower-dimensional extractions to $< 10^{-12}$). The flux normalisation enters as $\sigma \propto 1/\Phi$; correctly varying Φ per flux universe was the dominant systematic fix (§5).

3.5 Binning

The reported 2D binning ($14 p_T \times 16 p_{\parallel}$ bins; 205 of 224 reported) follows `minerva_paper_anc/bin_mapping.txt`; the 19 unreported diagonal cells have $p_T^{\text{hi}}/p_{\parallel}^{\text{lo}} > \tan 20^\circ$. The extension axes use E_{avail} : $[0, 0.1, 0.2, 0.4, 0.8, 1.5, 3.0, 10]$ GeV; q_3 : $[0, 0.2, 0.4, 0.6, 0.8, 1.2, 2.0, 100]$ GeV; W : $[0, 1.1, 1.4, 1.8, 2.2, 3.0, 100]$ GeV — each ending in a wide catch bin so the marginals capture the full CC-inclusive tails (a truncated top edge would break the marginalisation anchors of §7.3). Because the binning is applied only after unfolding, these choices can be revisited without re-running the unfold. The full edge table and reported-bin mapping are recorded in `2D_OMNIFOLD_REFERENCE.md` and App. B.

Although OmniFold uses no response matrix, the detector smearing the unfold must undo is conveniently summarised by the truth→reco migration on the analysis binning (Fig. 2): the diagonal purity is ~ 0.6 (median) on every axis, i.e. bin-to-bin migrations are substantial and a bin-by-bin correction would be badly biased — the regime unfolding is for.

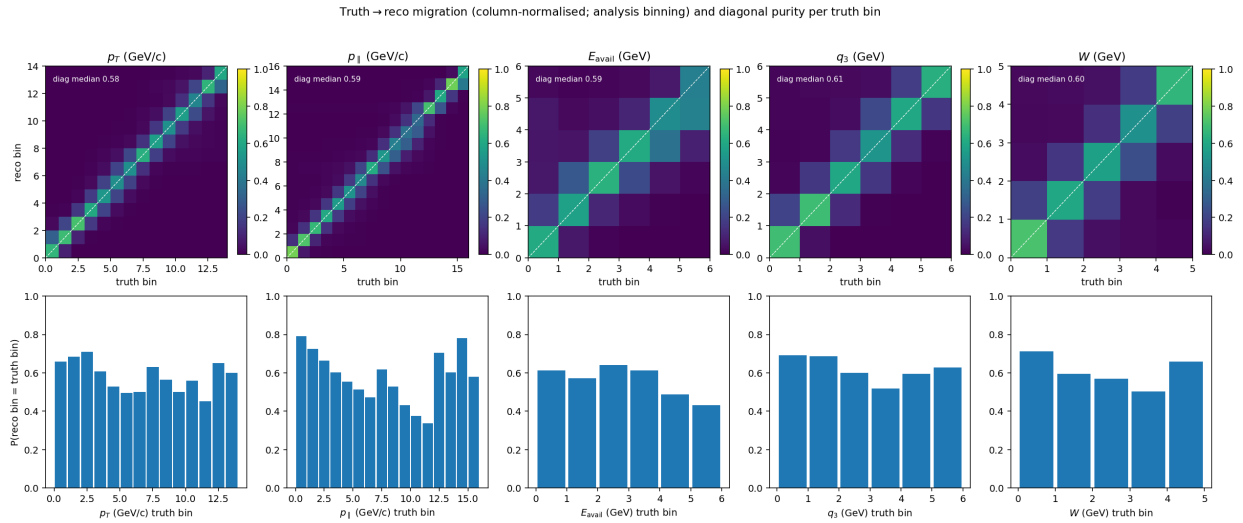


Figure 2: Truth→reco migration of the simulated signal on the analysis binning (top: column-normalised maps; bottom: diagonal purity per truth bin) for the five observables. Median diagonal purity is 0.58–0.61 per axis. Shown for illustration only — the unfold itself is unbinned.

4 Results

4.1 Double-differential cross section

Figure 3 shows the unfolded $d^2\sigma/(dp_T dp_{\parallel})$ and its 1D projections. The total cross section is $3.073 \times 10^{-38} \text{ cm}^2/\text{nucleon}$ (published: $3.039 \times 10^{-38} \text{ cm}^2/\text{nucleon}$).

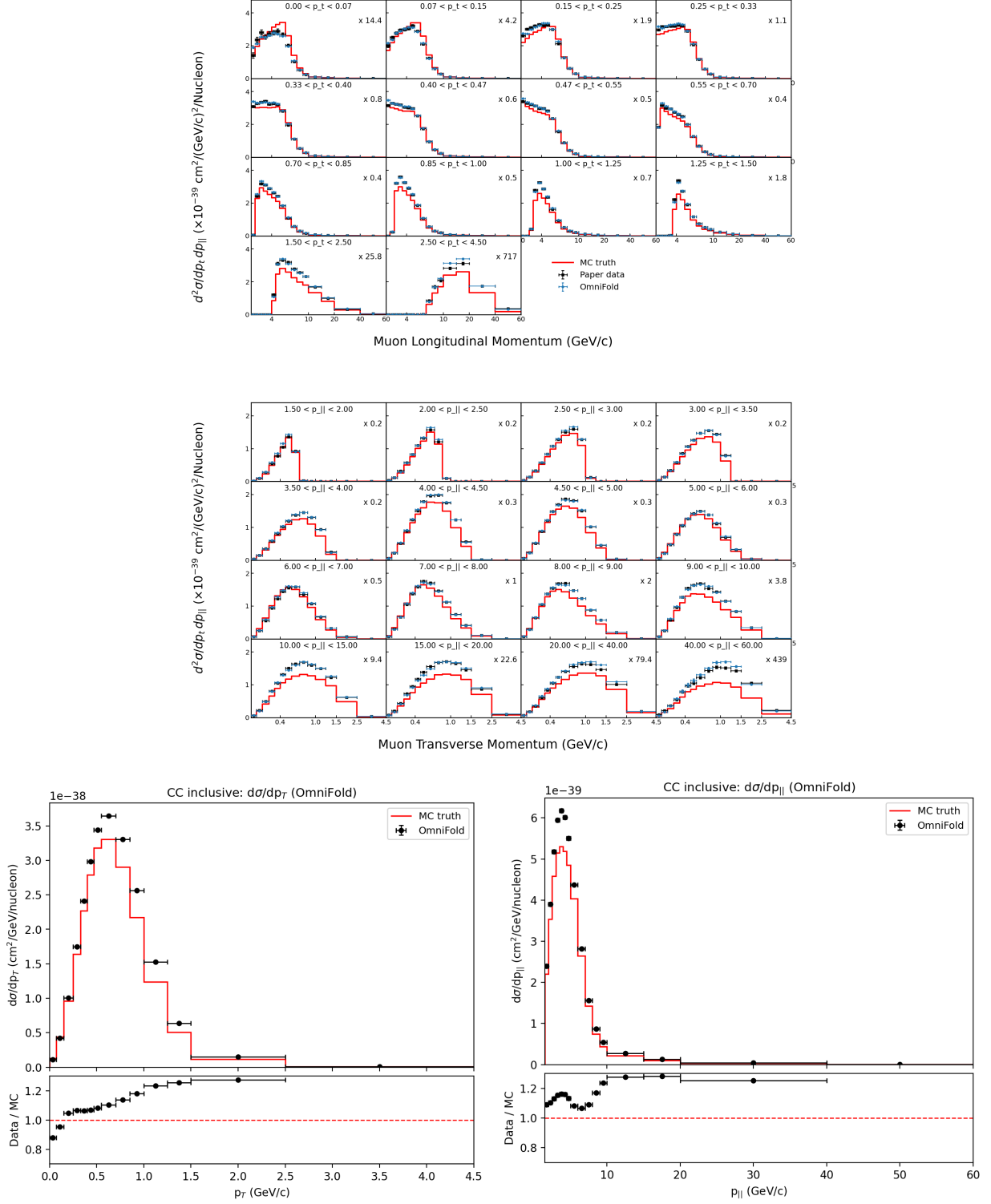


Figure 3: Unfolded 2D cross section (top, Fig. 13 style) and its $p_T/p_{||}$ projections (bottom).

Quantity (205 reported bins)	Value
$\sigma_{\text{tot}}(\text{ours})/\sigma_{\text{tot}}(\text{paper})$	1.011
Median bin ratio (ours/paper)	1.006
Bins within 5 %/10 %/20 %	77.6/94.1/98.5 %
χ^2/ndf , paper covariance	3.661
χ^2/ndf , combined covariance (paper + ours)	1.481
Pull mean / RMS (paper covariance)	0.089/0.598

Table 1: Agreement with arXiv:2106.16210. Per-bin pulls (Fig. 4) are sub- 1σ .

4.2 Comparison to the published measurement

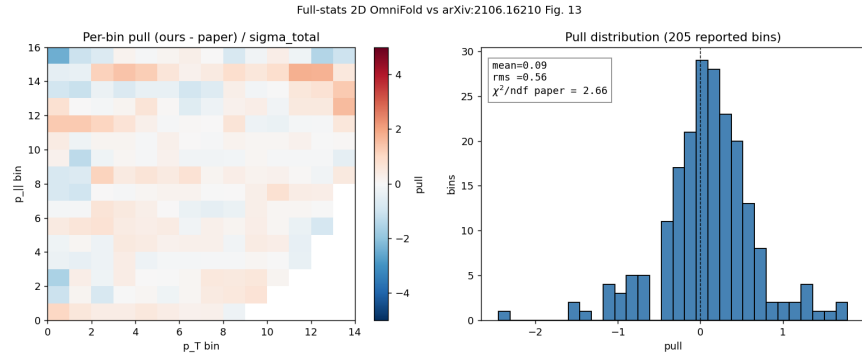


Figure 4: Per-bin pull (ours $-$ paper)/ σ on the 14×16 grid and its distribution.

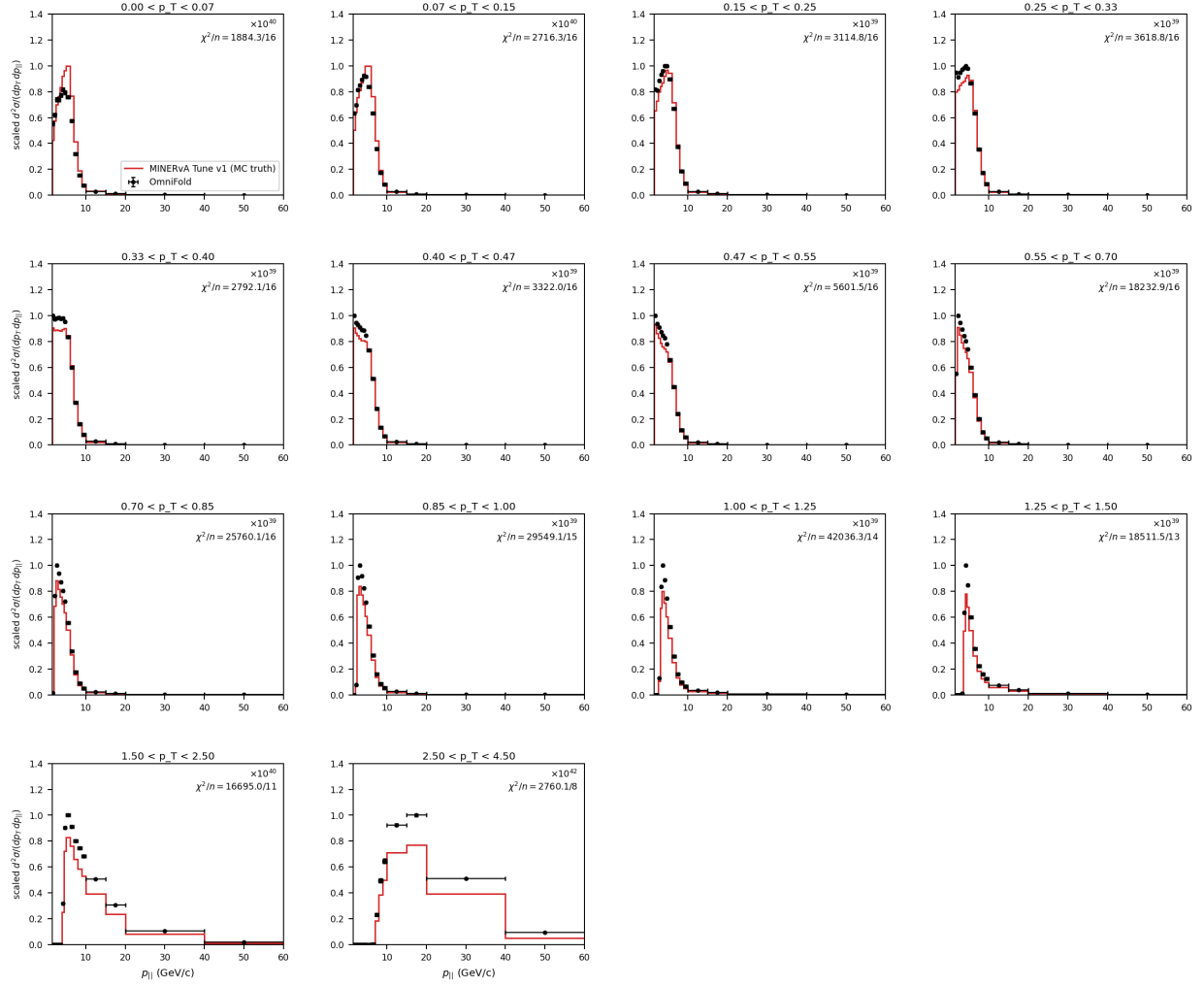
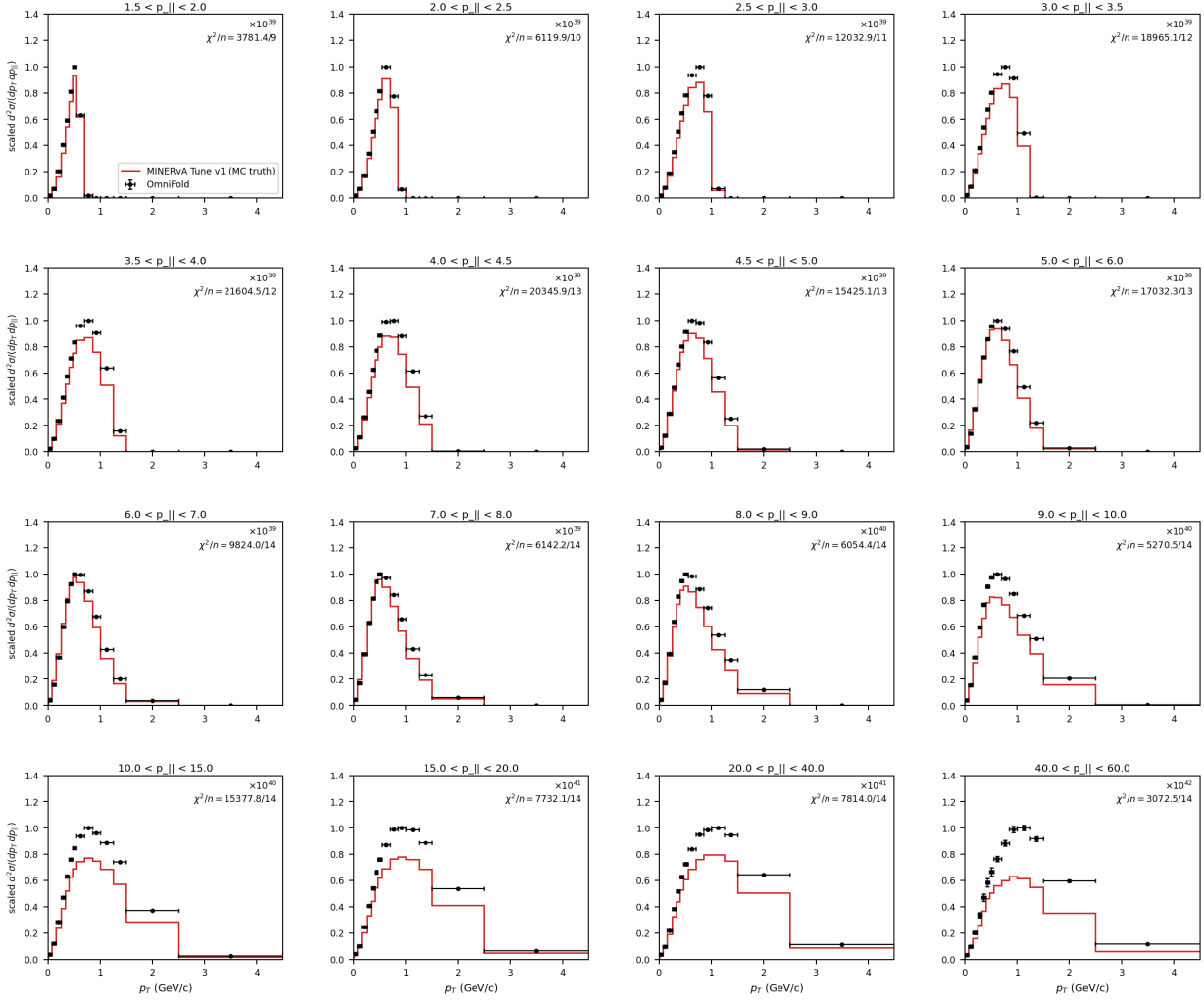


Figure 5: Bin-by-bin comparison to the published measurement: the unfolded cross section (points) overlaid on the published data (histogram) in slices of p_T versus $p_{||}$.

Figure 6: Same comparison as Fig. 5, sliced in $p_{||}$ versus p_T .

4.3 Anatomy of the $\chi^2 = 3.66$ tension

The central values agree (pull RMS 0.598, so a diagonal-only χ^2/ndf would be 0.37), yet the full paper-covariance χ^2/ndf is 3.66: a $10\times$ inflation living entirely in the off-diagonal structure of the published covariance. The eigenmode decomposition (Fig. 7) shows the tension is carried by moderate- λ modes with $5\text{--}6\sigma$ eigen-pulls (the 10 smallest- λ modes carry only 3%; keeping the 73 best-measured directions still gives 1.42), so it is *not* a small- λ inversion artifact. It is *shape*, not normalisation ($\chi^2_{\text{norm}} = 0.10$ vs $\chi^2_{\text{shape}} = 750$), localised to the low- p_T cross-section peak ridge where the flux and muon-energy bands dominate. A ~ 1 -unit methodological band comes from the GBDT estimator (exact-GBT 3.66 vs HistGBT 2.70 ± 0.04 vs LightGBM 2.65; iteration count is negligible, $5 \rightarrow 10$ iters move it by $+0.01$). Full treatment: App. B.5. The eigenmode-localisation maps and the additional model-comparison ratio maps are shown in Figs. 7–8.

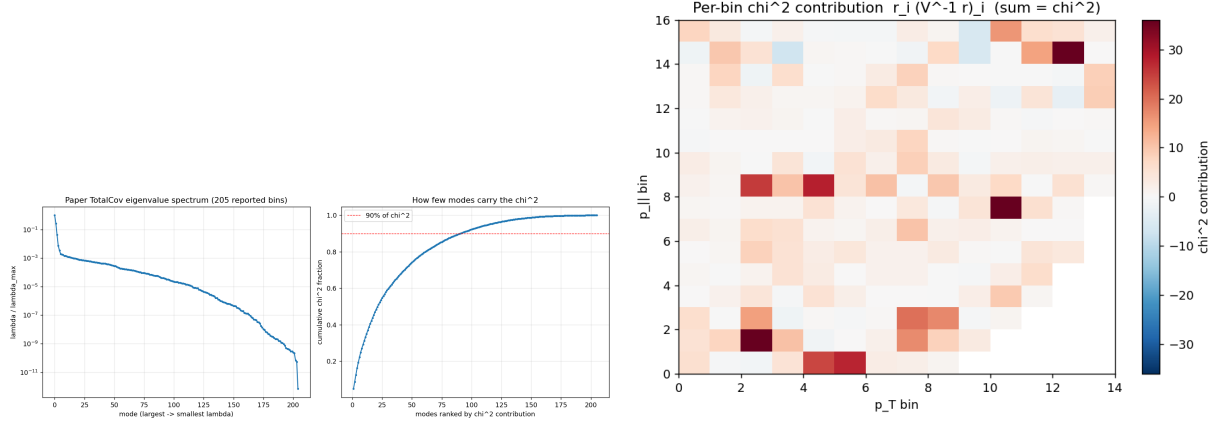


Figure 7: Tension anatomy: eigenvalue spectrum with cumulative χ^2 (left) and the per-bin χ^2 -contribution map (right).

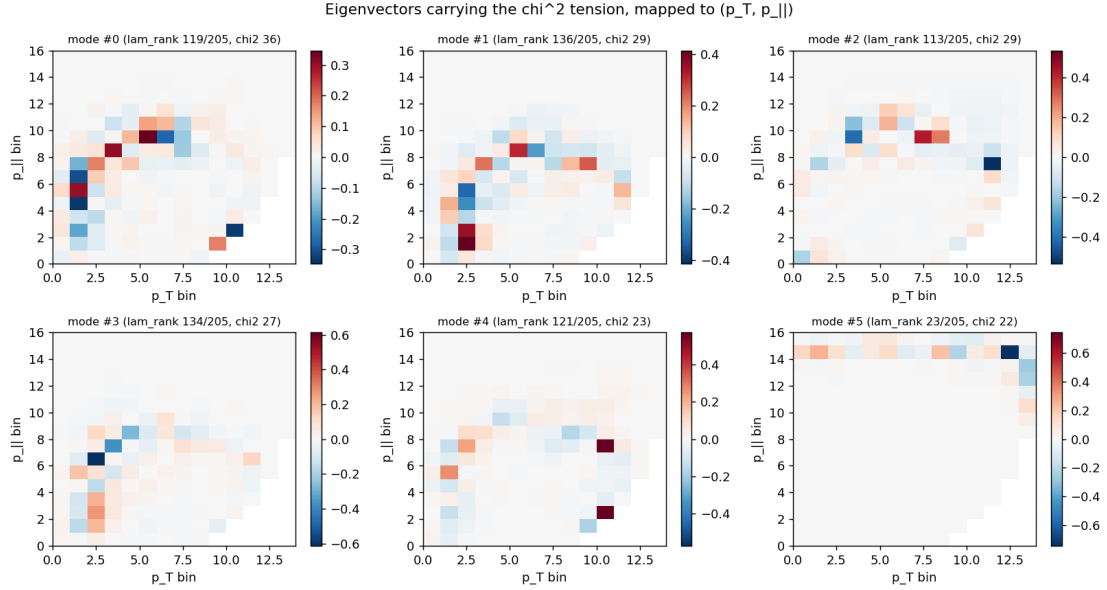


Figure 8: Bin-space maps of the leading tension-carrying eigenmodes of the published covariance: the 5–6 σ eigen-pulls live on the low- p_T cross-section-peak ridge where the flux and muon-energy bands dominate.

4.4 Comparison to GENIE MINERvA Tune v1

Both the published data and our unfolded result are compared to the GENIE 2.12.6 [17] MINERvA Tune v1 prediction shipped with the ancillary (Fig. 9). The tune under-predicts the total by $\sim 9\%$ ($\sigma_{\text{tune}}/\sigma_{\text{data}} = 0.91$), concentrated in the low- p_T peak, and is strongly disfavoured by the data covariance (χ^2/ndf : data vs tune 33.0, ours vs tune 26.5). Our unfolded result tracks the data far more closely than the model does. (`compare_to_models.py`.) The published analysis [1] additionally compares to NuWro [18], GiBUU [19] and NEUT [20]; because OmniFold delivers a per-event reweighted truth ensemble, any such generator prediction can be compared to our result

in *any* projection or dimension without re-unfolding — a capability we exploit for the 3D extension (§7).

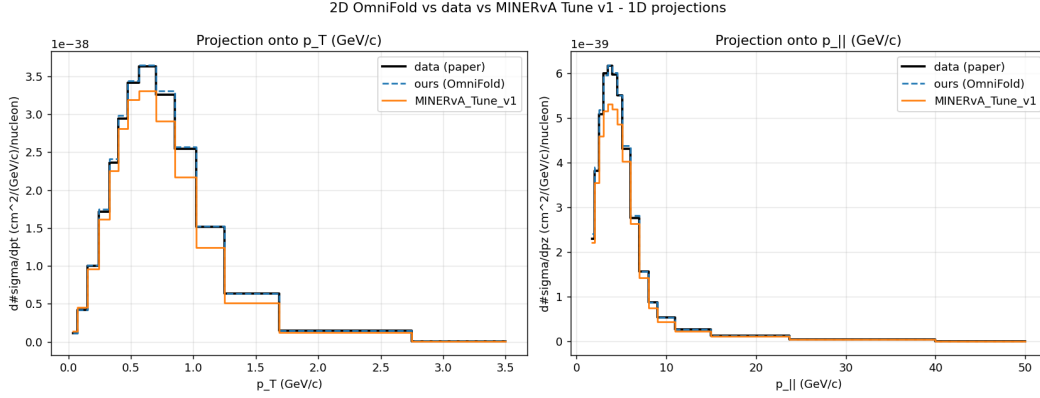


Figure 9: Our result (dashed) on top of the published data (black); MINERvA Tune v1 (orange) dips below in the peak.

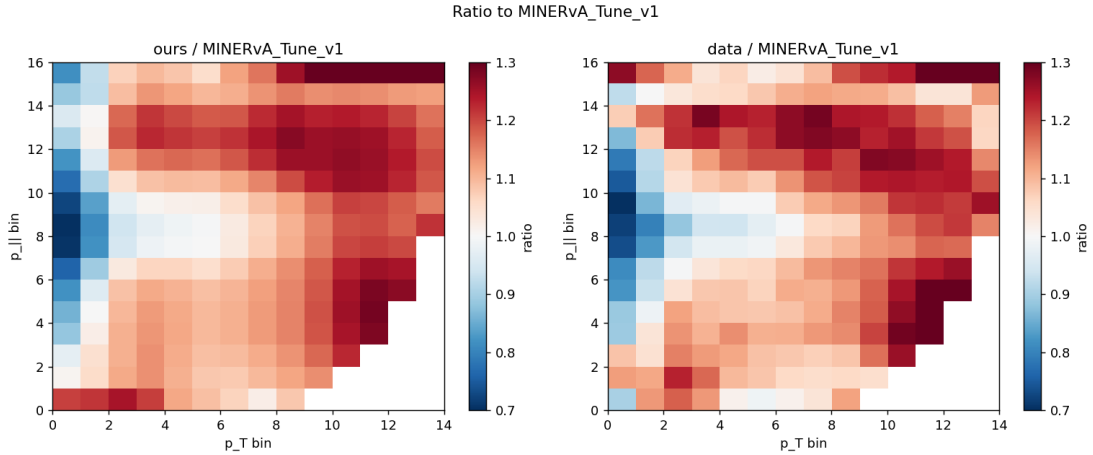


Figure 10: Bin-by-bin ratio maps on the $(p_T, p_{||})$ grid: ours/paper (left) and Tune v1/paper (right). The tune’s deficit is concentrated in the low- p_T peak; our result is flat near unity.

5 Uncertainty quantification

The uncertainty budget combines three independent blocks — systematic (universe), statistical (bootstrap), and ML (seedscan) — summed assuming independence (distinct RNGs / physics sources). This section describes each block and their combination for the 2D measurement; the 3D and 4D campaigns (§7.6, §7.8) repeat the identical procedure on the higher-dimensional grids. Full mathematical procedures, with worked examples on the reported numbers, are in App. B. The headline is that our *standalone* combined budget matches the published total.

Component	N	$\sqrt{\text{Tr } C}$	per-bin median rel. σ
ML noise (lgbm seedscan)	10	5.06×10^{-41}	0.166 %
Statistical (Poisson bootstrap)	300	1.82×10^{-40}	0.549 %
Systematic (universe sweep + 1.4 % norm)	187+1	3.21×10^{-39}	6.830 %
Combined (block sum)	—	3.22×10^{-39}	6.865 %
Paper TotalCov (reference)	—	2.68×10^{-39}	6.86 %

Table 2: Stage-2 UQ envelope on the 205 reported bins. The standalone combined budget (6.865 %) matches the published total (6.86 %).

5.1 Systematic universes

The detector and model systematics use the MAT-MINERvA universe machinery: 44 error bands (6 lateral — shifted reconstructed kinematics — and 38 vertical — per-event reweights), 187 universes in total, identical to the published analysis’s `GetStandardSystematics` set. Each universe is a *full re-unfold*: the event loop dumps the universe’s weights and (for lateral bands) shifted observables, and the entire OmniFold chain runs on them with the central value’s seeds and configuration, so the systematic propagates through the learned reweighting itself rather than through a frozen response. Pair ($\pm 1\sigma$) bands use the MINERvA sum-of-squares convention $\frac{1}{2}(\delta_+ \delta_+^\top + \delta_- \delta_-^\top)$; multi-universe bands (the 100-universe PPFX flux multisim) use the sample covariance. The construction is MAT-conformant (`MnvVertErrorBand::CalcCovMx`; verified byte-for-byte against MAT `MnvH1D::GetTotalErrorMatrix`, max rel. diff $\sim 10^{-17}$). Top bands: Flux 4.99 %, Muon_Energy_MINOS 2.31 %, MinosEfficiency 1.47 %, Muon_Energy_MINERvA 1.29 %. Figure 11 shows the grouped fractional-uncertainty projections in the published Fig. 6/7 style (the full 205×205 covariance projected exactly, $C_{1D} = PC_{2D}P^\top$, before dividing by the 1D central value).

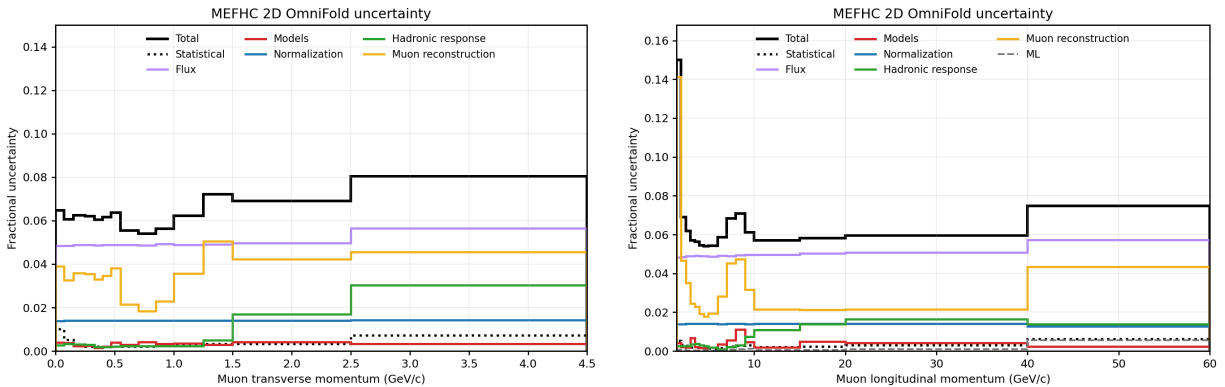


Figure 11: Fractional uncertainty on the 2D cross section projected onto p_T (left) and $p_{\parallel}$ (right), grouped by source in the style of Figs. 6–7 of Ref. [1]: flux, interaction models, normalisation, hadronic response, muon reconstruction, and the statistical band.

Flux $1/\Phi$ normalisation. The dominant flux term enters as $\sigma \propto 1/\Phi$; varying Φ per flux universe (rather than dividing every universe by the CV Φ) raised the Flux band from $\sim 1\%$ to 4.99 % and brought the standalone combined budget to 6.87 %. The 1.4 % target-nucleon normalisation is a fully-correlated rank-1 band. The 100-universe flux covariance has a participation-ratio effective

rank of ≈ 1 (one coherent normalisation mode carries 99.6 % of the variance), and its leading mode correlates at $\rho \approx 0.9$ with the rank-1 muon-energy-scale band — the rederived Bashyal [21] joint block (App. A, item 1).

5.2 Statistical uncertainty: Poisson bootstrap

The statistical block is a 300-replica per-event Poisson(1) weight bootstrap applied jointly to data and simulation, with two independent sub-RNGs (data and MC are independent variance sources), the MC reco and truth weights riding a *single* per-event draw (preserving the within-event migration correlation), and the GBDT seed pinned per replica so the bootstrap and seedscan variances are separable. Each replica is a full re-unfold; the covariance is the sample covariance over replicas (Fig. 12). Our bootstrap block is genuinely smaller than the published statistical block ($\approx 2.5\times$ by trace) — an unbinned statistical efficiency confirmed by the data-only stream of the data/MC split bootstrap, not a missing term (App. A, item 1).

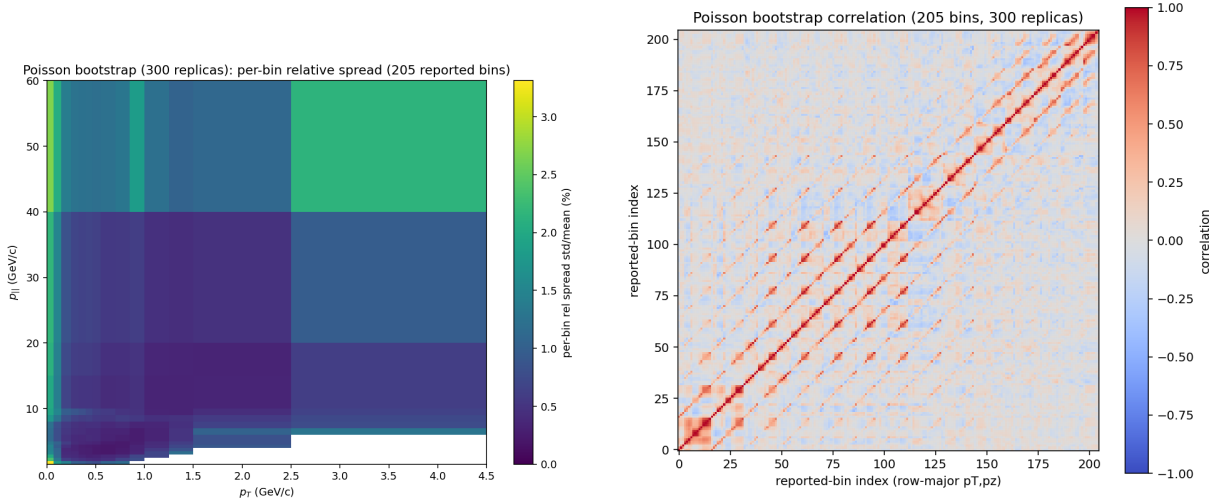


Figure 12: 300-replica Poisson bootstrap: per-bin relative spread of the unfolded 2D cross section (left) and the bootstrap correlation matrix on the 205 reported bins (right).

5.3 ML/optimisation uncertainty: seedscan

ML stochasticity is budgeted by re-unfolding with 10 independent estimator seeds at fixed data and fixed universe (Fig. 13); the sample covariance over seeds is the ML block. It is the smallest block (median 0.166 % per bin). Two refinements harden it: (i) a *train/test-split* seedscan, which also re-draws the internal training split, gives a band $1.24\times$ the pure-seed band, moving the combined 3D budget by only $+0.04\%$ in $\sqrt{\text{Tr}}$ — the published budgets use the split-inclusive band; (ii) the ensemble-mean central value (averaging the seed ensemble) agrees with the production single-run CV to 0.28 %.

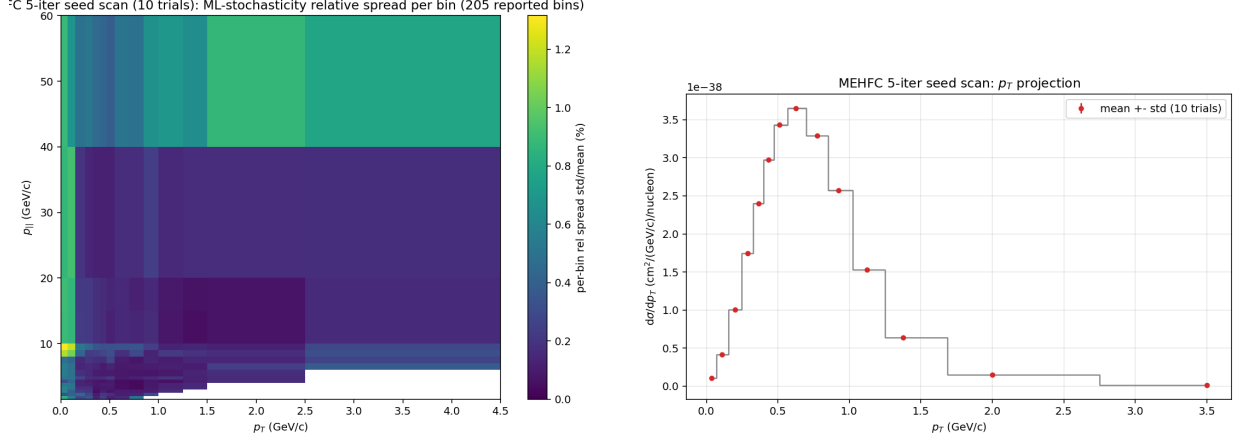


Figure 13: ML seedscan: per-bin relative spread over 10 estimator seeds (left) and the resulting fractional band projected onto p_T (right).

5.4 Combination and the unified-throw test

The published covariance is the block sum $C = C_{\text{syst}} + C_{\text{stat}} + C_{\text{ML}}$, which assumes the blocks are independent *and* that the unfolding responds linearly to simultaneous perturbations (so that per-band covariances add). The second assumption is tested directly by the *unified throw* (§7.8): joint re-unfolds with all reweightable knob bands and the flux multisim perturbed simultaneously per event. In 2D/3D the block sum is consistent with the throws; on the 4D grid it *underestimates* the vertical systematic by a factor ~ 2 , concentrated in the high- p_T , lowest- E_{avail} corner, and the published 4D covariance therefore adopts the unified-throw magnitude via a PSD-safe per-bin inflation (§7.8). This is a direct, quantitative validation (and, where broken, correction) of the linearity assumption underlying the standard block-sum construction.

χ^2 caveat. The combined paper+ours $\chi^2/\text{ndf} = 1.481$ double-counts systematics shared with the paper (flux/GENIE/MINOS); the ours-only inverse-covariance χ^2 is evaluated only with a truncated-spectral convention on the well-constrained subspace. We therefore quote per-bin pulls, the regularisation scan, and the truncated result together rather than treating an unconstrained pseudo-inverse as a standalone goodness-of-fit (App. A).

6 Validation

Closure. Three closure families, all passing (1A lgbm 5-iter; thresholds in parentheses):

- **Truth-reweight** (multiply truth and reco weights by the same factor): gaussian- p_T residual 0.046 %, tilt- $p_{\parallel}$ 0.013 % (thr 1.5 %) — injected truth recovered.
- **Hidden-variable** (bias on $\Delta p_T = p_T^{\text{reco}} - p_T^{\text{truth}}$, a resolution variable not in the feature set): leakage linear in amplitude, $A = 0.05/0.10/0.30 \rightarrow 0.57/1.14/3.43$ % (thr 3.0 %).
- **Alt-model** (train on CV, pseudo-data from an alternative universe): MaCCQE 0.068 %, Flux 0.376 % (thr 2.0 %).

Bottom-line test. Following the unbinned-unfolding guide [22], the unfolded result must recover the data’s deviation from the prior with a residual far smaller than the feature itself (the unfolding cannot manufacture discriminating power). Using the closure runs, in the bins carrying the injected feature the unfolded result tracks the injected truth to $\approx 10\times$ better than the feature size: 2D (gaussian-bump truth reweight) residual 1.69 % vs injected 17.2 % (ratio 0.098); 3D (+30 % E_{avail} bump, E_{avail} projection) 1.84 % vs 18.1 % (ratio 0.102). Both pass. (The naive data-vs-prior form—reco vs truth χ^2 —instead has truth>reco, as expected for a high-resolution measurement where detector smearing lowers the reco baseline and a stat-only diagonal omits the unfolding covariance; the proper full-covariance goodness-of-fit is the χ^2 -vs-paper of §4.3 and the generator comparisons.)

Unbinned goodness-of-fit. As a binning-free check that the unfold actually removes the data $\leftrightarrow$ MC mismatch, we use a classifier two-sample test (C2ST): a classifier is trained to separate the data from the (reweighted) simulation in the full unbinned feature space, and its test-set accuracy is compared to the chance value 0.5. *Before* unfolding the two are separable — accuracy 0.523, AUC 0.535, $z = 33.4$ ($p \approx 5 \times 10^{-244}$); *after* unfolding the classifier is driven to chance — accuracy 0.501, AUC 0.501, $z = 1.4$, $p = 0.17$. Data and unfolded simulation are thus statistically indistinguishable with no detectable residual structure: the unbinned analogue of a binned goodness-of-fit, and it passes.

Coverage. The UQ note records a 200-toy closure+bootstrap campaign: mean coverage 68.71 % vs Gaussian 68.27 %; mean $|\text{residual}|/\sigma$ 0.794 vs $\sqrt{2/\pi} = 0.798$; and 97.6 % of bins meeting the 65 % target. The current checkout retains only the 20-toy stage-1 summary under `uq/coverage/`; restore the 200 toy ROOTs before independently regenerating this number from disk.

Stability. 5-iter total σ within 0.026 % of 10-iter (per-bin shape RMS 1.54 %); HistGBT and exact-GBT agree on total σ to 0.04 %. Efficiency (Fig. 16) reproduces the published Fig. 5.

Classifier choice (GBDT vs. NN). Published OmniFold analyses use neural-network classifiers [3, 22]; we use gradient-boosted decision trees, whose output must still be a calibrated posterior for $w = p/(1 - p)$ to be valid. On the step-1 reco classifier (data vs. MC reco) the reliability curve is near-diagonal (Brier ≈ 0.25 at AUC ≈ 0.537 : data and reco MC are nearly indistinguishable, so the reweight is a modest correction). Across the $(p_T, p_{\parallel})$ bins the true data/MC ratio still spans $\approx 100\%$; the GBDT recovers it to 4.7 % median, at least as well as a dense MLP (20.9 %, untuned), and the two reweights agree with bin-wise correlation 0.92 — the learned reweight is robust to the classifier family. A full scalar keras-MLP OmniFold cross-check on the 3D inputs gives total ratio 1.0078 versus the GBDT reference and median projection differences of 0.7 % to 1.4 %, consistent with the assigned ML band. As a stronger family cross-check, a vendored PET (point-cloud transformer [23]) unfold was run on the actual per-hadron point clouds — the non-muon reconstructed clusters (`cluster_energy`, `cluster_pos`, `cluster_z`, filtered on `cluster_isMuontrack`) read by `CVUniverse::GetRecoClusters`, after an earlier attempt that used the wrong, mostly-empty `ExtraEnergyClusters_*` branch was discarded. Area-normalized against the frozen 4D GBDT result, the PET unfold reproduces the scalar shape to a median 2.3 % to 3.9 % per bin (p_T 3.9 %, $p_{\parallel}$ 2.4 %, E_{avail} 2.6 %, q_3 2.3 %; Fig. 15). In its role here this is a shape/method-robustness check: it shows the learned reweight is stable across both the classifier family (GBDT vs. MLP vs. transformer) and the input representation (engineered scalars vs. raw point cloud). Section 9 elevates the PET track to a full-statistics, absolutely-normalised cross section with its

own closure gate and systematic budget. Figure 14 shows the underlying classifier-calibration diagnostics (reliability curve and binned ratio recovery) for the GBDT and MLP step-1 classifiers.

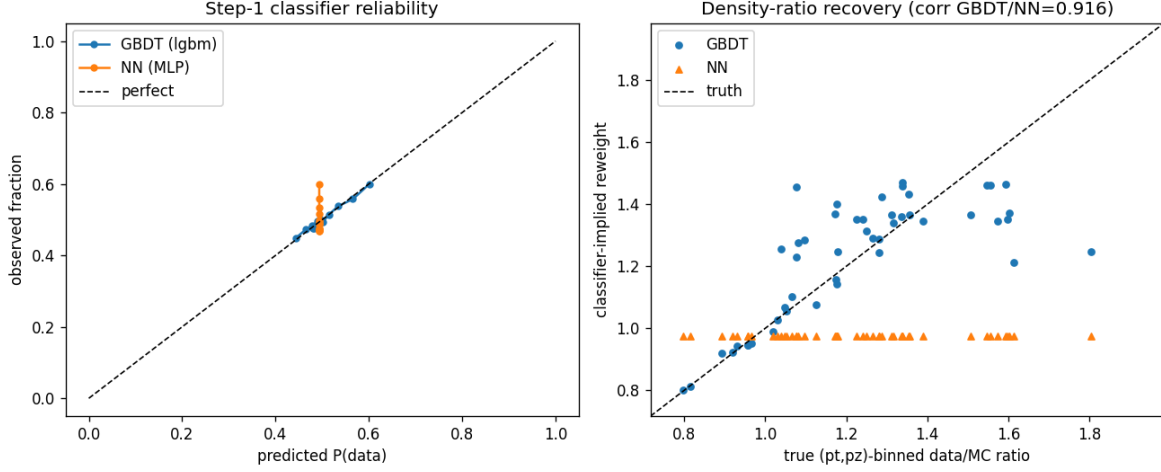


Figure 14: Classifier-calibration diagnostics on the step-1 (data vs. reconstructed MC) task: reliability curves, weight distributions, and binned data/MC ratio recovery for the production GBDT and an untuned dense MLP. The GBDT is near-diagonal in reliability and recovers the binned ratio to 4.7% median.

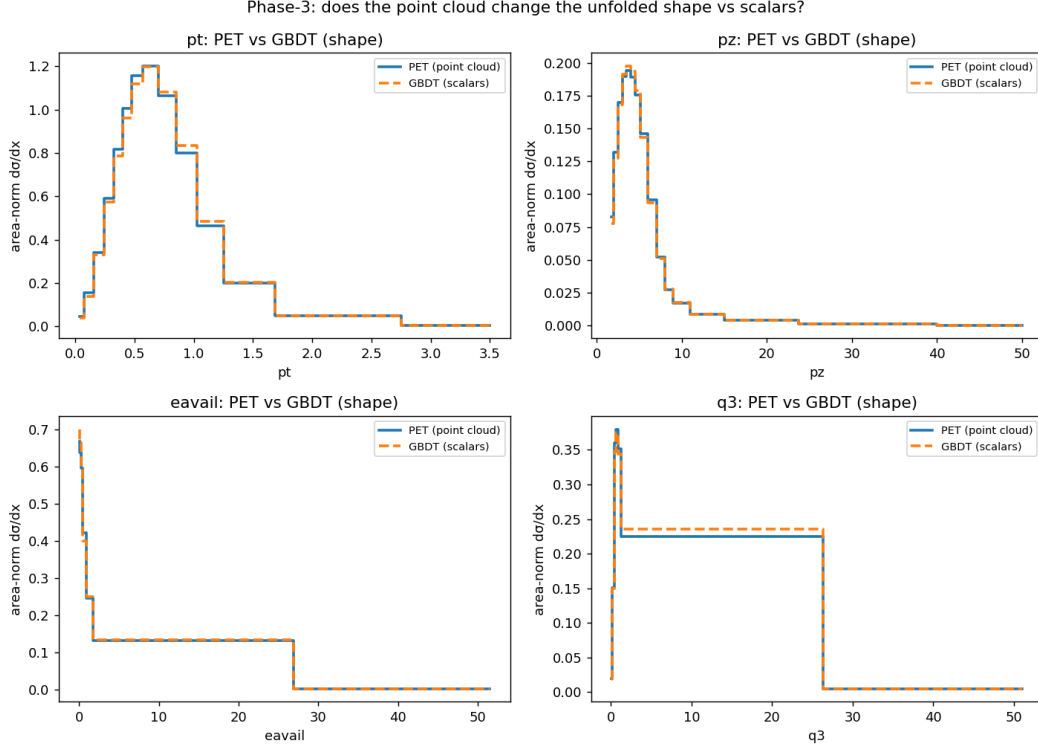


Figure 15: Method-robustness check: area-normalized 4D unfolded shapes from the PET point-cloud transformer (solid) versus the production scalar GBDT (dashed), projected onto each axis. Median per-bin shape differences are 2.3 % to 3.9 %; the only visible deviation is the wide q_3 catch bin, a subsample-statistics effect. Shape-only comparison (PET on a 2M-event subsample), not an independent physics result.

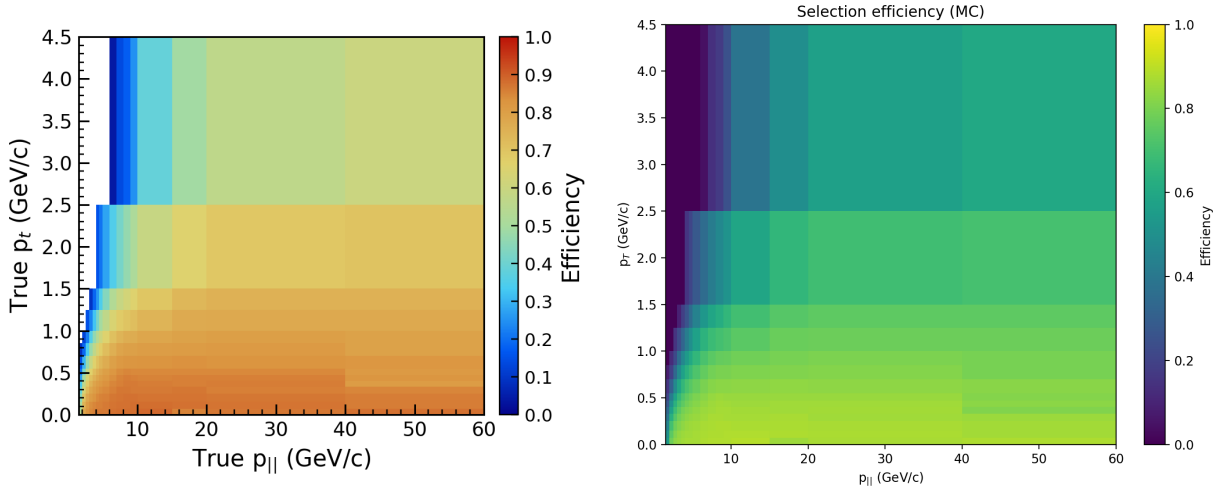


Figure 16: Selection efficiency: 1D projections in the published Fig. 5 style (left) and the full $(p_T, p_{||})$ efficiency map (right). Efficiency is diagnostic only — OmniFold’s miss treatment performs the acceptance correction natively (§3.1).

7 Higher-dimensional extension: available energy

The enabling property of unbinned OmniFold is that adding an observable is adding an input feature: the step-1/step-2 classifiers simply gain a column, with no migration-matrix or binning penalty. There is no practical D’Agostini IBU analogue, which makes this the primary physics motivation for the technique. We exercise it by extending the measurement to $d^3\sigma/(dp_T dp_{\parallel} dE_{\text{avail}})$ with available energy as a third axis.

The significance is clear from the measurement record: every published MINERvA differential cross section has unfolded at most two kinematic variables simultaneously (Fig. 17, Table 4), the double-differential measurements all relying on binned IBU. A simultaneous *three*-observable unfold of MINERvA data is, to our knowledge, without precedent, and is a direct consequence of the unbinned formulation.

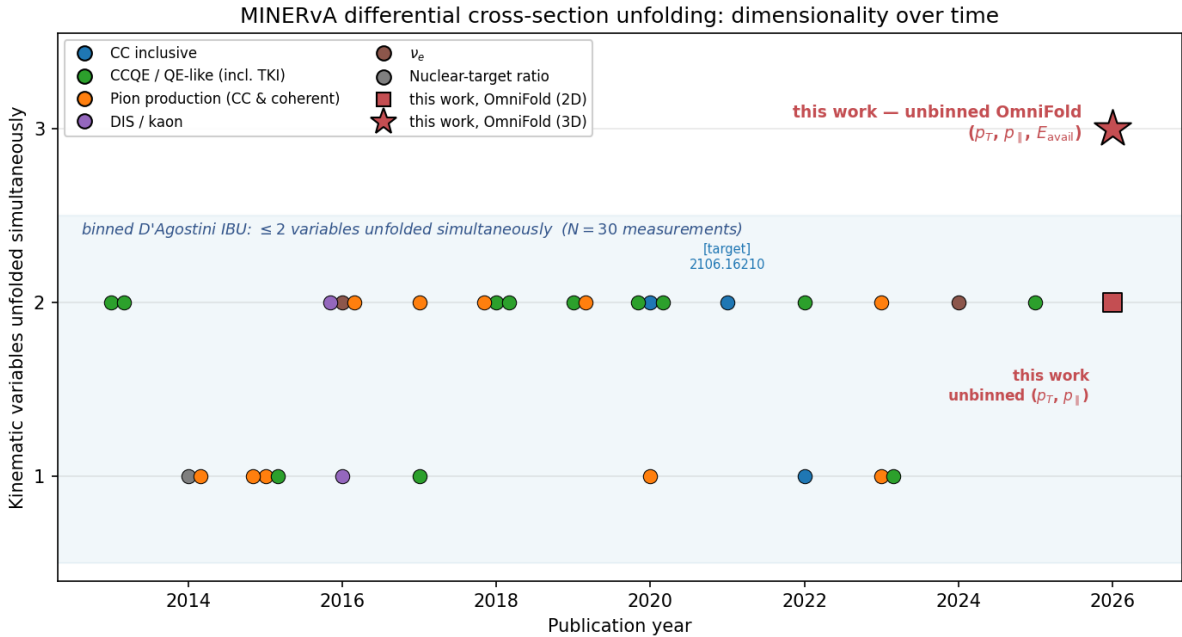


Figure 17: Dimensionality of the full set of MINERvA differential cross-section measurements over time (number of kinematic variables unfolded *simultaneously*; a double-differential $d^2\sigma/dx dy$ counts as two), coloured by channel. All $N = 30$ prior measurements use binned D’Agostini IBU and reach at most two variables; this work’s unbinned OmniFold adds a third. Source: the MINERvA data-release page; full list in Table 4. Dimensionality is classified from each analysis’s headline differential result; flux, reconstruction, generator-tuning and total-only entries are excluded.

7.1 Definitions and pipeline

Available energy follows the MINERvA low-recoil convention (Ref. [24], Eq. 4): truth $E_{\text{avail}} = \sum_p KE + \sum_{\pi^\pm} KE + \sum_{\pi^0, \gamma} E$ (excluding neutrons and the muon), via `CVUniverse::GetEAvailableTrue()`; reco E_{avail} is the calorimetric tracker+ECAL estimate scaled by 1.17, `CVUniverse::NewEavail()`. Both accessors were added to the event loop, which was re-run over all twelve ME FHC playlists to append the E_{avail} branches alongside the existing $p_T/p_{\parallel}$ branches. The unfolding driver is the 2D pipeline with a three-column feature stack $(p_T, p_{\parallel}, E_{\text{avail}})$; the cross-section extraction generalises to a 3D Jacobian $\Delta p_T \Delta p_{\parallel} \Delta E_{\text{avail}}$ with the flux Φ still applied per p_T (broadcast over $p_{\parallel}$ and

E_{avail}). The E_{avail} binning $[0, 0.1, 0.2, 0.4, 0.8, 1.5, 3.0, 100]$ GeV ends in a wide catch bin so that the E_{avail} -marginal captures the full CC-inclusive recoil tail; the global OmniFold completeness is $c = 1.0000$, as in 2D.

7.2 The available-energy spectrum

Figure 18 shows the unfolded $d\sigma/dE_{\text{avail}}$ (the 1D projection of the 3D result). It falls monotonically from $2.19 \times 10^{-38} \text{ cm}^2/\text{GeV}/\text{nucleon}$ at the low-recoil peak to 6.8×10^{-41} in the catch bin — the expected CC-inclusive shape, with most of the rate at $E_{\text{avail}} \lesssim 0.8 \text{ GeV}$.

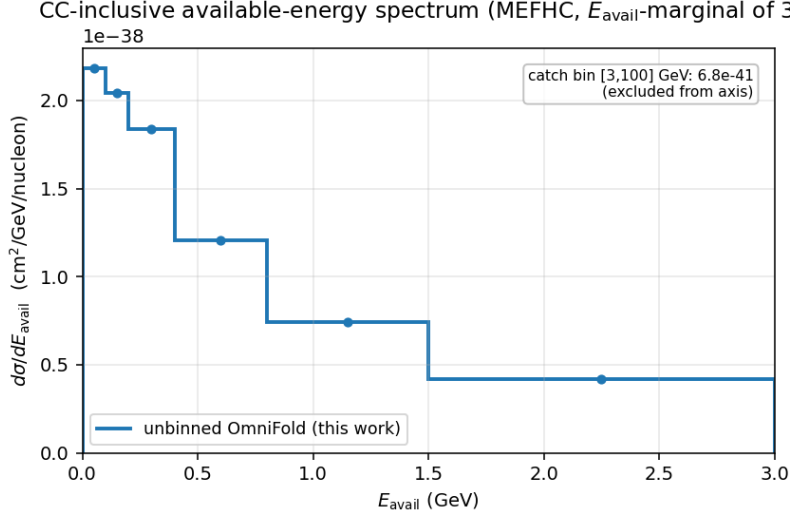


Figure 18: Unfolded available-energy spectrum $d\sigma/dE_{\text{avail}}$, the 1D projection of the 3D measurement. The wide $[3, 100]$ GeV catch bin (value annotated) is excluded from the axis.

7.3 Validation against the 2D measurement

With no published 3D reference, the validation anchor is that marginalising the 3D cross section over E_{avail} must reproduce the 2D result of §4. The marginal recovers the 2D *normalisation* closely: total σ agrees to $+0.95\%$ and the per-bin marginal/2D ratio has median 1.0016. The *shape* agreement is looser — the marginal versus the published covariance gives $\chi^2/\text{ndf} = 4.98$, above the frozen 2D values (3.66 default, 2.65 LightGBM-CV), driven by a $\sim 4.4\%$ per-bin scatter that the third axis injects into the $(p_T, p_{\parallel})$ marginal (Fig. 19).

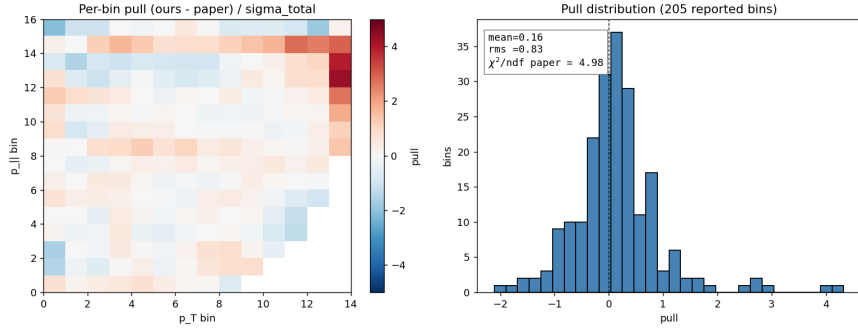


Figure 19: Per-bin pull of the E_{avail} -marginal against the published covariance, on the 14×16 grid and its distribution.

7.4 Closure: the scatter is physical, not a method bias

A reweight closure isolates whether that 4.4% scatter is method bias or real information. We inject a known +30% Gaussian bump in *truth* E_{avail} (centre 0.3 GeV, width 0.15 GeV) into the pseudo-data and require the 3D unfold to recover it. It does: the recovered step-2 mean weight is 1.048 against the injected 1.0485, and the residual of the unfold against the reweighted-truth reference is, on the E_{avail} -marginal, median 0.9999 with standard deviation 0.0006 (0.06%); the per-3D-bin residual standard deviation is 3.2% (MC statistics in sparse high-recoil corners). The closure marginal scatter is thus $\sim 70\times$ tighter than the 4.4% data-versus-2D scatter. The method is therefore unbiased on the new axis, and we read the elevated marginal χ^2 as *genuine data* $\leftrightarrow$ *MC structure that the available-energy dimension exposes* — information that is invisible to the 2D measurement — rather than a pipeline artefact (the total normalisation matches to $< 1\%$).

This establishes the 3D framework and its first closure.

7.5 Generator comparison in 3D

Because the published ancillary ships only binned 2D model histograms, a quantitative comparison of the 3D result (or of arbitrary projections) requires generating truth-level events ourselves. No detector simulation is needed — the unfolded result already lives at truth level — so a generator prediction is obtained by histogramming its truth $d^3\sigma$ in the same fiducial phase space and the same truth E_{avail} definition. We compare four predictions:

- **GENIE 2.12.10 central value** — generated with `gevgen` on a CH target with the MINERvA ME FHC flux and the DefaultPlusValenciaMEC configuration [17].
- **MINERvA Tune v1** — the analysis tune (base GENIE + RPA, low-recoil 2p2h, and non-resonant-pion reweights), extracted directly from the event-loop MC truth (the `w_truth` weights), which reproduces the shipped ancillary curve to 0.01% in normalisation.
- **NuWro 21.09** [18] — an independent generator (C target), the same flux and phase space.
- **GiBUU 2019** [19] — a transport-theory generator (C target), built and run *natively* on the compute system (no container), same flux and phase space.

NEUT [20] is not included (no accessible build at the time of writing). Each prediction is normalised by its own flux-averaged total CC cross section per nucleon ($\langle\sigma\rangle \approx 3.6\text{--}4.0 \times 10^{-38} \text{ cm}^2$).

Figure 20 overlays the four on the unfolded result, with the combined-covariance systematic band of §7.6 on the data. The flux-averaged totals in the measurement phase space order as GiBUU (2.22×10^{-38}) < NuWro (2.34×10^{-38}) < GENIE CV (2.52×10^{-38}) < Tune v1 (2.71×10^{-38}) < data ($3.08 \times 10^{-38} \text{ cm}^2/\text{nucleon}$): all four models under-predict the rate, by 12 % to 28 %. On $(p_T, p_{\parallel})$ the four track the data shape closely; the discriminating axis is E_{avail} , where NuWro, Tune v1 and GiBUU *suppress* the low-recoil quasielastic peak below bare GENIE (RPA and nuclear effects) — yet the data sit *above* all four in the lowest- E_{avail} bins. This low-available-energy excess is invisible to the 2D measurement and is a direct statement of the unbinned method’s high-dimensional reach.

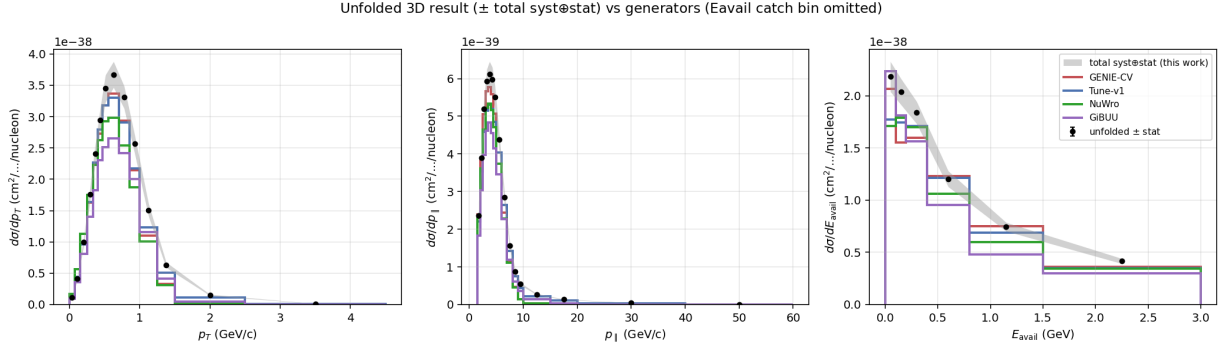


Figure 20: 1D projections of the unfolded 3D cross section (black, with the combined-covariance systematic band) versus four generator predictions produced at truth level: GENIE 2.12 CV, MINERvA Tune v1, NuWro 21.09, and GiBUU 2019. On p_T and $p_{\parallel}$ the models track the data shape; on E_{avail} they split, and all under-predict the data at low available energy. The wide E_{avail} catch bin is omitted from the axis.

Quantitatively, against the combined 3D covariance (§7.6) the integrated E_{avail} rate (catch bin dropped) places the models at -7.2% (GENIE CV, 1.3σ), -9.5% (Tune v1, 1.8σ), -15.3% (NuWro, 2.9σ) and -21.9% (GiBUU, 4.1σ) below the data. A full-3D goodness-of-fit over all 1431 bins, evaluated with a truncated-spectral χ^2 on the rank-247 constrained subspace of the combined covariance (a raw pseudo-inverse of the rank-deficient matrix is never taken), excludes all four at $p \approx 0$ but ranks them sharply: diagonal-only χ^2/ndf is Tune v1 **4.8**, GiBUU 32.4, GENIE CV 34, NuWro 36; the full correlated metric is Tune v1 131, NuWro 1287, GENIE CV 1512, with GiBUU worst of all because 23.5 % of its residual lies *outside* the constrained subspace (its 3D shape differs in directions the covariance barely constrains) on top of the largest normalisation offset. The truncation scan and per-generator eigen-mode residuals are shown in Fig. 21.

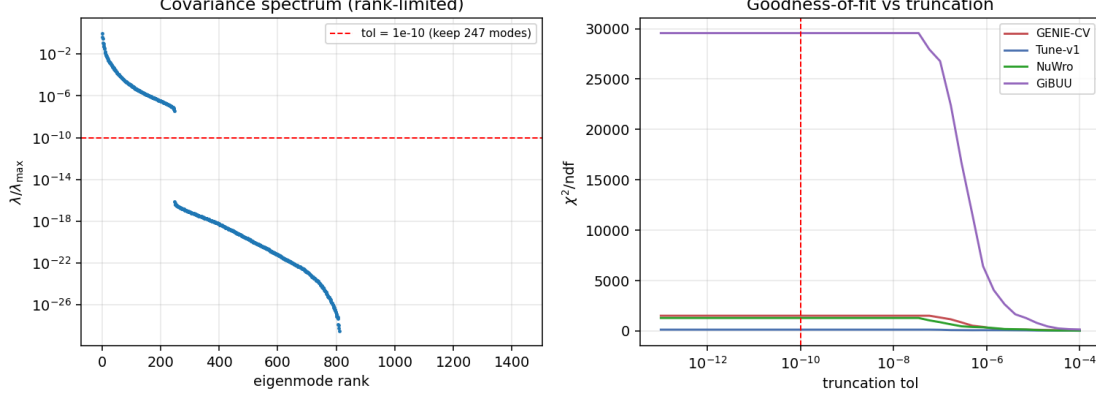


Figure 21: Full-3D truncated-spectral goodness-of-fit: per-generator χ^2/ndf versus eigenvalue-truncation tolerance, with the fraction of the residual captured by the constrained subspace. The ranking (Tune v1 best, GiBUU worst) is stable across the truncation choice.

7.6 The 3D systematic-uncertainty campaign

The band on Fig. 20 and the χ^2 above rest on a full 3D analogue of the 2D uncertainty budget of §5. We re-ran the event loop with the systematic universes dumped on the three-axis ($p_T, p_{||}, E_{\text{avail}}$) grid (a 187-universe sweep), propagated the bootstrap statistical and ML-seedscan covariances, and summed them into a single combined covariance $C_{\text{syst}} + C_{\text{stat}} + C_{\text{ML}}$ over all 1431 reported bins. The result has $\sqrt{\text{tr}} = 5.72 \times 10^{-39}$, a median per-bin relative uncertainty of 10.4 %, and numerical rank 247 (of 1431); it is flux-dominated, with the same systematic band ordering as the 2D measurement (Fig. 22). This combined covariance is what projects onto each axis for the per-bin bands and onto the full space for the goodness-of-fit, and it is the internally-correct reference for the “ours-only” χ^2 given that our bootstrap statistical block is genuinely smaller than the paper’s (an OmniFold efficiency, confirmed by the data/MC split; see App. A).

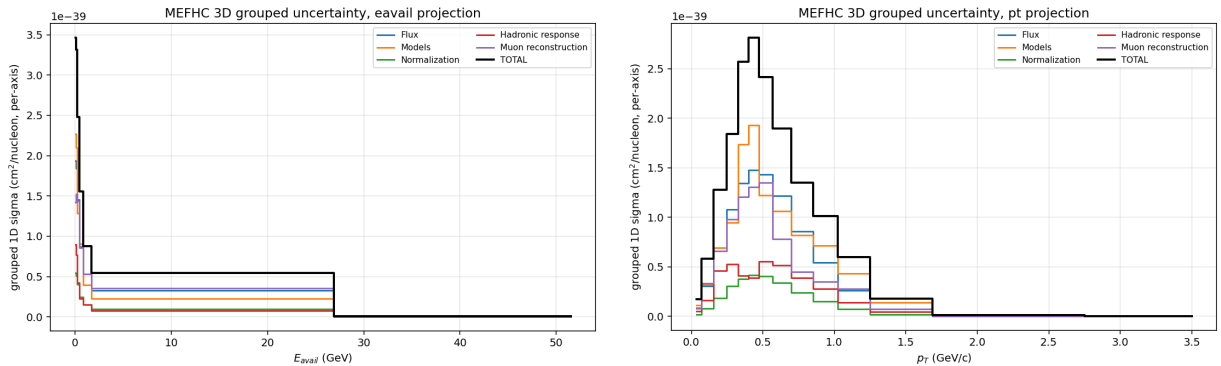


Figure 22: Grouped fractional systematic bands of the 3D combined covariance projected onto E_{avail} (left) and p_T (right): flux-dominated, with the same band ordering as the 2D budget.

7.7 The low-available-energy excess recovers the known low-recoil 2p2h deficit

The low- E_{avail} excess is the most physically interesting feature the third axis exposes, and the 3D framework lets us identify its origin directly by generating the relevant physics ourselves rather than

only reweighting. It sits in a well-established MINERvA low-recoil program: the enhancement of the event rate between the quasielastic and Δ peaks, with multi-nucleon final states, was first isolated in the low-three-momentum-transfer subsample [25] and is the effect the empirical MINERvA tune was built to absorb — our measurement re-exposes it on the muon-kinematics $\times E_{\text{avail}}$ axes.

Final-state interactions do not explain it. A `grwght1p` reweight pass on the 2M central-value GENIE events varying the pion-FSI dials shows both are sub-percent on $d\sigma/dE_{\text{avail}}$: `FrInel_pi` shifts the spectrum by $\leq 0.74\%$ (peak in 0.10–0.20 GeV) and `FrAbs_pi` by $\leq 0.82\%$ (peak in the lowest bin), with the integrated rate moving only ± 0.02 – 0.03% at $\pm 1\sigma$. A $< 1\%$ knob cannot bridge a 7–15% excess, so the excess points to the initial-state/nuclear model, not rescattering.

The shape and size match 2p2h. A subtlety drives the diagnosis: our base GENIE central value was generated with the default event-generator list, which contains *no* meson-exchange-current (2p2h/MEC) events (`mec = 0` for all 1.48×10^6 CC events; the modes are QE 11%, RES 22%, DIS 66%, COH 0.9%). Any 2p2h is therefore *added*, not reweighted. Decomposing the central-value $d\sigma/dE_{\text{avail}}$ by interaction mode (per-bin mode count-fractions $\times$ the bin value) and overlaying the data localises the excess to the quasielastic- Δ dip: the [0.10, 0.20] bin is $+3.9\sigma$ and [0.20, 0.40] is $+2.3\sigma$, while the QE-dominated [0, 0.10] already matches ($+0.8\sigma$) and [0.40, 1.50] match to $\pm 0.4\sigma$. 57% of the integrated deficit sits at $E_{\text{avail}} \leq 0.4$ GeV, exactly where 2p2h lives, and closing it requires a 2p2h component $\approx 43\%$ of the QE rate (62% locally) — the standard empirical/Valencia-MEC magnitude, in contrast to the sub-percent FSI dials (Fig. 23).

Enabling Valencia 2p2h confirms it. We regenerated the GENIE central value with the empirical 2p2h turned on (`--event-generator-list Default+CCMEC`, 2M events). Because the Nieves–Simo–Vacas MEC channel, though present in the splines, is absent from the `gsp12root` total-CC graph, the sample is normalised by the non-MEC CC count (anchoring QE+RES+DIS+COH to the known total and letting MEC add on top by $1/(1 - f_{\text{MEC}})$, with $f_{\text{MEC}} = 2.87\%$). The added 2p2h lands squarely in the dip and moves GENIE toward the data: the [0, 0.10] bin goes from $+0.8\sigma$ to -0.04σ (onto the data), [0.10, 0.20] from $+3.9\sigma$ to $+2.85\sigma$, and [0.20, 0.40] from $+2.3\sigma$ to $+1.0\sigma$ (Fig. 24). It fills 46% of the data–CV gap in the dip and 27% of the integrated deficit (CV $-7.2\% \rightarrow -5.2\%$), and improves the full-3D χ^2/ndf from 1512 to 1145 (-24%). It remains well short of Tune v1 (131), because Tune v1 layers MINERvA’s empirical low-recoil 2p2h *enhancement* and RPA on top of the stock Valencia MEC — i.e. stock Valencia 2p2h is real but under-strength for MINERvA low recoil, which is precisely the known MINERvA-tune story and is consistent with NuWro and GiBUU (native, un-enhanced 2p2h) sitting low as well. The one feature 2p2h does not touch is a separate $+2.2\sigma$ excess in the high- E_{avail} [1.50, 3.00] bin, a DIS-tail modelling issue (§A).

This low-available-energy deficit is the same feature MINERvA’s dedicated low-recoil inclusive measurement reports: Ascencio *et al.* [26], which measures $d^2\sigma/(dq_3 dE_{\text{avail}})$ for $q_3 < 1.2$ GeV at the same $\langle E_\nu \rangle \sim 6$ GeV, finds GENIE and NuWro underpredicting the data in the low- E_{avail} region absent an enhanced 2p2h component. Our excess appears in the muon-kinematics extension (integrated over all q_3) rather than at fixed q_3 , so the comparison is of the E_{avail} *shape* and direction rather than absolute normalisation; the consistency confirms that the OmniFold 3D extension recovers the established MINERvA low-recoil/2p2h feature as a by-product of the muon-kinematics unfolding. The overlay tooling (`3d-unfolding/genie/compare_ascencio_eavail.py`) is in place for a quantitative shape comparison once the [26] data release is loaded.

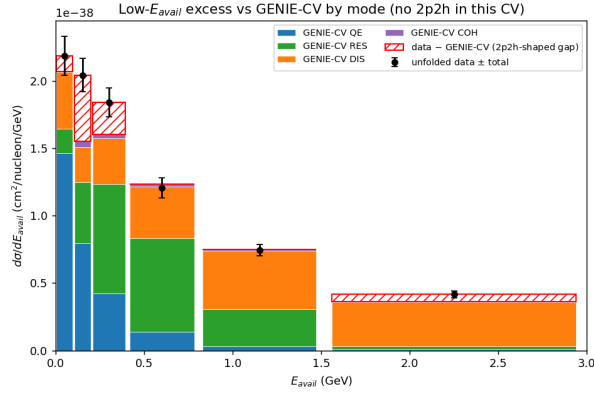


Figure 23: Central-value $d\sigma/dE_{\text{avail}}$ decomposed by interaction mode and overlaid with the unfolded data. The data excess is localised to the quasielastic- Δ dip ($E_{\text{avail}} \leq 0.4$ GeV), the shape and size of a missing 2p2h component.

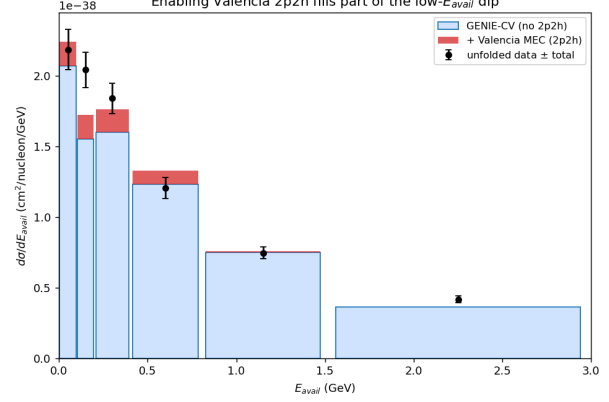


Figure 24: Re-generating GENIE with the empirical Valencia 2p2h enabled (red, stacked on the no-2p2h CV) fills part of the low- E_{avail} dip — about half the data-CV gap, the remainder being the MINERvA-tune enhancement.

7.8 Fourth axis: three-momentum transfer q_3

The same one-feature-per-observable property lets us add a fourth axis at no methodological cost. We choose the three-momentum transfer $q_3 = |\vec{q}|$ because (E_{avail}, q_3) is the plane in which the quasielastic, 2p2h and resonance processes separate, and because it makes the comparison to MINERvA’s dedicated low-recoil measurement (Ascencio *et al.* [26], $d^2\sigma/dq_3 dE_{\text{avail}}$) bin-compatible. Truth q_3 is the generator momentum transfer (`CVUniverse::Getq3True()`); reco q_3 is the calorimetric estimator built from the muon Q^2 and the recoil system (`CVUniverse::RecoQ3()`). Both accessors were added and the event loop re-run over all twelve playlists, exactly as for E_{avail} . Unlike E_{avail} , q_3 is *not* invariant under the lateral systematic shifts, so the shifted- q_3 universes are dumped per universe rather than reused from the central value. The driver feature stack becomes $(p_T, p_{\parallel}, E_{\text{avail}}, q_3)$; the q_3 binning $[0, 0.2, 0.4, 0.6, 0.8, 1.2, 2.0, 100]$ GeV is deliberately coarse (seven bins) and ends in a catch bin, for $14 \times 16 \times 7 \times 7 = 10\,976$ four-dimensional bins (4830 populated).

Marginal closure. Marginalising the 4D result over q_3 reproduces the 3D E_{avail} spectrum and its low-recoil excess (Fig. 26, right): the E_{avail} -marginal data/GENIE-CV ratio is 1.21 in the lowest bin, dips to 1.01 across the quasielastic- Δ region, and rises again to 1.16–1.23 in the high- E_{avail} DIS tail — the same shape established in 3D (§7.7), now recovered as a by-product of the higher-dimensional unfold (integrated data/GENIE-CV = 1.13). This is the 4D→3D analogue of the 3D→2D Jacobian anchor of §7.3.

What q_3 adds. The new content is the resolved (E_{avail}, q_3) plane (Fig. 26, left). Integrated over E_{avail} the q_3 -marginal discrepancy is mild (1.03–1.19, centre panel), but the joint map is structured: at fixed E_{avail} in the quasielastic- Δ dip the data/GENIE-CV ratio *rises* with q_3 , the central value over-predicting the low- q_3 , moderate- E_{avail} cells (ratio down to 0.66) while the data excess concentrates at low E_{avail} (all q_3) and toward high q_3 . This is precisely the correlation a muon-only or fixed- q_3 measurement cannot expose, and it is the natural input for the bin-compatible Ascencio comparison. We show it here at central-value level (the base GENIE event-generator list contains no 2p2h, as in §7.7); per-cell significances use the combined 4D covariance below.

Bin-identical comparison to the published low-recoil measurement. The Ref. [26] supplemental data (44-cell $d^2\sigma/(dE_{\text{avail}} dq_3)$ with its full covariance, distributed inside the public arXiv source package) enables a direct quantitative cross-check. Merging both measurements onto their maximal common grid — edges present in *both* binnings, with our $(p_T, p_{\parallel})$ marginalisation restricted to $p_{\parallel} < 20$ GeV/c to mirror the Ascencio muon gate ($\theta_{\mu} < 20^\circ$, $1.5 < p_{\mu} < 20$ GeV) — gives two super-cells, $E_{\text{avail}} < 0.4$ GeV in $q_3 \in [0.4, 0.6)$ and $[0.6, 1.2)$ GeV: the low-recoil region where the 2p2h physics lives. Our result sits 9.2 % and 6.3 % above Ascencio (pulls 1.3σ and 0.9σ against the summed covariances), with a full-covariance $\chi^2/\text{ndf} = 1.68/2$ ($p = 0.43$): **the two measurements are consistent** (Fig. 25). Caveats: the two share MINERvA flux/detector systematics, treated here as independent; the p_{μ} -vs- $p_{\parallel}$ gate differs by $\lesssim 6\%$ kinematic smearing at the 20 GeV boundary. A finer 44-cell comparison would require re-unfolding onto the Ascencio edges (trivial for the unbinned method) *and* a systematic sweep on that binning; given the agreement above it is not pursued.

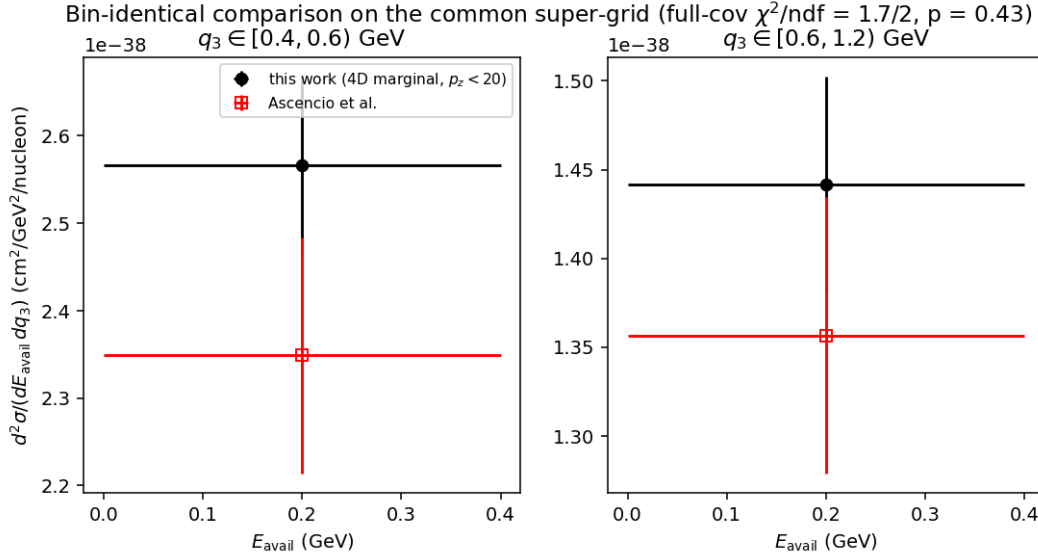


Figure 25: Bin-identical comparison to Ascencio *et al.* [26] on the maximal common (E_{avail}, q_3) grid, with each side’s full covariance (ours: the adopted 4D combined covariance marginalised with the Ascencio muon gate). Full-cov $\chi^2/\text{ndf} = 1.68/2$, $p = 0.43$.

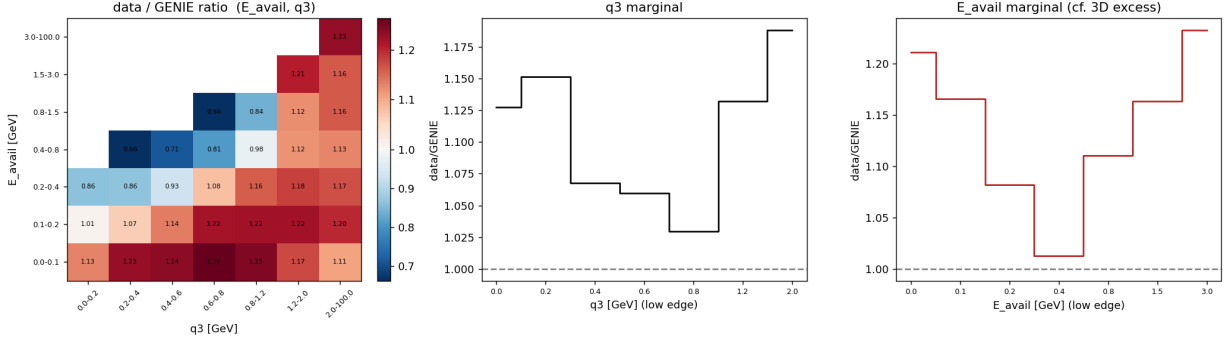


Figure 26: Data/GENIE-CV ratio of the 4D unfold projected onto the (E_{avail}, q_3) plane (left; blank cells are kinematically empty), with the q_3 (centre) and E_{avail} (right) marginals. The E_{avail} marginal reproduces the 3D low-recoil excess; the joint plane shows the data – MC discrepancy to carry genuine q_3 structure (the ratio rising with q_3 across the quasielastic- Δ dip). GENIE central value, no 2p2h in the base event-generator list.

Systematic covariance. A 187-universe systematic sweep on the 4D grid (the same band set as 2D/3D, the shifted- q_3 lateral universes dumped per universe) yields a syst-only covariance with $\sqrt{\text{tr}} = 2.09 \times 10^{-38}$, median per-bin relative uncertainty 13.4 % (84th percentile 22 %), and numerical rank 142 of 4830 reported bins. The rank is set by the number of independent throws, not the bin count: it is essentially the 3D syst-only value (139), since both budgets use the same ~ 40 one-sided bands plus the 100-universe flux multisim. Adding the q_3 axis multiplies the bins ($1431 \rightarrow 4830$) without adding throws, so the covariance is simply more rank-deficient ($9.7 \rightarrow 2.9$ % of bins spanned) and the truncated-spectral χ^2 of §7.5 becomes correspondingly more essential. The band *ordering* also shifts: where the 2D/3D budgets are flux-dominated, the 4D budget is model-dominated — the 2p2h band is the single largest contributor, followed by the M_A^{RES} , muon-energy-scale and M_A^{CCQE} bands, with flux sub-dominant (~ 2.8 % median). This is the expected consequence of binning in q_3 , the variable that most directly discriminates the interaction models, and it identifies where a q_3 -resolved measurement is both most informative and most systematics-limited. The combined budget $C_{\text{syst}} + C_{\text{stat}} + C_{\text{ML}}$ (folding in the 4D bootstrap and ML-seedscan blocks, which raise the rank into the ~ 250 range as in 3D) is complete (median per-bin 13.5 %).

Unified-throw cross-check and adoption. The block sum above adds each systematic band’s covariance independently; it omits the nonlinear cross-band term that arises because the OmniFold reweighter is not linear in the input perturbations. We quantify that term with a *unified throw*: 160 joint re-unfolds in which all 12 reweightable knob bands and the flux multisim are perturbed simultaneously per event ($w_{\text{cv}} \prod_b \rho_b^{g_b}$, $g_b \sim \mathcal{N}(0, 1)$) and the cross section re-extracted, with the OmniFold run-to-run jitter floor subtracted via a zero-shift null set. On the real $(p_T, p_{\parallel}, E_{\text{avail}}, q_3)$ binning the jitter-corrected unified covariance is $2.0\times$ the block sum in $\sqrt{\text{tr}}$ — i.e. the block sum *underestimates* the vertical systematic by a factor of two. The inflation is concentrated rather than diffuse (the per-bin σ median ratio is 1.00, but the top 1 % of bins by variance-excess carry 78 % of the trace increase), and those bins are the high- p_T , lowest- E_{avail} (< 0.1 GeV) corner where the migration matrix is most off-diagonal and bands couple most strongly. Because 160 throws cannot estimate a 4830×4830 matrix (a direct swap is badly rank-deficient and non-PSD), we *adopt* the result by transferring the throw’s per-bin fractional inflation $g_i = \max(\sigma_i^{\text{uni}}, \sigma_i^{\text{blk}})/\sigma_i^{\text{blk}} \geq 1$ onto the sweep’s own vertical block, $C \rightarrow (C - C^{\text{vert}}) + G C^{\text{vert}} G$, which is positive-semidefinite

by construction and conservative (never below the block baseline). The adopted combined 4D covariance carries $\sqrt{\text{tr}} = 3.85 \times 10^{-38}$ (median per-bin 13.5 % to 14.9 %). This refines the coarse 3D probe (1.16 median / 1.40 trace) and is the published 4D systematic object.

8 Fifth axis: the (E_{avail}, W) deep-inelastic-tail study

Enabling Valencia 2p2h resolves the low-recoil excess (§7.7) but leaves a separate $+2.2\sigma$ data excess in the $E_{\text{avail}} \in [1.50, 3.00)$ GeV bin, where 2p2h does not contribute and the sub-percent pion-FSI dials cannot reach. Quantified against the adopted combined 4D covariance (§7.8) marginalised to the E_{avail} axis, the discrepancy is severe: GENIE CV, GENIE+MEC, NuWro and GiBUU are incompatible with the unfolded $d\sigma/dE_{\text{avail}}$ at $> 21\sigma$ over the seven E_{avail} bins (22.4σ , 24.9σ , 21.9σ , 21.2σ respectively) and at $> 15\sigma$ in the DIS tail ($E_{\text{avail}} \geq 0.8$ GeV) alone. GiBUU spreads its deficit across the entire tail (data/generator 1.59 at $E_{\text{avail}} \in [0.8, 1.5)$, 1.36 at $[1.5, 3.0)$) rather than piling it in the deep catch bin like the GENIE variants — an independent, qualitatively distinct confirmation. To test the deep-inelastic hypothesis directly, we added the hadronic invariant mass W as a fifth OmniFold axis.

8.1 W as a fifth axis

Truth W is the experimenter’s invariant mass $W = \sqrt{M^2 + 2M(E_\nu - E_\mu) - Q^2}$ with M the struck-nucleon mass (`CVUniverse::GetTrueExperimentersW()`); reco W uses the same calorimetric energy transfer q_0 and muon-kinematic Q^2 as the reco q_3 estimator, with the average nucleon mass (`CVUniverse::RecoW()`). Like q_3 — and unlike E_{avail} — W is not invariant under the lateral systematic shifts, so the event loop dumps shifted- W lateral universes per universe, mirroring the q_3 treatment (§7.8); the dump also carries truth hadronic diagnostics (proton and pion multiplicities, leading-hadron angle) for later interpretation. The W binning $[0, 1.1, 1.4, 1.8, 2.2, 3.0, 100]$ GeV brackets the quasielastic peak, the Δ , the transition region, and the DIS tail. The five-axis unfold $(p_T, p_\parallel, E_{\text{avail}}, q_3, W)$ passes the same marginalisation anchors as its predecessors: the W -marginal recovers the frozen 4D cross section to 0.11 % in normalisation (5D/4D total = 1.0011), with per-axis median differences of 0.3 % to 1.5 %, and the $d\sigma/dW$ spectrum is finite and non-negative across all six bins.

8.2 Where the excess lives in the (E_{avail}, W) plane

Projecting the unfolded five-axis result onto (E_{avail}, W) and subtracting the GENIE central value localises the excess (Fig. 27): it is *predominantly deep-inelastic*. The two highest- E_{avail} bands carry 57 % of the positive excess; of all positive excess, high- E_{avail} (≥ 0.8 GeV) holds 67 %, and 83 % of *that* sits at $W \geq 1.8$ GeV, with the single largest cell the deep-DIS corner ($E_{\text{avail}} > 3$ GeV, $W > 3$ GeV) at 22 %. A secondary low- W (< 1.1 GeV, QE-like) excess and a small Δ -region ($W \simeq 1.4$ – 1.8 GeV) deficit are also present.

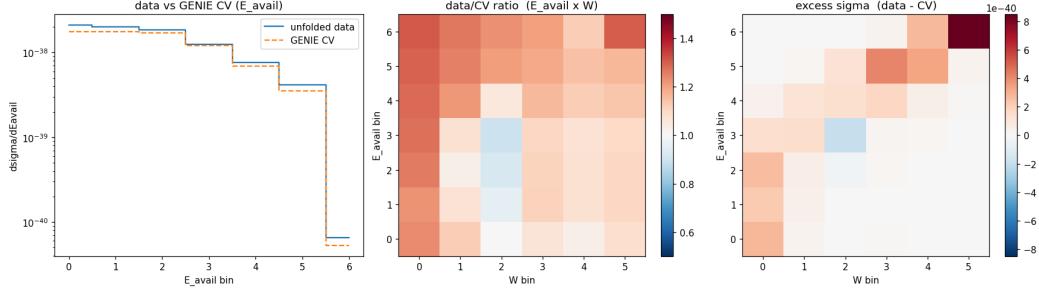


Figure 27: Unfolded-data minus GENIE-CV excess across the (E_{avail}, W) plane. The positive excess concentrates at high E_{avail} and high W — the deep-inelastic region — with a secondary low- W (QE-like) component and a small Δ -region deficit.

8.3 Generator band: the corner is generator- and tune-robust

We confirm this against independent spline-normalised (E_{avail}, W) predictions from GENIE CV, GENIE+Valencia 2p2h/MEC, NuWro, and GiBUU (Fig. 28). All four underpredict the high- $E_{\text{avail}} \times$ high- W corner by 54 % to 58 % (data/generator = 1.54, 1.58, 1.56). Crucially, enabling Valencia 2p2h does *not* close the corner — it slightly worsens it (1.54 $\rightarrow$ 1.58), since 2p2h populates low W — and the independent NuWro and GiBUU generators miss it by the same or a larger margin (GiBUU is the most deficient overall, integrated $\sigma = 2.22 \times 10^{-38} \text{ cm}^2/\text{nucleon}$ against the data's 3.07×10^{-38}); at $W \in [2.2, 3.0]$ GeV all four sit 23 % to 25 % below the data.

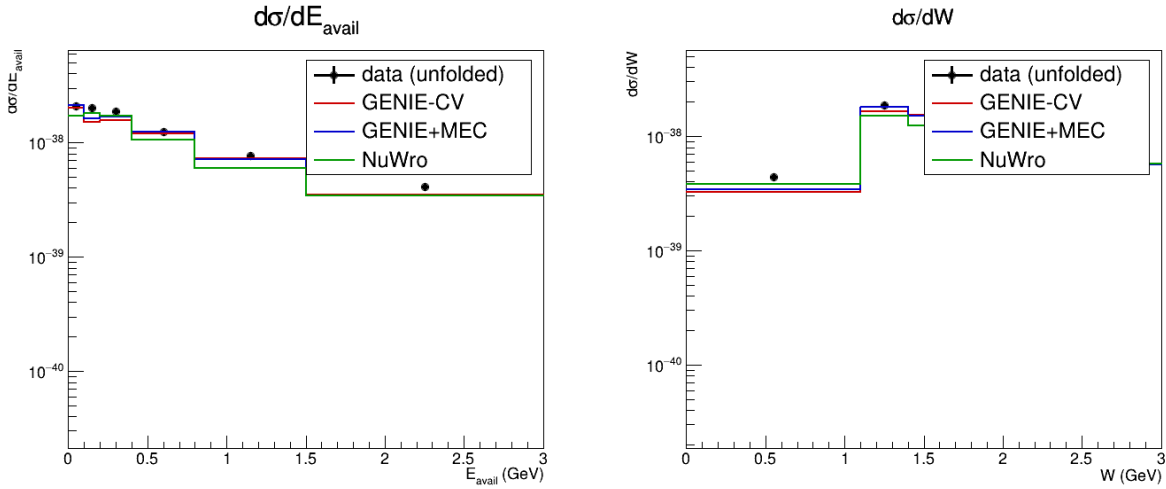


Figure 28: Unfolded (E_{avail}, W) cross section versus the four-generator band (GENIE CV, GENIE+MEC, NuWro, GiBUU). Every generator underpredicts the high- W , high- E_{avail} DIS corner; adding 2p2h does not help there because 2p2h populates low W .

8.4 The (E_{avail}, W) covariance and significances

To turn the band into a number we build a dedicated 42-bin (E_{avail}, W) covariance, $C = C_{\text{sys}} + C_{\text{stat}} + C_{\text{lateral}}$, by a *frozen-reweighter* block sum on the merged five-axis `universes_full` sample: the central-value OmniFold push-weights are held fixed and only the per-event truth reweight

ratios change per systematic universe (the same methodology as the PET budget, §9.2; no per-universe re-inference). The covariance carries 13 knob bands $\pm 1\sigma$, the 100-universe flux multisim, a Poisson statistical diagonal, and the detector (lateral) bands transferred from the 4D combined covariance — marginalised to E_{avail} and spread over W by the central-value shape, a documented approximation (W -resolved laterals would need reconstructed-space re-inference; the laterals are subdominant at $\sim 4\%$). The result has a median relative uncertainty of 14.8% per bin and is validated by reproducing the frozen five-axis product’s E_{avail} and W marginals to 0.1%.

Generator	full plane (42 bins)		high- W DIS corner (12 bins)	
	χ^2/ndf	$N\sigma$	χ^2/ndf	$N\sigma$
GENIE CV	412.7/42	16.7	116.9/12	9.0
GENIE+MEC	390.5/42	16.1	121.1/12	9.2
NuWro	1148.4/42	31.2	149.6/12	10.5
GiBUU	1930.2/42	> 37	381.1/12	18.2

Table 3: Full-covariance incompatibility of each generator with the unfolded (E_{avail}, W) cross section, over the full plane and restricted to the high- W DIS corner ($E_{\text{avail}} \geq 0.4 \text{ GeV}$, $W \geq 1.8 \text{ GeV}$).

Over the full plane the four generators are incompatible with the data at 16–37 σ ; restricted to the twelve-bin high- W DIS corner they miss at 9.0–18.2 σ (Table 3), with GiBUU again the most deficient. The excess is thus a genuinely DIS-region ($W \gtrsim 1.8 \text{ GeV}$) feature — a *modelling* deficit that no current generator or tune reproduces — and not a low- W resonance artefact. The complementary cross-check against MINERvA’s published low-recoil measurement is now done in the (E_{avail}, q_3) plane (§7.8): on the maximal common grid the two measurements agree at $\chi^2/\text{ndf} = 1.68/2$ ($p = 0.43$).

9 Point-cloud track: PET on the raw reconstructed clusters

The full promise of OmniFold is unfolding on the event’s *raw* representation rather than on engineered scalar summaries. This section documents that track: a point-cloud transformer (PET, the architecture of OmniLearn [23], vendored with the upstream OmniFold implementation) unfolds the same ME FHC data using, on the reconstructed side, the actual calorimetric recoil clusters — per-cluster energy, transverse position and z (`cluster_energy`, `cluster_pos`, `cluster_z`, filtered on `cluster_isMuontrack` via `CVUniverse::GetRecoClusters()`) alongside the muon kinematics. An earlier attempt that read the wrong, mostly-empty `ExtraEnergyClusters*` branches was discarded and the chain rebuilt from the corrected dump. **Maturity:** this is a validated *method milestone* — a real, absolutely-normalised, full-statistics cross section with its own closure gate and a first systematic budget — but it is not yet at the production GBDT result’s maturity (its training is subsample-limited and its budget wider; see below).

9.1 From shape check to absolute cross section

The PET network trains on a GPU on a 2M-event subsample (2 OmniFold iterations); the learned step-2 truth reweight is then applied to the *full* 32.8M-event generator sample (`MultiFold.reweight`), binned on the 4D grid, and pushed through the standard extraction (`extract_cross_section_nd` with the GBDT run’s completeness — completeness is reweight-independent — and the same flux/POT/nucleon constants). Three gates validate the absolute path:

1. **Full-statistics reweight sanity:** the pushed weights over the 32.8M generator events are finite with mean 1.028;
2. **Closure (decisive):** unfolding MC-reco-as-pseudo-data recovers MC truth to total 0.9884 with uniform per-axis median residuals of $\sim 1.1\%$ — the absolute machinery is unbiased;
3. **Cross-check vs GBDT:** on data, the PET absolute total is $2.796 \times 10^{-38} \text{ cm}^2/\text{nucleon}$ versus the GBDT 3.066×10^{-38} , ratio 0.912, with per-axis median differences of 6.5% to 9.9% (Fig. 29).

Because the closure is exact to $\sim 1\%$, the $\sim 9\%$ data-side gap is a genuine *training-configuration* difference (PET trained on the 2M subsample with 2 iterations versus the full-statistics 5-iteration GBDT), not a normalisation bug: an under-iterated unfold pushes the result less far from the prior. A higher-epoch retrain moved the total by -1.6% (ratio 0.897), confirming the training-config sensitivity; closing the gap toward the ML band needs more events/iterations and is deferred.

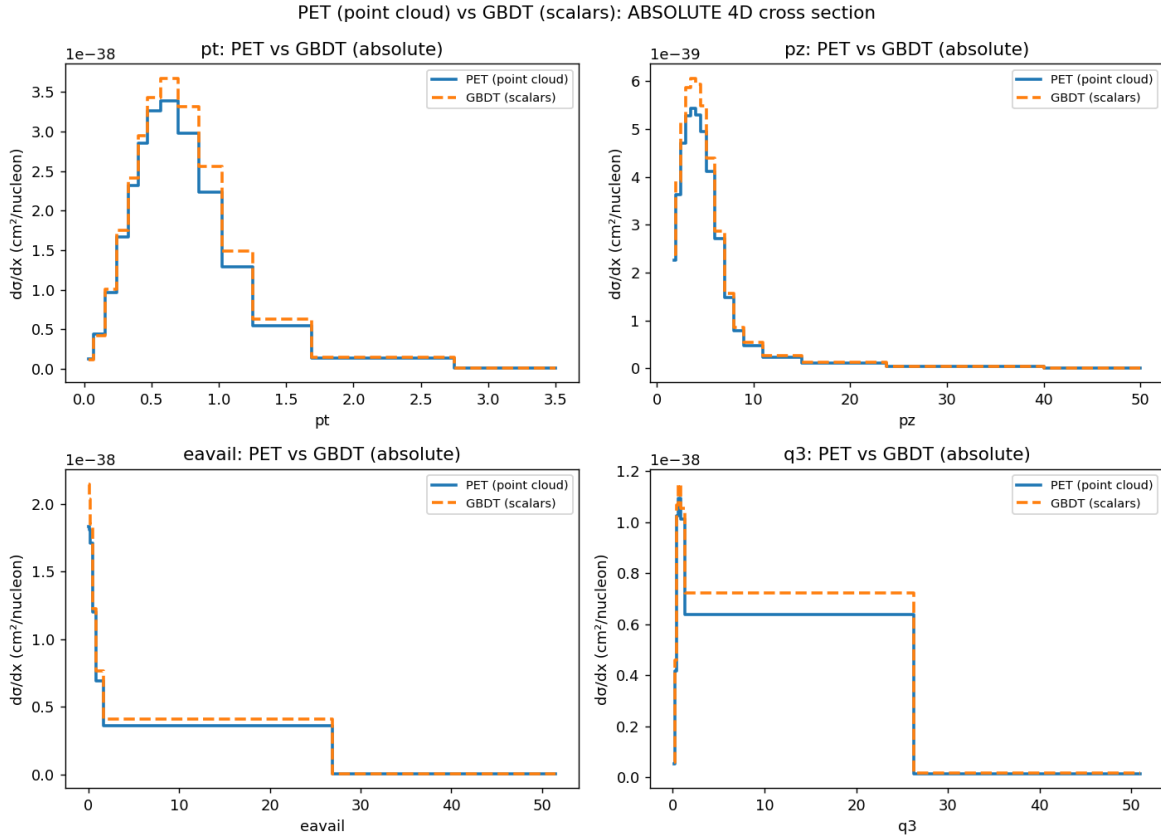


Figure 29: Absolutely-normalised 4D cross section from the PET point-cloud unfold (full-statistics reweight of the generator sample) versus the production scalar GBDT result, projected on each axis. The PET/GBDT total ratio is 0.912; the closure gate (text) shows the absolute machinery itself is unbiased to $\sim 1\%$.

9.2 PET systematic budget (frozen reweighter)

A first full uncertainty budget is attached with the *frozen-reweighter* method: the PET step-2 weights are held fixed while each systematic universe’s truth reweight ratios re-bin the generator

sample — 13 knob bands, the 100-universe flux multisim, a 100-replica Poisson bootstrap (statistical), an alternative-training difference (ML), and the detector (lateral) bands transferred from the GBDT 4D covariance by fractional projection. The combined budget on the PET reported bins has median per-bin fractions of 18.3 % (systematic), 4.2 % (statistical), 3.3 % (ML), 4.0 % (lateral, transferred), totalling 23.0 % — wider than the GBDT 4D budget’s 13.5 %, as expected for the subsample-trained network. The residual gap to a fully self-contained PET budget is per-lateral re-inference on shifted reconstructed clouds, which requires re-running the PET network per lateral universe and is deferred.

9.3 What the PET track demonstrates

The point-cloud track closes the methodological loop begun by the scalar GBDT/MLP cross-checks (§6): the unfolded physics is stable across classifier family (GBDT, MLP, transformer) *and* input representation (engineered scalars, raw reconstructed clusters), now including the absolute normalisation. It also establishes the infrastructure — point-cloud DataLoader, GPU training, full-sample reweighting, frozen-reweighter budgets — for unfolding directly on low-level MINERvA data, the natural endpoint of the unbinned program.

10 First full-phase-space results

Maturity: central-value measurement with a three-prior dependence envelope; the systematic (universe) campaign is in production at the time of writing. Decision memo and job trail: `nd-unfolding/FPS_PILOT.md`

The published phase space ($\theta_\mu < 20^\circ$, $1.5 < p_\parallel < 60 \text{ GeV}/c$, $p_T < 4.5 \text{ GeV}/c$) hides a third of the signal: weighting the true ν_μ CC rate in the tracker fiducial volume, 66.4 % lies inside the cuts, 22.3 % at $p_\parallel < 1.5 \text{ GeV}/c$, 11.3 % at $\theta_\mu > 20^\circ$, and the high- $p_T/p_\parallel$ tails are negligible (Fig. 30). Because OmniFold’s miss treatment performs acceptance correction natively, removing the truth-level kinematic cuts is a configurational change (an event-loop switch enlarging the truth denominator from 32.8M to 49M events; the reconstruction-level selection, including the MINOS match, is untouched), and the unfold extrapolates into the unmeasured regions through the step-1 miss regressor. The honest question is not *whether* it can extrapolate but *at what prior cost*; this section quantifies that.

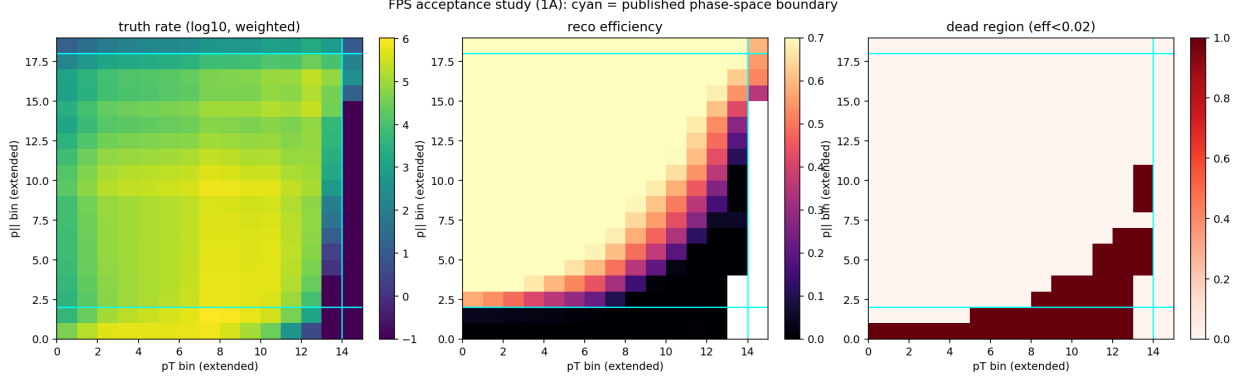


Figure 30: Full-phase-space acceptance study (all twelve playlists): weighted truth rate (left), reconstruction efficiency (centre), and the dead region with $\epsilon < 2\%$ (right) on the extended $(p_T, p_{||})$ grid. Cyan lines mark the published phase-space boundary. Efficiency is ~ 0.65 and flat through the published region, degrades smoothly across the $\theta > 20^\circ$ wedge, and dies in the $p_{||} < 1.5 \text{ GeV}/c$ strip and the high- p_T corner; the dead region carries 27.7% of the total truth rate.

Validation battery. Three results establish the extension (Fig. 31):

1. **Anchor.** Restricted to the published-phase-space sub-block, the full-phase-space unfold reproduces a control unfold (standard omnifile, paper grid, identical settings) to 0.9994 in integral, with a median per-cell ratio of 1.0013 and median deviation 0.57% over all 185 reported cells — opening the phase space does not damage the measurement where it was already made. The control itself reproduces the frozen production total ($3.073 \times 10^{-38} \text{ cm}^2/\text{nucleon}$) exactly.
2. **Closure.** A self-closure on the extended grid (simulated reconstruction as pseudo-data) returns recovered/truth = 1.0000 in every cell including the extension — no bookkeeping or normalisation bias in the enlarged-denominator configuration.
3. **Three-prior envelope.** Repeating the unfold with three priors — MnvTune v1, bare GENIE, and a NuWro-shaped prior (built by reweighting the truth $(p_T, p_{||})$ shape to two million raw NuWro events, which carry no phase-space cut) — spans the total cross section by only $\pm 1.5\%$. Per cell, the half-spread/mean is median 2.9% inside the published region but median 7.9% (90th percentile 62%) in the extension, concentrated in the zero-acceptance cells (Fig. 32).

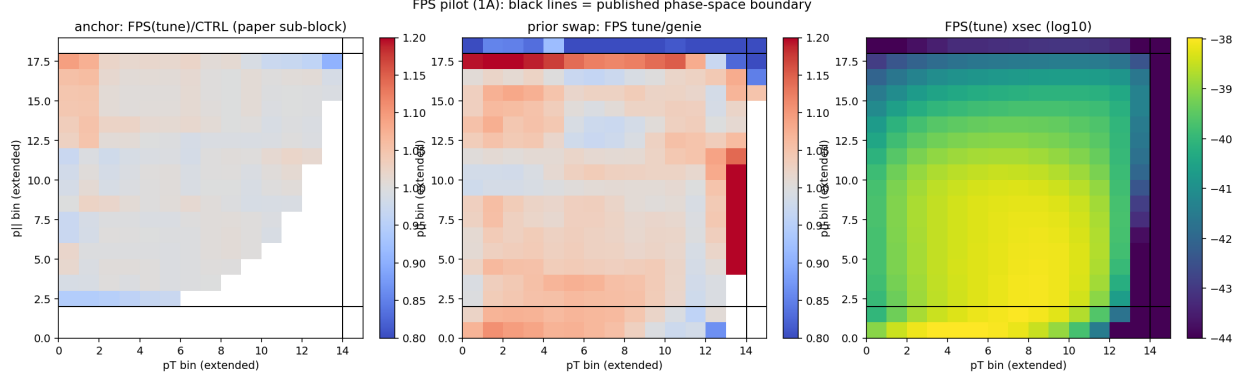


Figure 31: Full-phase-space honesty battery (all playlists): anchor ratio in the published sub-block (left; median 1.0013), the MnvTune/bare-GENIE prior-swap ratio (centre), and the full-phase-space cross section (right, log scale). Black lines mark the published boundary.

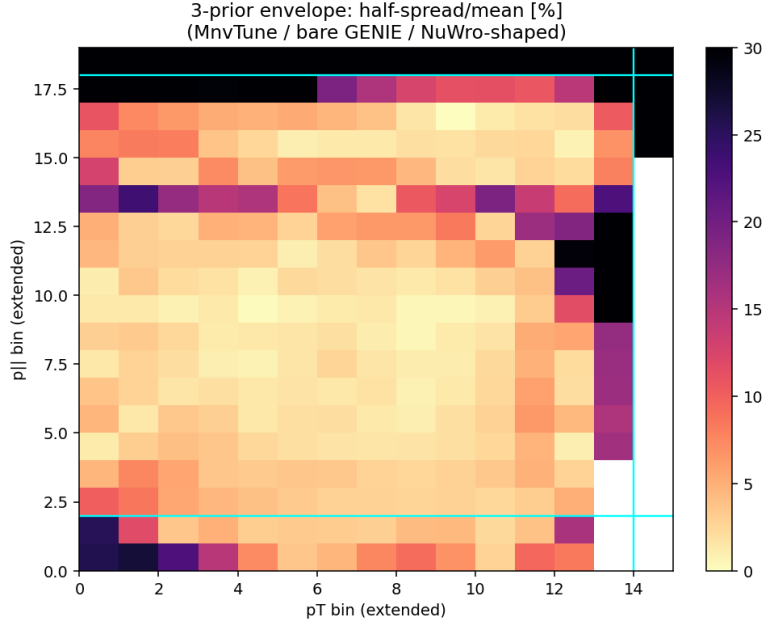


Figure 32: Three-prior dependence envelope (half-spread/mean per cell, in percent). Prior dependence is at the few-percent level wherever the detector has acceptance and reaches tens of percent only in the dead cells — the quantitative basis for the two-tier reporting.

Result and reporting convention. The full-phase-space CC-inclusive cross section on the tracker fiducial volume is

$$\sigma_{\text{FPS}} = 4.502 \times 10^{-38} \text{ cm}^2/\text{nucleon} \quad (\text{CV; systematic campaign in production}),$$

46% above the restricted-phase-space $3.073 \times 10^{-38} \text{ cm}^2/\text{nucleon}$. We adopt **two-tier reporting**: *tier 1* (the measurement) covers cells with $\epsilon \gtrsim 2\%$ — the published region plus the measurable extension ($\sim 6\%$ of the rate, chiefly the $\theta > 20^\circ$ wedge and the $p_{||} \in [0.75, 1.5] \text{ GeV}/c$ edge); *tier 2* (prior-extrapolated) covers the dead cells ($\sim 28\%$ of the rate), published but flagged, carrying the

three-prior envelope as an explicit additional band. The in-production systematic campaign repeats the full universe/bootstrap/ML budget on the extended grid, with the unified-throw construction (§5.4) mandatory: the migration-heavy corner that broke the 4D block sum by $\times 2$ sits *inside* this measurement.

11 Summary and outlook

Summary. Unbinned OmniFold with gradient-boosted trees reproduces the published MINERvA ME FHC CC-inclusive $d^2\sigma/(dp_T dp_{\parallel})$ (σ_{tot} ratio 1.011; 94 % of bins within 10 %; pull RMS 0.60) with a standalone uncertainty budget that matches the published total (6.87 % vs 6.86 % median per bin) and passes a full validation suite: three closure families, a bottom-line test, frequentist coverage, binned and unbinned (C2ST) goodness-of-fit, and classifier-family cross-checks. On that foundation we performed the first simultaneous three-, four- and five-observable unfolds of MINERvA data ($+E_{\text{avail}}$, $+q_3$, $+W$), each validated by recovering its lower-dimensional predecessor under marginalisation. The physics output:

- the low- E_{avail} excess over all four truth-level generators recovers the established MINERvA low-recoil/2p2h deficit through an independent method, with the expected FSI insensitivity and Valencia 2p2h response (§7.7);
- the (E_{avail}, q_3) plane resolves the deficit’s momentum-transfer structure and shifts the systematic budget from flux- to model-dominated (§7.8);
- the 4D unified-throw study found the block-sum covariance underestimates the vertical systematic by $\sim 2\times$, and the published 4D covariance adopts the corrected magnitude (§5.4) — a methodological result in its own right;
- the (E_{avail}, W) study localises the remaining high- E_{avail} excess to the deep-inelastic region: all four generators miss the high- W corner at 9–18 σ on a dedicated covariance (§8);
- the PET point-cloud track yields a full-statistics, absolutely-normalised cross section from the raw reconstructed clusters, closure-validated and budgeted (§9).

Outlook: completing the full-phase-space measurement. The full-phase-space extension is no longer prospective: §10 reports the validated central value ($\sigma_{\text{FPS}} = 4.502 \times 10^{-38} \text{ cm}^2/\text{nucleon}$, 46 % above the restricted phase space) with its anchor, closure, and three-prior envelope. What remains is its uncertainty campaign — the full universe/bootstrap/ML budget on the extended grid with the mandatory unified-throw construction (§5.4), plus hidden-variable closure and coverage toys exercised specifically in the extension regions — all in production at the time of writing. Pending items inherited from this note: NEUT as a fifth generator, per-lateral PET re-inference, the finer 44-cell Ascencio comparison (the maximal-common-grid version is done and consistent, §7.8), and the remaining collaborator confirmations of App. A.

Acknowledgments

[TODO: acknowledgments — advisor, MINERvA collaborators consulted, NERSC (Perlmutter, project m3246), MAT/MINERvA-101 framework authors]

A Resolution of the open questions from internal review

An earlier draft listed six items for collaboration input. The 2026-06-03 audit (scripts under 2d-unfolding/uq/ and 3d-unfolding/genie/; full log in LITERATURE_NOTES.md) closes or reduces all of them. Each item below states its resolution and what, if anything, still needs a collaborator. Technical detail is in App. B.4.

1. **Flux $\leftrightarrow$ Muon_Energy joint block and the rank gap — *resolved*.** The premise of the original question (“the paper sums more bands than we propagate”) was wrong. Our systematic-only covariance is rank 140/205; the paper’s (TotalCov – StatOnly) is rank **120** — *lower* — so we propagate at least as many independent systematic directions, using the same `GetStandardSystematics`. The paper’s V is full-rank only because it includes the *statistical* block; we match that with $\mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}}$ (rank 201). The one named cross-band correlation, the Bashyal flux $\leftrightarrow$ muon-energy-scale block [21], *is* rederivable and now built (`uq/rederive_flux_muonE_cross.py` $\rightarrow$ `uq/flux_muonE_cross.root`): the muon-energy-scale systematic is rank-1, so the cross term is rank-2, with $\rho = 0.900$ (two independent estimators, 0.87–0.90). It is physically real but adding it leaves the rank unchanged (140 $\rightarrow$ 140) — its directions already lie in $\text{range}(\mathbf{C}^{\text{syst}})$. The residual stat-block difference (our bootstrap $2.5\times$ smaller by trace) is explained by the data/MC split bootstrap: it is present in the *data-only* stream ($\mathbf{C}_{\text{data}}/\text{StatOnly} = 0.356$), i.e. genuine unbinned-OmniFold statistical efficiency, not a missing MC term. *Remaining (optional)*: the fitted $(\Phi_{\text{norm}}, \Phi_{\text{shape}}, \text{MuonE}_{\text{scale}})$ posterior from MAT-MINERvA would give a fully paper-independent ρ .
2. **FrInel_pi exclusion — *resolved (negligible)*.** An independent `grwght1p` pass on the 2M CV events shows both pion-FSI dials are sub-percent on $d\sigma/dE_{\text{avail}}$: $\text{FrInel_pi} \leq 0.74\%$ and $\text{FrAbs_pi} \leq 0.82\%$ (§7.7). A $< 1\%$ knob cannot bridge the 7–15% low- E_{avail} excess, and we are consistent with MAT in excluding it. *Remaining (immaterial)*: a one-line collaborator confirmation that the MAT exclusion is still current guidance.
3. **PPFX effective rank — *resolved*.** The 100-universe flux band covariance has hard rank 99 but a participation-ratio effective rank of ≈ 1 : a single coherent normalization mode carries 99.6% of the variance (9 modes for 99.9%). So the PPFX flux uncertainty is, to excellent approximation, a fixed-rank (rank-1, at most rank-9) object; treating the 100 universes as 100 independent directions over-counts the flux dimensionality. This is also why the leading flux mode correlates $\rho \approx 0.9$ with the rank-1 muon-energy-scale mode (item 1). A PCA / fixed-rank flux covariance is therefore both justified and recommended.
4. **Ours-only χ^2 — *method resolved; precedent open*.** The obstruction is statistical, not rank: $\mathbf{C}^{\text{ours}} = \mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}} + \mathbf{C}^{\text{ML}}$ is rank 201 and invertible; we report the ours-only goodness-of-fit on it via a truncated-spectral χ^2 (retain $\lambda > 10^{-10}\lambda_{\text{max}}$), the same convention as the 3D comparison, and lead with per-bin pulls plus the regularization scan (§4.3). *Remaining*: published precedent from other MAT-based OmniFold/unbinned analyses for the ours-only χ^2 convention.
5. **3D systematic covariance — *method resolved; precedent open*.** The available-energy extension carries a full combined covariance (universe+bootstrap+ML) over all 1431 reported bins, rank 247 (§7.6); the goodness-of-fit is the rank-247 truncated-spectral χ^2 (§7.5), the recommended treatment for a rank-deficient unbinned covariance. *Remaining*: whether there

is MINERvA precedent for *publishing* a 3D systematic covariance from low-recoil samples, and endorsement of this GoF.

6. **High- E_{avail} DIS-tail excess — *localized in W , generator- and tune-robust; modelling open.* (This study is presented fully in §8; the audit record is kept here.) Enabling Valencia 2p2h resolves the low-recoil excess but leaves a separate $+2.2\sigma$ data excess in the $[1.50, 3.00)$ GeV bin (§7.7), where 2p2h does not contribute and the (sub-percent) FSI dials of item 2 cannot reach. To test the deep-inelastic hypothesis directly we added the hadronic invariant mass W as a validated fifth OmniFold axis (its W -marginal recovers the frozen 4D cross section to 0.11 %) and compared the unfolded (E_{avail}, W) cross section to the GENIE central value. The excess is *predominantly deep-inelastic*: the two highest- E_{avail} bands carry 57 % of the positive excess; of all positive excess, high- E_{avail} (≥ 0.8 GeV) holds 67 % and 83 % of *that* sits at $W \geq 1.8$ GeV, with the single largest cell the deep-DIS corner ($E_{\text{avail}} > 3$ GeV, $W > 3$ GeV) at 22 %. A secondary low- W (< 1.1 GeV, QE-like) excess and a small Δ -region ($W \simeq 1.4$ – 1.8 GeV) deficit are also present. So the missing strength behaves like a deep-inelastic-tail *modelling* deficit, exactly the hypothesis the W axis was added to test. We confirm this against a *generator band*: independent spline-normalised (E_{avail}, W) predictions from GENIE CV, GENIE + Valencia 2p2h/MEC, NuWro, and GiBUU all underpredict the high- E_{avail} × high- W corner by 54 % to 58 % (data/generator = 1.54, 1.58, 1.56). Crucially, enabling Valencia 2p2h does *not* close it — it slightly *worsens* the corner ($1.54 \rightarrow 1.58$), since 2p2h populates low W — and the independent NuWro and GiBUU generators miss it by the same or a larger margin (GiBUU is the most deficient overall, $\sigma = 2.22 \times 10^{-38}$ cm²/nucleon); at $W \in [2.2, 3.0)$ GeV all four sit 23 % to 25 % below the data. The DIS-tail excess is therefore *generator- and tune-robust*: no current model reproduces it. Attaching a number, we marginalise the adopted combined 4D covariance (§7.6, unified-throw systematic) to the E_{avail} axis and evaluate the full-covariance χ^2 of the unfolded data against each generator’s $d\sigma/dE_{\text{avail}}$: all four are incompatible with the data at $> 21\sigma$ over the seven E_{avail} bins and $> 15\sigma$ in the DIS tail ($E_{\text{avail}} \geq 0.8$ GeV) alone — GENIE CV 22.4σ , GENIE + MEC 24.9σ , NuWro 21.9σ , GiBUU 21.2σ . GiBUU spreads its deficit across the *entire* DIS tail (data/generator 1.59 at $E_{\text{avail}} \in [0.8, 1.5)$, 1.36 at $[1.5, 3.0)$) rather than piling it in the deep-DIS catch bin like the GENIE variants, an independent and qualitatively distinct confirmation. Resolving the excess in W confirms and sharpens this picture: we build the full (E_{avail}, W) covariance (42 bins; `C_syst+C_stat+C_lateral`, median 14.8 % per bin) by a frozen-reweighter block sum on the 5D `universes_full` sample, validated by reproducing the frozen 5D marginals to 0.1 %. Over the full (E_{avail}, W) plane the four generators are incompatible with the data at GENIE CV 16.7σ ($\chi^2/\text{ndf} = 412.7/42$), GENIE + MEC 16.1σ (390.5/42), NuWro 31.2σ (1148.4/42), and GiBUU $> 37\sigma$ (1930.2/42). The deficit is *localised in the high- W DIS corner*: restricting to the twelve bins with $E_{\text{avail}} \geq 0.4$ GeV and $W \geq 1.8$ GeV, all four miss the data at 9.0–18.2 σ — GENIE CV 9.0 σ (116.9/12), GENIE + MEC 9.2 σ (121.1/12), NuWro 10.5 σ (149.6/12), and GiBUU 18.2 σ (381.1/12, again the most deficient). The excess is thus a genuinely DIS-region ($W \gtrsim 1.8$ GeV) feature, not a low- W resonance artefact. *The low-recoil cross-check is done*: the bin-identical full-covariance comparison to Ref. [26] on the maximal common (E_{avail}, q_3) grid gives $\chi^2/\text{ndf} = 1.68/2$, $p = 0.43$ (§7.8).**

B Statistical methods

This appendix is the complete uncertainty-quantification methodology of the measurement — the seedscan, bootstrap and universe-covariance procedures, the MAT-conformant covariance construction, the $1/\Phi$ flux normalisation, the comparison conventions (χ^2 , pulls, shape-only), the ill-conditioning/rank analysis, the tension anatomy, the coverage toys, and the closure tests. It was developed as (and supersedes) the standalone companion document `docs/uq_statistical_methods.tex`; each procedure gives the recipe, the math, the implementation pointer, and a worked example using a reported number. (The companion’s introductory statistics primer is omitted here; only the analysis-specific numerical conventions are retained. The χ^2 convention referenced throughout is Eq. (10) in §B.3.2.)

B.1 Numerical conventions: inverting a covariance (Cholesky, ridge jitter, pseudo-inverse)

Equation (10) requires inverting $\mathbf{C}$. Three practical issues come up in this analysis.

Positive-definiteness check (Cholesky). A matrix $\mathbf{C}$ is positive-definite (PD) iff there is a lower-triangular $\mathbf{L}$ with $\mathbf{C} = \mathbf{L}\mathbf{L}^\top$ (the Cholesky factorisation). If Cholesky succeeds, $\mathbf{C}$ is invertible and Eq. (10) is well-posed.

Ridge jitter. Numerical noise can leave a PD matrix barely non-PD. The standard fix is to add an isotropic ridge $\varepsilon \mathbf{I}$. The `uq/analyze_uq.py` script does exactly this:

$$\varepsilon = 10^{-12} \frac{\text{Tr}\mathbf{C}}{n}, \quad (1)$$

where the trace sets the natural scale of the eigenvalues. Twelve orders of magnitude below the mean diagonal is far below any real signal in $\mathbf{C}$ and safely above floating-point round-off.

Rank deficiency at $N \leq n$. If you estimate $\mathbf{C}$ from N replicas in a space of n bins with $N \leq n$, the resulting sample covariance has rank at most $N - 1$: there are not enough independent “directions” to fill the full $n \times n$ matrix. This was the regime of the Stage-1 bootstrap ($N = 50$ replicas, $n = 205$ bins): Cholesky failed without jitter, and the recorded `jitter = 1.6e-94` in the run log was a numerical artefact of the rank deficiency rather than a real near-singularity. The Stage-2 bootstrap ($N = 300 > 205$) is full-rank. The *systematic* covariance suffers a more stubborn version of this problem that survives at any universe count — see §B.4.

Pseudo-inverse. For the χ^2 comparison against the paper covariance, `compare_to_paper_fullcov.py` replaces $\mathbf{C}^{-1}$ with the Moore–Penrose pseudo-inverse $\mathbf{C}^+$, computed via SVD. Writing $\mathbf{C} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ with $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_n)$,

$$\mathbf{C}^+ = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^\top, \quad \Sigma_{ii}^+ = \begin{cases} 1/\sigma_i & \sigma_i > \tau, \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

with a small threshold τ (NumPy’s default is $\tau = n \cdot \sigma_{\max} \cdot \varepsilon_{\text{machine}}$). This silently drops singular modes that are numerically indistinguishable from zero, replacing them with 0 in the inverse rather than ∞ . The result is a stable quadratic form even when $\mathbf{C}$ has zero or near-zero eigenvalues.

B.2 Variance-source procedures

Each subsection below follows the same template: *What it estimates* · *Procedure* · *Math* · *Code pointer* · *Worked example*.

B.2.1 ML stochasticity: the seedscan

What it estimates. The variance in the unfolded cross section due to random choices inside the GBDT (gradient-boosted decision tree) optimiser. OmniFold is deterministic given the input weights, the network architecture, and the random seed; flipping only the seed produces slightly different fits to the per-event reweighting functions, and that spread is the “ML noise floor”.

Procedure. Run $n = 10$ full 5-iteration MEFHC unfolds with everything fixed except the GBDT `random_state`. The driver `sbatch_unfold_2d_MEFHC_5iter_seedscan.sh` pins three seeds per trial via `--seed N`:

$$(\text{step-1 classifier}, \text{step-2 classifier}, \text{miss regressor}) = (N, N + 1, N + 2), \quad N \in \{1, \dots, 10\}. \quad (3)$$

The canonical seedscan uses the `lgbm` (LightGBM) backend, so it is backend-matched to the production unfold; an earlier HistGBT seedscan gave a slightly larger floor and is superseded.

Math. For each reported bin i , build the n -vector $X_i^{(1)}, \dots, X_i^{(n)}$ across trials and compute the unbiased sample standard deviation

$$\sigma_i^{\text{ML}} = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (X_i^{(k)} - \bar{X}_i)^2}, \quad \bar{X}_i = \frac{1}{n} \sum_k X_i^{(k)}. \quad (4)$$

The headline diagnostic is the per-bin *relative* spread $\sigma_i^{\text{ML}}/\bar{X}_i$, summarised by its median over the 205 reported bins.

Code pointer. `seedscan/analyze_seedscan.py` (rollup); `sbatch_unfold_2d_MEFHC_5iter_seedscan.sh` (driver); covariance output `uq/seedscan_lgbm_ml/uq-covariance_ml.root`.

Worked example (STATUS, UQ envelope). With $n = 10$ `lgbm` trials on the 205-bin set:

- Total- σ relative spread: 0.0055%.
- Per-bin median relative spread: 0.166%.
- Component trace $\sqrt{\text{Tr} \mathbf{C}^{\text{ML}}} = 5.061 \times 10^{-41}$.

The paper’s per-bin total-uncertainty median is 6.86%; the ML noise is therefore $\approx 2.4\%$ of the paper’s quoted uncertainty and far subdominant to data-stat and systematics. Going from $n = 4$ to $n = 10$ trials moved the headline numbers within rounding, so the seedscan envelope is converged.

B.2.2 Statistical uncertainty: the Poisson bootstrap

What it estimates. The combined data-statistical and MC-statistical variance, propagated *coherently* through the full OmniFold pipeline. Coherent means: if some data event happens to be “up-weighted” in a replica, the same event flows through both the step-1 classifier training and the final reweighted-MC cross-section construction; if an MC event is up-weighted, the same up-weight is applied to its reco-tree and truth-tree copies, preserving the response matrix.

Procedure. For each replica $k = 1, \dots, N$:

1. Draw per-event Poisson(1) factors $b_i^{\text{data},(k)}$ for every data event and $b_i^{\text{MC},(k)}$ for every MC event, using two independent sub-RNGs (a data RNG seeded from k , and an MC RNG seeded from $k + 10^7$). The two sources are uncorrelated by construction.
2. Multiply data weights by b^{data} and MC weights (both reco and truth copies of each MC event) by the *same* b^{MC} . This keeps the migration matrix self-consistent.
3. Run OmniFold (5 iterations) with the GBDT seed *pinned*. Pinning the seed isolates the bootstrap signal from the ML noise of §B.2.1.
4. Save `hXSec2D` (and 1D projections).

Stage-1 used $N = 50$ replicas (rank-deficient on 205 bins); Stage-2 scaled to $N = 300 > 205$ for a full-rank statistical covariance.

Caveat (seed-varying replicas). The available $N = 300$ covariance was produced with replicas that varied *both* `--bootstrap-seed N` and `--seed N`, so it carries a small ML-stochastic contribution on top of the pure Poisson statistics. The drivers are now patched to pass a fixed `--seed 1` with only `--bootstrap-seed N` varying; a pure-Poisson rerun should replace this block when available. The leak is numerically small relative to the systematic block (the ML floor is 0.166% vs the stat 0.564%), but the statistical covariance is not yet purely Poisson.

Math. Flatten each replica’s 14×16 histogram into a 205-vector $\mathbf{X}^{(k)}$ over reported bins (row-major over $(p_T, p_{\parallel})$). The bootstrap covariance is the unbiased sample covariance:

$$(\mathbf{C}^{\text{boot}})_{ij} = \frac{1}{N-1} \sum_{k=1}^N (X_i^{(k)} - \bar{X}_i)(X_j^{(k)} - \bar{X}_j), \quad \bar{X}_i = \frac{1}{N} \sum_k X_i^{(k)}. \quad (5)$$

This is exactly `np.cov(Xrep, rowvar=False, ddof=1)` in `uq/analyze_uq.py`. The rank-deficiency check is the Cholesky-plus-jitter pattern from §B.1.

Why Poisson(1) is the right primitive. A non-parametric bootstrap with replacement assigns event i a multinomial count n_i with $\mathbb{E}[n_i] = 1$, $\text{Var}[n_i] = (N-1)/N \rightarrow 1$, and $\mathbb{E}[\sum n_i] = N$. In the large- N limit the n_i become independent and each is Poisson(1) (Appendix B.9). For a weighted analysis you multiply $w_i \rightarrow w_i n_i$; since n_i may equal zero, some events are dropped from a given replica, exactly mimicking the “with replacement” resample.

Code pointer. Bootstrap weight draws and seed pinning: `unfold_2d_omnifold_unbinned.py` (`--bootstrap-seed`, `--seed`).

Covariance rollup: `uq/analyze_uq.py`.

Output: `uq/bootstrap_MEFHC_300/uq.covariance.boot300.root`.

Worked example (STATUS, $N = 50$ vs $N = 300$, MEFHC lgbm).

Metric (205 bins)	$N = 50$	$N = 300$
Total σ relative std	0.068%	0.061%
Per-bin median relative spread	0.532%	0.564%
Per-bin p_{84}	1.235%	1.215%

The per-bin median is stable from $N = 50$ to $N = 300$ ($0.53\% \rightarrow 0.56\%$); the $N = 300$ block is the one used in the headline because it is full rank ($N > 205$), whereas $N = 50$ was rank-deficient and its Cholesky needed jitter 1.6×10^{-94} . The bootstrap median is $\approx 3.4\times$ the seedscan median (0.166%), as expected for a sample of $\mathcal{O}(10^6)$ data events. Across datasets the per-bin median scales as $\sqrt{N_{\text{events}}}$: playlist 1A ($\approx N/12$ of MEFHC) gives 1.46% , consistent with $\sqrt{12} \times 0.56\% \approx 1.9\%$ at the same backend.

B.2.3 Systematic uncertainty: universes

What it estimates. Model uncertainty, by re-running the full unfold under each of several physically-motivated reweightings of the MC. Where the bootstrap shakes the *events*, the universes shake the *model*: a different flux $\rightarrow$ different per-event weights $\rightarrow$ different unfolded cross section.

Procedure. The C++ event loop (`runEventLoopOmniFold.cpp`, env `MNV101_DUMP_UNIVERSES`) dumps, for each MC event, alternative weight branches $w_{B,u}^{\text{truth}}$ and $w_{B,u}^{\text{reco}}$ for every (band B , universe index u) pair. The publication-grade Stage-2 sweep has **187 universes across 44 bands**: 6 *lateral* bands (BeamAngleX/Y, MuonResolution, GEANT_{Neutron,Pion,Proton}, Muon_Energy_MINERvA) that require re-running the event loop with shifted detector quantities, plus 38 *vertical* bands that only re-weight. The dominant ensemble is the 100-universe PPFX flux band; almost all other bands are $\pm 1\sigma$ pairs.

Band (examples)	Variation	N_u
Flux	PPFX hadron-production sampling	100
Muon_Energy_MINOS	Muon energy scale, $\pm 1\sigma$	2
Muon_Energy_MINERvA	Muon energy scale (lateral), $\pm 1\sigma$	2
MinosEfficiency	MINOS muon-match efficiency, $\pm 1\sigma$	2
MaRES, MvRES, MaCCQE	GENIE knobs, $\pm 1\sigma$	2 ea.
GEANT_{Neutron,Pion,Proton}	FSI response (lateral), $\pm 1\sigma$	2 ea.

For each (B, u) , `unfold_2d_omnifold_unbinned.py --universe B:u` reweights all MC events (truth and reco) by the matching branch and runs OmniFold with a fixed seed. A separate matched “CV” job runs without `--universe`, with the *same* omnifile, backend, seed, iteration count, and flux, so that ML noise cancels between CV and universe.

Math (MAT-conformant, the current convention). For universe (B, u) , define the per-bin value $\mathbf{X}^{(B,u)}$, a 205-vector over reported bins. MINERvA’s MAT framework (`PlotUtils/MnvVertErrorBand::Calc`; see §B.2.4) builds the per-band covariance as the *mean-centered, biased* sample covariance, referenced to the **universe mean** (not the CV) and normalised by $1/N_u$:

$$\mathbf{C}^B = \frac{1}{N_u} \sum_{u=1}^{N_u} (\mathbf{X}^{(B,u)} - \langle \mathbf{X} \rangle_B) (\mathbf{X}^{(B,u)} - \langle \mathbf{X} \rangle_B)^\top, \quad \langle \mathbf{X} \rangle_B = \frac{1}{N_u} \sum_u \mathbf{X}^{(B,u)}. \quad (6)$$

This single formula is applied *uniformly* to all $N_u \geq 2$ — there is **no special-case $\pm 1\sigma$ pair formula**. For a $\pm 1\sigma$ pair ($N_u = 2$) it collapses to the rank-1 $\frac{1}{4}(\mathbf{X}_+ - \mathbf{X}_-)(\mathbf{X}_+ - \mathbf{X}_-)^\top$; mean-centering kills the symmetric component, so a one-sided or asymmetric pair correctly contributes near-zero variance rather than the rank-2 contribution the legacy CV-centered formula would give (Appendix B.10).

Target-nucleon normalization (rank-1 add-on). MINERvA quotes a 1.4% overall target-nucleon normalization uncertainty (Aliaga NIM, arXiv:1305.5199) that lives outside the `MnvVertErrorBand` machinery. We add it as a fully-correlated rank-1 outer product on top of the band sum:

$$\mathbf{C}^{\text{norm}} = (\sigma_N \mathbf{X}^{\text{CV}})(\sigma_N \mathbf{X}^{\text{CV}})^\top, \quad \sigma_N = 0.014, \quad (7)$$

via `analyze_universes.py --add-norm 0.014`.

Total systematic covariance. Bands are assumed independent (different physics sources, different RNGs), so

$$\mathbf{C}^{\text{syst}} = \sum_B \mathbf{C}^B + \mathbf{C}^{\text{norm}}. \quad (8)$$

Independence is a modelling assumption with one known correlated exception — the Bashyal flux $\leftrightarrow$ Muon_Energy_MINOS joint block (§B.4), which we have since rederived ($\rho = 0.900$, `uq/flux_muonE_cross.r`) it is real but rank-preserving, so for the 2D budget the block sum above is retained as the headline and the cross term is carried as a diagnostic.

There is, however, a second and larger correction that Eq. (8) omits: the band sum is linear in the per-band covariances, whereas the OmniFold reweighter is *nonlinear* in the input perturbations, so jointly perturbing all bands and re-unfolding produces a cross term beyond any pairwise correlation. We measure this directly with a *unified throw* — N joint re-unfolds in which every reweightable band and the flux multisim are perturbed simultaneously per event ($w_{\text{CV}} \prod_B \rho_B^{g_B}$, $g_B \sim \mathcal{N}(0, 1)$) and the cross section re-extracted, with the OmniFold run-to-run jitter floor subtracted via a zero-shift null set (`nd-unfolding/unified_throw_cov.py`). On the 4D ($p_T, p_\parallel, E_{\text{avail}}, q_3$) analysis binning the jitter-corrected unified covariance is **2.0** $\times$ the block sum in $\sqrt{\text{tr}}$: the block sum *underestimates* the vertical systematic by a factor of two, concentrated in the high- p_T /lowest- E_{avail} corner where migration is most off-diagonal (top 1% of bins carry 78% of the trace excess; the per-bin median ratio is 1.00). This supersedes the earlier single-throw superposition probe, whose cross terms sat at the jitter floor and so falsely leaned block-sum-valid. For the 4D budget we therefore *adopt* the unified result, transferring its per-bin fractional inflation $g_i = \max(\sigma_i^{\text{uni}}, \sigma_i^{\text{blk}})/\sigma_i^{\text{blk}} \geq 1$ onto the sweep’s own vertical block, $\mathbf{C}^{\text{syst}} \rightarrow (\mathbf{C}^{\text{syst}} - \mathbf{C}^{\text{vert}}) + \mathbf{G} \mathbf{C}^{\text{vert}} \mathbf{G}$ ($\mathbf{G} = \text{diag } g$), which is positive-semidefinite by construction (a direct rank- N swap is not) and never falls below the block baseline.

Legacy formula (forensic only). The pre-2026-05-28 rollup used a CV-centered pair estimator $\frac{1}{2}(\Delta_+ \Delta_+^\top + \Delta_- \Delta_-^\top)$ with $\Delta = \mathbf{X} - \mathbf{X}^{\text{CV}}$. It agrees with Eq. (6) only for symmetric pairs and overcounts asymmetric ones; moving to the MAT formula dropped the total universe trace by 1.9%. It is reachable via `--legacy-pair-formula` for reproduction of the superseded $\chi^2/\text{ndf} = 1.473$ headline only.

Code pointer. Event-loop dump: `runEventLoopOmniFold.cpp` (env `MNV101_DUMP_UNIVERSES`).

Per-universe unfold: `unfold_2d_omnifold_unbinned.py --universe BAND:IDX`.

Per-band rollup: `uq/analyze_universes.py (--add-norm 0.014)`.

Output: `uq/universe_stage2_MEFHC_full_matcorr/uq_universe_covariance_full_matcorr.root`.

Worked example (STATUS, full matcorr sweep). Top contributing bands, per-bin median relative σ_i/X_i^{CV} over 205 reported bins:

Band	Per-bin median rel. σ
Muon_Energy_MINOS	2.305%
Muon_Energy_MINERvA	1.285%
MinosEfficiency	1.472%
Flux (PPFX, $N_u = 100$)	1.008%
MaRES	0.548%
MvRES	0.377%
MaCCQE	0.357%
Systematic total (44 bands + norm)	4.783% (median); $\sqrt{\text{Tr}\mathbf{C}} = 2.463 \times 10^{-39}$

Grouped by category the medians are: Muon reconstruction 3.366%, Models 2.034%, Flux 1.008%, Hadronic response 0.758%, Statistical 0.563%, Normalization 0.081%. Muon energy scale (MINOS+MINERvA) and the MINOS-efficiency band dominate; these are exactly the bands the Ruterbories paper flags as flux-correlated (§B.4). *(These are the MAT-conformant matcorr medians; the pre-2026-05-28 legacy build gave larger per-band values, e.g. Muon_Energy_MINOS 2.571%.)*

B.2.4 How MINERvA builds its published covariance

To compare honestly we had to establish what MINERvA actually does — which is *not* spelled out in the paper but lives in the MAT C++ framework. The takeaway: **MAT samples per-band exactly as we do**, using the mean-centered $1/N$ sample covariance, and sums bands element-wise.

What the Ruterbories 2021 paper says (Section V). Three uncertainty categories — flux, detector response, neutrino-interaction model — evaluated by “re-extracting the cross section using modified simulations”, i.e. the universe method, but with *no covariance formula given*. Specifics quoted: flux below 4% across most of phase space thanks to the ν - e -scattering constraint (Valencia, arXiv:1906.00111); 1.4% target-nucleon normalization; in-situ test-beam hadron response with per-particle GEANT inelastic variations; muon energy scale 1% constrained by the low-recoil fit of Bashyal (JINST 16 P08068), with the explicit statement that “*the fit causes the flux and muon energy scale uncertainties to be correlated and that correlation is propagated to the final result*” — the only non-block-diagonal piece the paper admits. The only covariance that appears in equations is V in the model-test χ^2 (Eq. 1, “the measurement covariance matrix”), with 205 degrees of freedom (Table I), and a log-normal χ^2 variant motivated by the highly-correlated flux normalization (Peelle’s Pertinent Puzzle).

What the paper does *not* say. No per-band covariance construction; no $\pm 1\sigma$ pair formula; no band-independence/block-sum statement; nothing about rank, condition number, or regularization of V ; no bootstrap or coverage validation (the paper uses D’Agostini iterative unfolding). All of this is delegated to MAT or simply unstated.

Per-band covariance: `MnvVertErrorBand::CalcCovMx`. Reading the MAT source establishes three facts, encoded in Eq. (6):

- the reference is the **universe mean** $\langle x \rangle_u$, not the central value X_{CV} ;
- the normalization is $1/N$ (biased), not Bessel’s $1/(N-1)$ — a 1% offset for PPFX ($N = 100$);
- there is **no special case for $N = 2$** — the mean-centered N -universe form is applied uniformly. The “MINERvA-101 $\pm 1\sigma$ pair” formula $\frac{1}{2}(\Delta_+ \Delta_+^\top + \Delta_- \Delta_-^\top)$ is a folk re-derivation that agrees only for CV-symmetric shifts.

We verified our rollup against MAT byte-for-byte: populating a `PlotUtils::MnvH1D` with the same 44 bands / 188 universes and comparing `MnvH1D::GetTotalErrorMatrix` against our `--add-norm 0` rollup gives a max element-wise relative difference of 5.5×10^{-17} (machine ε), with $\sqrt{\text{Tr}}$ matching to seven digits (`uq/verify_matcorr_vs_mnvh1d.py`).

Total covariance: `MnvH1D::GetTotalErrorMatrix`. A straight element-wise sum across bands plus the statistical matrix. Band-independence is assumed; cross-band correlations exist only if encoded inside a single combined band (e.g. the Bashyal joint fit, handled outside this loop). MAT does *no* ridge / shrinkage / SVD truncation on the published total cov and no special handling of rank deficiency — the published V is full-rank because it includes the *statistical* block, not because it sums richer systematics (its systematic-only rank is $120 < 140$; see §B.4).

Upstream references that set each band’s $\pm 1\sigma$ amplitude.

Input	Reference
Flux / PPFX (100 universes)	Aliaga, PRD 94 092005 (2016), arXiv:1607.00704
ν - e flux constraint ($< 4\%$)	Valencia, PRD 100 092001 (2019), arXiv:1906.00111
Muon-E scale $\leftrightarrow$ flux fit	Bashyal, JINST 16 (2021) P08068
Test-beam / GEANT hadron response	Aliaga, NIM A 789 28 (2015), arXiv:1501.06431
GENIE reweight knobs	Andreopoulos, NIM A 614 87 (2010); GENIE manual arXiv:1510.05494
1.4% normalization (detector mass)	Aliaga, NIM A 743 130 (2014), arXiv:1305.5199

B.2.5 Block-sum combination

What it estimates. The total uncertainty covariance, combining the three statistically independent components above.

Independence assumption. The bootstrap samples *events* (with two distinct sub-RNGs for data and MC). The universes vary *model weights*, with the GBDT seed pinned. The seedscan varies the *GBDT seed*, with all weights fixed. The three Monte Carlos are independent by construction:

$$\mathbf{C}^{\text{tot}} = \mathbf{C}^{\text{ML}} + \mathbf{C}^{\text{stat}} + \mathbf{C}^{\text{syst}}. \quad (9)$$

The combined-cov hook in `compare_to_paper_fullcov.py` (`--omnifold-cov ROOT:HIST`, repeatable) adds OmniFold- derived covariances on top of the paper’s `TotalCovariance` through exactly this rule. $\mathbf{C}^{\text{ML}}$ is more than an order of magnitude below the other two; the headline includes it but it is numerically negligible.

Worked example (STATUS, UQ envelope). Component traces and per-bin median relative spreads:

Component	$\sqrt{\text{TrC}}$	Per-bin median rel. σ
ML noise (lgbm seedscan, $n = 10$)	5.061×10^{-41}	0.166%
Statistical (Poisson bootstrap, $N = 300$)	1.828×10^{-40}	0.564%
Systematic (matcorr sweep + norm)	2.463×10^{-39}	4.783%
Combined (block sum)	2.470×10^{-39}	4.822%
Paper <code>TotalCov</code> (reference)	2.676×10^{-39}	6.86%

The combined OmniFold envelope is $\approx 70\%$ of the paper’s per-bin median uncertainty, dominated by the systematic block. (The combined $\sqrt{\text{Tr}}$ is the quadrature sum of the three independent component traces; the per-bin median 4.822% is taken directly from the matcorr rollup summary.)

B.3 Comparing to the paper

B.3.1 The 205-bin selection

The paper reports $d^2\sigma/(dp_T dp_{\parallel})$ on a 14-bin p_T axis $\times$ 16-bin $p_{\parallel}$ axis = 224 cells (edges in `minerva_paper_anc/bin_mapping.txt`). Of those, 205 have $x^{\text{paper}} > 0$ in `data_result.ptpl_2D_minerva_inclusion`; the remaining 19 are paper-unreported cells that the paper’s analysis suppresses via an efficiency cut. We drop those 19 from both sides of every χ^2 and pull — that is the unambiguous “apples-to-apples” choice.

A secondary diagnostic mask, used only as a sensitivity check, is the strict-interior set: cells where $p_T^{\text{high}}/p_{\parallel}^{\text{low}} \leq \tan 20^\circ$. Some cells in the 205-bin set straddle the $\theta_\mu < 20^\circ$ kinematic cut and have wildly suppressed paper efficiencies that we do not replicate; pulls on those cells run to $\mathcal{O}(100)\sigma$ and dominate the χ^2 . The 185-bin interior cut removes them; we report it only for sensitivity. **The canonical metric is the 205-bin set** (commit 65fb085, *Seeds can: use the 205 paper-reported bins everywhere*).

B.3.2 Covariance-aware χ^2/ndf

Math. With $\Delta \equiv \mathbf{x}^{\text{ours}} - \mathbf{x}^{\text{paper}}$ on the 205-bin set and $\mathbf{C}$ the chosen covariance,

$$\chi^2 = \Delta^\top \mathbf{C}^+ \Delta, \quad \text{ndf} = n_{\text{reported}} = 205, \quad \chi^2/\text{ndf} = \chi^2/205. \quad (10)$$

The implementation (`chi2_with_cov` in `compare_to_paper_fullcov.py`) builds the 205×205 block, inverts with `np.linalg.pinv` (§B.1), and forms the quadratic form.

Code pointer. `compare_to_paper_fullcov.py` (headline); `--omnifold-cov ROOT:HIST` (repeatable) adds each OmniFold covariance via Eq. (9); `--subtract-stat` removes the paper’s stat block to avoid double-counting the bootstrap (§B.4).

Worked example: comparison to paper (STATUS headline).

Metric (205 bins)	Paper	OmniFold (combined)
Width-weighted total σ , $\sqrt{\mathbf{w}^\top \mathbf{C} \mathbf{w}}/(\mathbf{w}^\top \mathbf{x})$	4.61%	2.26%
Per-bin median relative σ	6.86%	4.822%
χ^2/ndf	3.661 (paper-only)	1.699 (paper+ours)
Log-normal χ^2/ndf	—	1.688

The $3.661 \rightarrow 1.699$ drop (-54%) is the central result of the UQ campaign: once the OmniFold-side error budget is honestly included, the agreement with the paper improves to $\chi^2/\text{ndf} \approx 1.7$. (A perfectly correct model with correct uncertainties gives exactly 1; 1.7 tolerates small residual shape mismatch in high- $p_{\parallel}$ tails, see §B.3.4.) **This combined χ^2 double-counts systematics shared with the paper** (flux, GENIE, RPA, 2p2h, MINOS); see §B.4 for why it is still the honest headline and why an ours-only χ^2 is not yet quotable. The log-normal variant (matching the paper’s Table I treatment of the correlated flux normalization) gives 1.688, essentially identical.

Edge-bin sensitivity ($p_{\parallel}$ -min cut). `compare_to_paper_fullcov.py` ships a side-table that recomputes χ^2/ndf on progressively shrinking subsets, using the paper covariance only (no OmniFold add):

$p_{\parallel} \geq (\text{GeV}/c)$	N_{bins}	χ^2	χ^2/ndf	median ratio
1.5	205	750.49	3.661	1.0064
2.0	196	685.36	3.497	1.0059
2.5	186	545.87	2.935	1.0059
3.0	175	474.84	2.713	1.0059
3.5	163	452.95	2.779	1.0073
4.0	151	428.72	2.839	1.0068

The residual is concentrated in the low- $p_{\parallel}$, near-edge cells; the high- $p_{\parallel}$ tails carry sub-2% shape disagreement. This is a robustness check, not a separate method.

B.3.3 Pull distribution

Math. For each of the 205 reported bins,

$$\text{pull}_i = \frac{x_i^{\text{ours}} - x_i^{\text{paper}}}{\sqrt{C_{ii}^{\text{tot}}}}, \quad \mathbb{E}[\text{pull}_i] = 0, \quad \text{Var}[\text{pull}_i] = 1 \text{ under the null.} \quad (11)$$

Summary diagnostics: pull mean, RMS, the 14×16 pull heatmap, and a pull histogram (in `compare_to_paper_fullcov.py`).

Worked example (STATUS headline). With the combined covariance (paper TotalCov + matcorr universe + bootstrap-300 + ML):

- Pull mean: 0.069 (centred on zero; no global offset).
- Pull RMS: 0.466.

Pull RMS < 1 means the combined error band is *conservative* relative to the actual disagreement — the band is wider than it needs to be to absorb the observed scatter. No bins have $|\text{pull}| > 3$ in the reported set. Compare with paper-cov-only: mean 0.089, RMS 0.598 — similar mean (no bias), slightly wider spread before we include our own uncertainties.

B.3.4 Shape-only χ^2 with Jacobian propagation

What it estimates. Whether the *shape* of the $(p_T, p_{\parallel})$ cross section matches the paper, independent of absolute normalisation. Useful because the absolute normalisation depends on the flux integral, target nucleon count, and POT (all of which can drift by $\mathcal{O}(\%)$), whereas the shape is a pure unfolding diagnostic.

Definition. With bin widths $w_j = (\Delta p_T)_j (\Delta p_{\parallel})_j$ and total $T(\mathbf{x}) = \sum_{j \in \text{mask}} w_j x_j$, the unit-area shape is

$$s_i(\mathbf{x}) = \frac{x_i}{T(\mathbf{x})}. \quad (12)$$

By construction $\sum_i w_i s_i = 1$.

Jacobian. We need the covariance of $\mathbf{s}$ given the covariance of $\mathbf{x}$. Differentiating,

$$\frac{\partial s_i}{\partial x_j} = \frac{\delta_{ij}}{T} - \frac{x_i w_j}{T^2} = \frac{1}{T}(\delta_{ij} - s_i w_j). \quad (13)$$

In matrix form (in `normalize_xsec_shape.py`),

$$\mathbf{J} = \frac{1}{T}(\mathbf{I} - \mathbf{s} \mathbf{w}^\top), \quad \mathbf{C}^{\text{shape}} = \mathbf{J} \mathbf{C}^{\text{paper}} \mathbf{J}^\top. \quad (14)$$

Rank reduction. $\mathbf{J}$ is rank $n-1$: the unit-area constraint $\sum_i w_i s_i = 1$ kills one degree of freedom. Algebraically, $\mathbf{w}^\top \mathbf{J} = \mathbf{w}^\top (\mathbf{I} - \mathbf{s} \mathbf{w}^\top)/T = (\mathbf{w}^\top - \mathbf{w}^\top)/T = \mathbf{0}^\top$, so the “all-bins-up” direction is the kernel of $\mathbf{J}$ and lives in the null space of $\mathbf{C}^{\text{shape}}$. Accordingly:

- $\mathbf{C}^{\text{shape}}$ has one zero eigenvalue and rank $n-1$.
- $\chi_{\text{shape}}^2 = (\mathbf{s}^{\text{ours}} - \mathbf{s}^{\text{paper}})^\top (\mathbf{C}^{\text{shape}})^+ (\mathbf{s}^{\text{ours}} - \mathbf{s}^{\text{paper}})$ (pseudo-inverse drops the constrained direction).
- $\text{ndf} = n_{\text{bins}} - 1 = 204$ on the 205-bin set.

Code pointer. `normalize_xsec_shape.py`: `shape_jacobian`, `shape_chi2`.

Worked example (STATUS headline). Shape χ^2/ndf on 205 bins with the paper’s `TotalCovariance` (Jacobian-propagated): 3.596. This is essentially the absolute-normalisation χ^2/ndf of 3.661 *minus* the contribution of overall scale. Total- σ ratio is $\sigma_{\text{tot}}^{\text{ours}}/\sigma_{\text{tot}}^{\text{paper}} = 1.0111$ — 1% scale difference. Shape disagreement at the $\sim 1\%$ level (median bin ratio 1.0064, 94% of bins within 10% of the paper) accounts for the remaining shape residual.

B.4 Ill-conditioning, rank deficiency, and open questions

The obstruction. For publication we would like an *ours-only* χ^2/ndf — the data-to-model agreement evaluated with our own covariance $\mathbf{C}^{\text{ours}} = \mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}} + \mathbf{C}^{\text{ML}}$ rather than the paper’s V . That requires inverting $\mathbf{C}^{\text{ours}}$, and it is rank-deficient:

$$\text{rank}(\mathbf{C}_{\text{sample}}^{\text{syst}}) = 140/205, \quad \text{cond}(\mathbf{C}^{\text{syst}}) = 8.7 \times 10^8. \quad (15)$$

The deficiency is set by **band count, not universe count**: under the MAT formula Eq. (6) each $\pm 1\sigma$ pair is rank 1 and the 100-universe PPF band is rank ≤ 99 (and *effectively* rank-1: a single normalization mode holds 99.6% of its variance, item 4 below), so the 44-band sum cannot fill the 205-dimensional bin space. *This is not a missing-bands problem.* The paper’s own systematic-only covariance (`TotalCov` – `StatOnlyCov`) is rank 120 — *lower* than our 140 — so we propagate at least as many independent systematic directions as the published analysis (we use the same `GetStandardSystematics`). The paper’s V is full-rank (rank = 204, cond = 1.5×10^{12}) purely because it **includes the statistical block** (`StatOnlyCov`, rank 205, near-diagonal), not because it sums richer systematics. Adding our 300-replica bootstrap does the same for us: $\text{rank}(\mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}}) = 201$, cond = 3.8×10^{14} . (An earlier, lower-replica bootstrap left the rank near 140 — the origin of the now-superseded “bootstrap barely changes the picture” note.)

The real obstruction is statistical, not rank. Even at rank 201 a defensible single ours-only χ^2 is out of reach — but for a *statistical-precision* reason, not a missing-band one. Our OmniFold bootstrap stat block is $2.5\times$ smaller by trace than the paper’s stat block ($\sqrt{\text{Tr} \mathbf{C}^{\text{boot}}} = 1.8 \times 10^{-40}$

vs $\sqrt{\text{Tr StatOnlyCov}} = 4.6 \times 10^{-40}$), so it regularises the small systematic modes $2.5\times$ less. The naive pseudo-inverse then gives $\chi^2/\text{ndf} = 252$ for $\mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}}$, dominated by poorly-determined directions, against the paper’s 3.66 with its larger stat floor. The lever toward an ours-only χ^2 is therefore the *statistical* block — not the hunt for more systematic bands. The gap is *not* a missing MC-statistical term: our bootstrap already draws independent Poisson(1) multipliers on *both* data and MC per replica (§B.2.2; Practical Guide 2507.09582 §5.1), so data-stat and MC-stat contribute jointly. With that excluded, the smaller stat covariance most likely reflects genuinely higher unbinned-OmniFold statistical efficiency relative to the published binned D’Agostini unfold (whose iterative regularisation amplifies statistical fluctuations), or a difference in what the paper’s StatOnlyCov encompasses — **the split-bootstrap test below now resolves this**, and either way it is a more productive question than counting bands.

Resolution (data/MC split bootstrap). We ran a 200+200-replica *split* bootstrap that Poisson-fluctuates the data weights only (`boot_data`) or the MC weights only (`boot_mc`) with independent RNGs, so in expectation $\mathbf{C}^{\text{boot}} = \mathbf{C}_{\text{data}}^{\text{boot}} + \mathbf{C}_{\text{mc}}^{\text{boot}}$. Three findings settle the question. (i) *Closure holds*: $\sqrt{\text{Tr}(\mathbf{C}_{\text{data}} + \mathbf{C}_{\text{mc}})}/\sqrt{\text{Tr}\mathbf{C}^{\text{boot}}} = 1.07$, with the variance split 77% data / 23% MC. (ii) The *data stream alone* gives $\sqrt{\text{Tr}\mathbf{C}_{\text{data}}}/\sqrt{\text{Tr StatOnly}} = 0.356$ ($\mathbf{C}^{\text{boot}}/\text{StatOnly} = 0.392 = 1/2.55$): the $2.5\times$ deficit is present in the data-only stat error, so it is *not* an MC-statistics omission — it reflects genuinely higher unbinned-OmniFold statistical efficiency than the binned D’Agostini unfold. (iii) There is *also* a structure difference: our stat block is correlated (mean $|\text{off-diag } \rho| = 0.108$, leading-eigenvector overlap with StatOnly 0.46) whereas the paper’s is near-diagonal (0.009); scaling our block to the paper trace still leaves $\chi^2/\text{ndf} = 76.7$, so the gap is shape as well as magnitude. **Consequence:** the correct ours-only χ^2 uses our *own* $\mathbf{C}^{\text{ours}}$, not the paper’s stat floor. The truncation-dependence below remains (the inversion is still regularizer-set), so we continue to lead with per-bin pulls and the scan; but with the stat block understood, the ours-only goodness-of-fit is now reported on $\mathbf{C}^{\text{ours}}$ via a truncated-spectral χ^2 (retain $\lambda > 10^{-10}\lambda_{\text{max}}$), the same convention carried into the 3D generator comparison (§B.4 is the 2D anchor; the 3D campaign applies it on all 1431 reported bins).

Regularization scans (all cutoff-dependent). On $\mathbf{C}^{\text{ours}}$ over the 205 reported bins, no regularizer gives a stable single number:

Method	Setting	Result
Eigenvalue truncation	$\lambda/\lambda_{\text{max}} > \{10^{-3}, 10^{-4}, 10^{-5}\}$	rank $\{19, 49, 82\}$; $\chi^2/205 = \{0.13, 1.12, 3.08\}$
Ridge $\mathbf{C} + \varepsilon\lambda_{\text{max}}\mathbf{I}$	$\varepsilon = \{10^{-3}, 10^{-4}, 10^{-5}\}$	$\chi^2/205 = \{0.35, 1.38, 4.30\}$
Diagonal-target shrinkage	$\lambda = 0.05$	rank $178 \rightarrow 204$, cond 1.3×10^{11} , $\chi^2/205 = 3.29$

Each method spans roughly two decades of $\chi^2/205$ across a two-decade scan of its knob, with no defensible single cutoff. The useful conclusion is diagnostic, not a number: the naive inverse is dominated by tiny, poorly-determined modes, and in the retained high-variance modes the central-value difference is not large. **Quote direct per-bin pulls (§B.3.3) and this scan, not a single unqualified ours-only χ^2 .**

Bootstrap/paper-stat double-count. The combined headline (§B.3.2) adds $\mathbf{C}^{\text{boot}}$ on top of the paper’s TotalCov, which already contains the paper’s statistical block — a double count. **--subtract-stat** replaces the paper baseline by $\text{TotalCov} - \text{StatOnlyCov}$ before adding our covariances, driving χ^2/ndf $1.699 \rightarrow 23.96$. That *over-corrects*: our $\sqrt{\text{Tr}\mathbf{C}^{\text{boot}}} = 1.83 \times 10^{-40}$ is far

smaller than the paper’s published stat block ($\sim$ the 2.7×10^{-39} scale of `TotalCov`), so subtracting the latter without a comparable replacement removes too much. The headline therefore keeps the double-counted 1.699 as the well-defined “what does OmniFold UQ add on top of the published V ” statement, with the de-double-counted number on file as a secondary diagnostic.

Open questions for MINERvA collaborators.

1. **Statistical block, not missing bands — RESOLVED.** We previously read the 140/205 rank as missing systematic bands; direct comparison refutes this — the paper’s systematic-only covariance is rank $120 < 140$, so our band set is at least as rich. The paper’s full rank is its statistical block, which we match with the bootstrap ($\mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}}$: rank 201). The residual — our bootstrap stat covariance $2.5\times$ smaller by trace than the paper’s stat block — is now *explained* by the data/MC split bootstrap (§B.4, “Resolution”): the deficit is present in the *data-only* stream ($\mathbf{C}_{\text{data}}/\text{StatOnly} = 0.356$), so it is not an MC-statistics omission but genuine unbinned-OmniFold statistical efficiency, with an additional correlation-structure difference (ours $\rho_{\text{off}} = 0.108$ vs the paper’s near-diagonal 0.009). The actionable conclusion: quote the ours-only χ^2 on our *own* $\mathbf{C}^{\text{ours}}$ — do not inflate the stat block to the paper’s. *No collaborator input required; retained here as the worked example.*
2. **Bashyal flux $\leftrightarrow$ Muon Energy MINOS joint block** (JINST 16 P08068) — the only cross-band correlation the paper names. We *rederived* it: `MuonEnergyScale` is rank-1, so the cross term is rank-2, $\mathbf{C}_{\times} = \rho(\mathbf{d}_f \mathbf{d}_e^{\top} + \mathbf{d}_e \mathbf{d}_f^{\top})$, with $\rho \approx 0.87\text{--}0.90$ recovered two independent ways from the shipped blocks (Cauchy–Schwarz in `TotalCov` = +0.90; subtraction residual = +0.87). It is real and now quantified, but adding it leaves our rank unchanged (its directions already lie in $\text{range}(\mathbf{C}^{\text{syst}})$). For a paper-independent ρ , is the fitted $(\Phi_{\text{norm}}, \Phi_{\text{shape}}, \text{MuonE}_{\text{scale}})$ posterior available from MAT-MINERvA?
3. **FrInel_pi — resolved (negligible).** It is commented out in `MAT-MINERvA/universes/GenieSystematics.c` with the note that “this knob should not currently be evaluated.” We are consistent with MAT in excluding it, and an independent `grwght1p` pass confirms both it ($\leq 0.74\%$ on $d\sigma/dE_{\text{avail}}$) and `FrAbs_pi` ($\leq 0.82\%$) are sub-percent — a $< 1\%$ knob cannot affect the budget. The only residual is a one-line confirmation that the exclusion is still current guidance (immaterial to the result).
4. **PPFX effective rank — resolved.** The 100-universe flux band covariance has hard rank 99 but participation-ratio effective rank ≈ 1 : a single coherent normalization mode carries 99.6 % of the variance (9 modes for 99.9 %). The PPFX flux uncertainty is therefore a fixed-rank (rank-1, at most rank-9) object; a PCA / fixed-rank flux covariance is justified, and treating the 100 universes as independent directions over-counts the flux dimensionality (it is also why the leading flux mode correlates $\rho \approx 0.9$ with the rank-1 muon-energy-scale mode, item 2).
5. **Publication precedent.** Have other MAT-based OmniFold / unbinned analyses hit this rank issue, and what did they publish for an ours-only χ^2 ?

What we do regardless. The publication-grade headline uses the combined covariance for $\chi^2/\text{ndf} = 1.699$ with the double-count caveat documented; the per-band rollup is MAT-conformant and byte-for-byte verified (§B.2.4); the 1.4% normalization is included as a rank-1 band; and the ours-only cov is reported via the truncated-spectral χ^2 on $\mathbf{C}^{\text{ours}}$ (rank 201) with the regularization scan disclosed. With the Bashyal joint block now rederived ($\rho = 0.900$, rank-preserving) and the stat-block difference explained by the split bootstrap, no missing-band item remains; the only collaborator-facing question is publication precedent for the ours-only convention.

B.5 Anatomy of the paper-covariance tension

The paradox. The central values agree well: $\sigma_{\text{tot}}^{\text{ours}}/\sigma_{\text{tot}}^{\text{paper}} = 1.011$, median bin ratio 1.006, and the *per-bin pull RMS* against the paper covariance is only 0.598 (§B.3.3). A diagonal-only χ^2 — ignoring all off-diagonal correlations — is therefore

$$\chi_{\text{diag}}^2/\text{ndf} = \frac{1}{n} \sum_i \frac{\Delta_i^2}{C_{ii}} = \text{RMS}(\text{pull})^2 + \text{mean}(\text{pull})^2 \approx 0.366, \quad (16)$$

yet the full-covariance $\chi^2/\text{ndf} = 3.661$. The tension is a $10\times$ inflation that lives *entirely* in the off-diagonal / small-eigenvalue structure of the published `TotalCov` (`diagnose_tension.py`). The rest of this section dissects it.

Eigenmode decomposition. Diagonalise the reported-block paper covariance, $\mathbf{C} = \sum_k \lambda_k \mathbf{u}_k \mathbf{u}_k^\top$ ($\lambda_1 \geq \dots \geq \lambda_n$), and project the residual Δ onto each eigenvector, $c_k = \mathbf{u}_k^\top \Delta$. Then

$$\chi^2 = \Delta^\top \mathbf{C}^+ \Delta = \sum_k \frac{c_k^2}{\lambda_k}, \quad \text{“eigen-pull”}_k \equiv \frac{c_k}{\sqrt{\lambda_k}}, \quad (17)$$

verified to reproduce the `pinv` $\chi^2 = 750.49$ to 3×10^{-9} . The decomposition shows the tension is **neither a few pathological null directions nor the absolute smallest eigenvalues**:

- The top contributing modes sit at $\lambda_k/\lambda_{\text{max}} \sim 10^{-4}$ – 10^{-5} — the lower-middle of the spectrum by magnitude, but still ~ 7 – 8 decades above the cond-floor ($\lambda/\lambda_{\text{max}} \sim 10^{-12}$) — with genuine eigen-pulls of 5 – 6σ .
- The ten *smallest*- λ modes carry only 3% of the χ^2 ; about 90 modes are needed to reach 90%. The χ^2 is broadly distributed, not localised to the ill-conditioned tail.

Regularisation robustness. A `pinv-rcond` scan confirms the tension is not a numerical artifact: retaining only the 73 best-determined directions ($\lambda_k/\lambda_{\text{max}} > 10^{-4}$) still gives $\chi^2/\text{ndf} = 1.42$; 139 directions ($> 10^{-6}$) give 2.79. A rank-truncation scan keeping the r largest- λ modes rises smoothly $0.69 (r=50) \rightarrow 2.35 (100) \rightarrow 3.30 (180) \rightarrow 3.66 (205)$ — no late jump, i.e. the tension does not ride on the last few least-determined modes.

It is shape, not normalisation. Splitting Δ into a component along the measured spectrum $\mathbf{x}^{\text{paper}}$ (an overall scale) and the orthogonal remainder, the scale piece contributes $\chi_{\text{norm}}^2 = 0.10$ and the shape piece $\chi_{\text{shape}}^2 = 750.1$. The +1.1% normalisation offset is statistically irrelevant to the tension; it is a correlated *shape* difference. (Consistent with the Jacobian shape-only $\chi^2/\text{ndf} = 3.60$, §B.3.4.)

Where it lives. Mapping the dominant eigenvectors and the per-bin contribution $\Delta_i(\mathbf{C}^+ \Delta)_i$ back onto the 14×16 grid (`tension_mode_maps.png`, `tension_chi2_map.png`) localises the tension to the **low- p_T cross-section peak ridge** ($p_T \lesssim 0.4$ GeV/c, p_T -bins 2/7/10 carry 16/11/12%) and the low- $p_{\parallel}$ near-edge cells ($p_{\parallel}$ -bins 0/1/8). The carrier modes are sign-alternating *shape* oscillations across the peak, not vertical p_T -columns (which a flux $1/\Phi(p_T)$ normalisation error would produce) and not the $\theta_\mu < 20^\circ$ acceptance boundary. This is exactly where the flux and Muon_Energy bands — the two largest, and the only named cross-band correlation (Bashyal, §B.4) — dominate the covariance: tight bin-to-bin correlation there leaves little freedom to absorb a coherent ~ 1 – 2% shape difference, so a small OmniFold-vs-IBU shape difference becomes a large χ^2 .

A methodological component (estimator and iteration). Part of the tension is regularisation-dependent — the same data, same normalisation ($\Sigma\sigma$ within 0.1%), unfolded with a different GBDT backend moves the paper-cov χ^2 by ~ 1 unit while central values barely change (iteration count, by contrast, is negligible):

OmniFold configuration	paper-cov χ^2/ndf	pull RMS
exact-GBT, 5 iter (frozen production)	3.66	0.598
HistGBT, 5 iter (10-seed mean)	2.70 ± 0.04	0.56
LightGBM, 5 iter	2.65	0.563
LightGBM, 10 iter	2.66	0.565

Two things separate cleanly. (i) **Iteration count is negligible:** LightGBM 5 $\rightarrow$ 10 iters moves the χ^2 by +0.01 (2.645 $\rightarrow$ 2.657), so OmniFold is well-converged at 5 iters and the tension is *not* an under-iteration artifact. (ii) **The GBDT estimator carries a ~ 1 -unit band:** the frozen exact-GBT production (3.66) is the outlier, while both histogram-binned backends agree at 2.65–2.70 (HistGBT is a 10-seed mean, range 2.63–2.74; the ± 0.04 ML-seed jitter is $\sim 25\times$ smaller than the 0.96 exact $\rightarrow$ hist gap, so that gap is genuinely the estimator, not seed noise). The exact-gradient trees fit finer shape structure (less implicit smoothing) and so deviate more from the paper’s IBU-regularised result. This is direct evidence that ~ 1 χ^2 -unit of the tension is OmniFold’s implicit regularisation rather than a physical data–MC disagreement (`tension_iter/`).

Conclusion. The tension is (i) a genuine, broadly-distributed, *correlated shape* difference — not a normalisation offset and not a small- λ inversion artifact — concentrated in the low- p_T peak where the flux/Muon_Energy bands dominate, plus (ii) a ~ 1 - χ^2 -unit methodological band from the GBDT estimator/iteration regularisation. Both point to the same next step: the missing Bashyal flux $\leftrightarrow$ Muon_Energy joint block (§B.4, open question 1) is exactly the covariance structure governing the region that carries the tension.

B.6 Calibration: coverage toys

What it estimates. Whether the bootstrap covariance band actually covers the truth at the rate it claims. A bootstrap band that is wrong — too tight or too loose — would give the right shape but the wrong χ^2 . Coverage toys provide an independent calibration check.

Procedure (`sbatch_coverage_toys_MEFHC_200.sh`). Generate $N_{\text{toy}} = 200$ closure unfolds, each one combining two existing modes of `unfold_2d_omnifold_unbinned.py`:

- `--closure`: pseudo-data are MC reco events (weighted), so the reference truth marginal $\mathbf{X}^{\text{truth}}$ is known exactly and is written to `hTruthXSec2D` in every output file.
- `--bootstrap-seed  $k$` (seeds chosen to avoid collision with the production replicas): per-event Poisson(1) draws, same as the production bootstrap.

Each toy is independent.

Math. Stack the toys into an $N_{\text{toy}} \times 14 \times 16$ array $\mathbf{U}^{(t)}$. For each reported bin (where $\bar{X}_i^{\text{truth}} > 0$, i indexing the 205 reported cells):

$$\sigma_i = \text{std}(\mathbf{U}_i^{(t)}; \text{ddof} = 1) \quad (\text{spread across toys}), \quad (18)$$

$$r_i^{(t)} = \frac{U_i^{(t)} - \bar{X}_i^{\text{truth}}}{\sigma_i} \quad (\text{per-toy, per-bin standardised residual}), \quad (19)$$

$$c_i = \frac{1}{N_{\text{toy}}} \sum_{t=1}^{N_{\text{toy}}} \mathbf{1}\{|r_i^{(t)}| \leq 1\} \quad (\text{per-bin coverage}). \quad (20)$$

The expected value of c_i under a well-calibrated Gaussian band is the central- σ probability of a unit normal:

$$c^{\text{expected}} = \Phi(1) - \Phi(-1) = \text{erf}(1/\sqrt{2}) \approx 0.6827. \quad (21)$$

A second diagnostic is the mean absolute pull, $\langle |r| \rangle$, which for a unit normal is $\sqrt{2/\pi} \approx 0.798$.

Code pointer. `uq/coverage_toys.py` (rollup); driver `sbatch_coverage_toys_MEFHC_200.sh`; outputs `uq/coverage/2d_xsec_MEFHC_5iter_lgbm_coverage_toy*.root` and `uq/coverage/coverage_summary.tx`.

Worked example (STATUS, coverage). With $N_{\text{toy}} = 200$ on the 205 reported bins:

- Mean coverage: 68.71% (target 68.27%; $\Delta = +0.44\%$; median 68.50%).
- $\langle |r| \rangle = 0.794$ (target 0.798; within 0.5%).
- Signed residual mean: $+0.006 \pm 0.082 \sigma$ (no bias).
- Fraction of bins above the 65% target: 97.6% — only 5 of 205 bins fall below.

The agreement of $\langle c \rangle$ and $\langle |r| \rangle$ with their expected values at the half-percent level is the cleanest demonstration that the Poisson-bootstrap band is properly calibrated. The 200-toy ensemble (vs the Stage-1 20 toys) shrinks the per-bin binomial uncertainty to $\sqrt{0.68 \cdot 0.32/200} \approx 3.3\%$, which is why 97.6% of bins now clear the 65% target.

B.7 Closure tests: detecting bias

The variance methods of §B.2 answer the question “how much does the unfold wobble.” Closure tests answer a different question: “is the unfold biased — does it return a value that is consistently offset from the right answer when we inject a known truth?” Bias does not show up in the variance procedures; you have to look for it.

The three closure tests inject a known deformation into the MC, unfold, and check that the answer reproduces the deformation. Each test has its own “acceptance threshold,” set comfortably above the ML-noise floor of 0.166%.

B.7.1 Truth-reweight closure

Procedure. For a known shape $f(p_T, p_{\parallel})$, multiply both the MC reco weights *and* the MC truth weights by f at the event level. Unfold the reweighted pseudo-data against the unchanged (CV) response. The reference is the reweighted truth marginal — the unfolded answer should match it.

Two shapes tested.

- **gauss_pt:** $f = 1 + A \exp[-((p_T - p_T^0)/\sigma)^2]$ with $A = 0.2$, $p_T^0 = 0.4$ GeV/c, $\sigma = 0.1$ GeV/c.
- **tilt_pz:** $f = 1 + \alpha(p_{\parallel} - p_{\parallel}^{\text{ref}})$ with $\alpha = 0.1$, $p_{\parallel}^{\text{ref}} = 5$ GeV/c.

Acceptance. Median per-bin $|\text{residual}| \leq 1.5\%$.

Code pointer. `uq/closure/closure_truth_reweight.py`.

Worked example (1A lgblm 5-iter).

- `gauss_pt`: median $|\text{residual}| = \boxed{0.046\%}$ (PASS).
- `tilt_pz`: median $|\text{residual}| = \boxed{0.013\%}$ (PASS).

Both are $\sim 100\times$ tighter than the threshold; OmniFold recovers the injected truth shape essentially perfectly.

B.7.2 Hidden-variable closure

Procedure. This is the subtle one. The OmniFold feature set sees $(p_T, p_{||})$ on reco and truth, but *not* the resolution variable $\Delta p_T \equiv p_T^{\text{reco}} - p_T^{\text{truth}}$. So we inject a bias on Δp_T only — a kinematic axis the network cannot see — and ask: how much of that bias leaks into the measured $(p_T, p_{||})$ marginal?

The pseudo-data weight bump is

$$w^{\text{reco}} \rightarrow w^{\text{reco}} \cdot \left(1 + A \exp\left[-((\Delta p_T - c)/\sigma_{\Delta p_T})^2\right]\right), \quad (22)$$

with $c = +0.1 \text{ GeV}/c$, $\sigma_{\Delta p_T} = 0.05 \text{ GeV}/c$. Truth weights are *unchanged*, so the reference is the unaltered CV truth marginal. The amplitude is scanned: $A \in \{0.05, 0.10, 0.30\}$.

Leakage coefficient. Define the signed residual $\bar{r}(A)$ as the median of the per-bin signed residual at amplitude A . The leakage coefficient is the slope of a linear fit:

$$\bar{r}(A) \approx \kappa_{\text{leak}} A \quad \Rightarrow \quad \kappa_{\text{leak}} = \frac{\partial \bar{r}}{\partial A}. \quad (23)$$

κ_{leak} tells you what fraction of an unseen Δp_T -shaped bias contaminates the visible answer.

Acceptance. Median per-bin $|\text{residual}| \leq 3.0\%$ (looser than truth- reweight because Δp_T and p_T are weakly correlated, so the unfolding can never fully kill the leak).

Code pointer. `uq/closure/closure_hidden_var.py`.

Worked example (1A lgblm 5-iter, 205 reported bins).

- Per-amplitude median residuals: $0.57\%/1.14\%/3.43\%$ for $A = 0.05/0.10/0.30$ — linear in A , as expected for small perturbations.
- Linear fit slope: $\kappa_{\text{leak}} = \boxed{0.113}$.

About 11% of a hidden Δp_T bias leaks into the measured $(p_T, p_{||})$ marginal — under the 3% threshold for the largest realistic deformation ($A = 0.30$). Larger leakage would indicate that Δp_T ought to be added to the feature set; the present number is small enough to be safely ignored.

B.7.3 Alternative-model closure

Procedure. Train the OmniFold response on the CV weights. Build the pseudo-data from an alternative-universe reweighting (e.g. `MaCCQE:0 +1 $\sigma$` , or `Flux:50` PPFX index). The reference is the in-acceptance-extrapolated alt truth:

$$\tilde{X}_i^{\text{ref}} = X_i^{\text{CV,truth}} \cdot \frac{X_i^{\text{alt,in-acc}}}{X_i^{\text{CV,in-acc}}}. \quad (24)$$

This scales the CV truth marginal by the ratio of in-acceptance counts between alt and CV, isolating the unfolding’s response to a model-shape change from acceptance/efficiency effects. `MaCCQE` probes an interaction-model shape change; `Flux` probes flux reweighting — two distinct kinds of model dependence.

Acceptance. Median per-bin $|\text{residual}| \leq 2.0\%$.

Code pointer. `uq/closure/closure_alt_model.py`.

Worked example (1A lgbm 5-iter).

- `MaCCQE:0 (+1 $\sigma$ )`: median $|\text{residual}| = \boxed{0.068\%}$ (PASS); in-acceptance bias -1.68% folded out by the extrapolation.
- `Flux:50` (PPFX idx 50): median $|\text{residual}| = \boxed{0.376\%}$ (PASS); in-acceptance bias -7.32% .

Both pass with $\gg 5\times$ headroom. The unfolding is robust to model variations on the order of $\pm 1\sigma$ of GENIE and full PPFX flux universes.

B.8 How the pieces fit together

We can now read the `2D_OMNIFOLD_STUDY_STATUS.md` headline in one sentence: *across 205 paper-reported bins, the combined OmniFold + paper error budget gives $\chi^2/\text{ndf} = 1.699$ and pull mean/RMS 0.069/0.466, with a pull-RMS- < 1 band that has been independently checked at the 0.5% level by 200 coverage toys, on top of a 0.166% ML-noise floor that is more than an order of magnitude below the systematic envelope.*

Audit trail of the headline

Number	Section	Procedure
$\chi^2/\text{ndf} = 1.699$	§B.3.2	Eq. (10) with $\mathbf{C}^{\text{paper}} + \mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{stat}} + \mathbf{C}^{\text{ML}}$
$\mathbf{C}^{\text{syst}}$	§B.2.3	187 universes, MAT-conformant Eq. (6) + norm
$\mathbf{C}^{\text{stat}}$	§B.2.2	Poisson bootstrap, 300 replicas, Eq. (5)
$\mathbf{C}^{\text{ML}}$	§B.2.1	lgbm seedscan, $n = 10$ trials
$\mathbf{C}^{\text{paper}}$	STATUS.md	Paper’s published <code>TotalCovariance</code>
Pull RMS = 0.466	§B.3.3	Per-bin standardised residual on combined $\mathbf{C}$
Combined median rel. = 4.822%	§B.2.5	$\sqrt{\text{diag} \mathbf{C}^{\text{tot}}} / \mathbf{X}^{\text{CV}}$
Mean coverage = 68.71%	§B.6	Coverage toys, $N_{\text{toy}} = 200$

Standing assumptions

1. **Gaussian approximation.** Pulls and χ^2 treat each bin as Gaussian. Justified because per-bin entries are weighted sums over $\mathcal{O}(10^4)$ MC events; central limit theorem applies.
2. **Independence of error sources.** Eq. (9) requires no cross-correlations between ML, statistical, and systematic. The bootstrap pins the GBDT seed (so ML noise cancels at first order); universes pin the seed too; the seedscan pins the bootstrap. Cross-terms are higher-order. (The available $N = 300$ bootstrap block carries a small residual ML leak, §B.2.2.)
3. **Bootstrap calibration.** §B.6’s mean coverage = 68.71% certifies $\mathbf{C}^{\text{stat}}$ to half a percent.
4. **Band independence in $\mathbf{C}^{\text{syst}}$.** Different physics sources are taken as uncorrelated; the one known exception, the Bashyal flux $\leftrightarrow$ muon-E joint block, is not yet propagated (§B.4).
5. **Combined- χ^2 double-count.** The headline shares systematics with the paper’s V and is the “what OmniFold UQ adds on top of V ” statement, not an independent ours-only test (§B.4).
6. **Closure passes are valid for the perturbations tested.** Nothing guarantees they pass for unphysical deformations far outside the training distribution.

B.9 Multinomial $\rightarrow$ independent Poisson bootstrap limit

A non-parametric bootstrap with replacement draws N events from the original N -event sample. Let n_i be the number of times event i is drawn. Then $(n_1, \dots, n_N) \sim \text{Multinomial}(N, \mathbf{p})$ with $p_i = 1/N$ uniform, $\sum_i n_i = N$.

The marginal $n_i \sim \text{Binomial}(N, 1/N)$, whose mean is 1 and variance $(N - 1)/N \rightarrow 1$. For large N , the binomial converges to Poisson(1) (Poisson limit theorem: many trials, small success probability, fixed expected count).

The full multinomial has the constraint $\sum n_i = N$, which introduces $\mathcal{O}(1/N)$ correlations between the n_i . In the large- N limit those correlations vanish and the n_i become *independent* Poisson(1) variables. This is the Poissonisation of the multinomial: the conditioned N -event multinomial is a Poisson process *conditioned* on hitting N counts, and the unconditional version is independent Poissons.

In practice, the independent-Poisson form is what we implement (one `np.random.default_rng().poisson(1.0)` draw per event), and the $\sum n_i = N$ constraint is dropped. For $N \gtrsim 10^6$ (our regime), the relative error from the dropped constraint is $\mathcal{O}(1/\sqrt{N})$ on the total event count — far smaller than the per-bin Poisson variance.

B.10 The $\pm 1\sigma$ pair covariance estimator: MAT vs legacy

For a single universe band with two universes $u = 0, 1$ at $\pm 1\sigma$, write the two value vectors as $\mathbf{X}_0$ and $\mathbf{X}_1$, with universe mean $\langle \mathbf{X} \rangle = \frac{1}{2}(\mathbf{X}_0 + \mathbf{X}_1)$. The **MAT-conformant** estimator Eq. (6) at $N_u = 2$ is

$$\mathbf{C}_{\text{MAT}}^B = \frac{1}{2} \sum_{u=0,1} (\mathbf{X}_u - \langle \mathbf{X} \rangle)(\mathbf{X}_u - \langle \mathbf{X} \rangle)^\top = \frac{1}{4}(\mathbf{X}_0 - \mathbf{X}_1)(\mathbf{X}_0 - \mathbf{X}_1)^\top, \quad (25)$$

a rank-1 outer product of the *half-difference*. It is mean-centered, so any common offset of the pair away from X_{CV} contributes nothing.

Contrast the **legacy** CV-centered pair estimator that the pre-2026-05-28 rollup used, with $\Delta_u = \mathbf{X}_u - \mathbf{X}_{\text{CV}}$ and decomposition $\Delta_0 = \mathbf{m} + \mathbf{a}$, $\Delta_1 = \mathbf{m} - \mathbf{a}$:

$$\mathbf{C}_{\text{legacy}}^B = \frac{1}{2}(\Delta_0 \Delta_0^\top + \Delta_1 \Delta_1^\top) = \mathbf{m} \mathbf{m}^\top + \mathbf{a} \mathbf{a}^\top. \quad (26)$$

The two agree *only* for a CV-symmetric pair ($\mathbf{m} = 0$), where both reduce to $\mathbf{a}\mathbf{a}^\top$. For an asymmetric or one-sided pair the legacy form adds a spurious rank-1 $\mathbf{m}\mathbf{m}^\top$ term (treating the common offset as a second variance direction), which MAT correctly collapses to near zero. This is why moving to the MAT formula *lowered* the total universe trace by 1.9% (the BeamAngle, MuonResolution, and one-sided NC bands had asymmetric shifts), and it is the single methodological correction behind the χ^2/ndf shift $1.473 \rightarrow 1.699$. The legacy estimator remains available via `--legacy-pair-formula` for forensic reproduction only; the earlier claim that the CV-centered convention is “consistent with the published MINERvA error budget” was incorrect — MAT is mean-centered (§B.2.4).

C The MINERvA measurement-dimensionality survey

The full classification behind Fig. 17: every MINERvA differential cross-section measurement, the number of kinematic variables its headline result unfolds simultaneously, and the reference.

Table 4: Full MINERvA differential cross-section survey: the number of kinematic variables unfolded simultaneously (Dim.), classified from each analysis’s headline differential result. Channels: CC-inclusive (incl.), CCQE/QE-like incl. transverse kinematic imbalance (QE), pion production CC and coherent (π), deep-inelastic / kaon (DIS), electron-neutrino (ν_e), nuclear-target ratio (ratio).

Year	Channel; observables	Dim.	Ref.
2013	QE; ν_μ CCQE $p_T, p_\parallel$	2	[27]
2013	QE; $\bar{\nu}_\mu$ CCQE $p_T, p_\parallel$	2	[28]
2014	ratio; CC on C/Fe/Pb to CH	1	[29]
2014	π ; coherent $\pi^\pm$ (total & diff.)	1	[30]
2015	π ; CC $1\pi^+$ (T_π, θ_π)	1	[31]
2015	QE; μ +p final states	1	[32]
2015	π ; $\bar{\nu}_\mu$ single π^0	1	[33]
2016	ν_e ; CCQE-like on CH	2	[34]
2016	π ; $\nu_\mu/\bar{\nu}_\mu$ pion prod.	2	[35]
2016	DIS; partonic nuclear effects	2	[36]
2016	DIS; CC K^+ production	1	[37]
2017	π ; CC single π^0	2	[38]
2017	QE; nuclear-dependence (multi-target)	1	[39]
2018	QE; $\bar{\nu}_\mu$ QE-like $p_T, p_\parallel$	2	[40]
2018	QE; μ -p mesonless TKI	2	[41]
2018	π ; coherent $\pi^\pm$ on C	2	[42]
2019	QE; quasielastic-like ν_μ	2	[43]
2019	π ; $\bar{\nu}_\mu$ single π^-	2	[44]
2020	incl.; CC-incl $\langle E_\nu \rangle \sim 3.5$ GeV $p_T, p_\parallel$	2	[45]
2020	QE; nuclear binding & TKI	2	[46]
2020	QE; high-statistics QE-like	2	[47]
2020	π ; CC π^0 nuclear effects	1	[48]
2021	incl.; CC-incl $\langle E_\nu \rangle \sim 6$ GeV $p_T, p_\parallel$ (target)	2	[1]
2022	QE; QE-like muon & proton kinematics	2	[49]
2022	incl.; CC-incl low momentum transfer	1	[26]
2023	π ; CC $1\pi^+$ (multi-target)	2	[50]
2023	π ; coherent π^+ (multi-target)	1	[51]
2023	QE; $\bar{\nu}_\mu$ multi-neutron low- E_{avail}	1	[52]

continued...

Year	Channel; observables	Dim.	Ref.
2024	ν_e ; low- Q^2 (E_{avail} , p_T)	2	[24]
2025	QE; A -dependence TKI (multi-target)	2	[53]
2026	this work ; unbinned $p_T, p_{\parallel}$	2	—
2026	this work ; unbinned $p_T, p_{\parallel}, E_{\text{avail}}$	3	—
2026	this work ; unbinned $p_T, p_{\parallel}, E_{\text{avail}}, q_3$	4	—

References

- [1] D. Ruterbories et al. “Measurement of inclusive charged-current ν_{μ} cross sections as a function of muon kinematics at $\langle E_{\nu} \rangle \sim 6$ GeV on hydrocarbon”. In: *Phys. Rev. D* 104.9 (2021), p. 092007. DOI: 10.1103/PhysRevD.104.092007. arXiv: 2106.16210 [hep-ex].
- [2] G. D’Agostini. “A Multidimensional unfolding method based on Bayes’ theorem”. In: *Nucl. Instrum. Meth. A* 362 (1995), p. 487. DOI: 10.1016/0168-9002(95)00274-X.
- [3] A. Andreassen et al. “OmniFold: A Method to Simultaneously Unfold All Observables”. In: *Phys. Rev. Lett.* 124.18 (2020), p. 182001. DOI: 10.1103/PhysRevLett.124.182001. arXiv: 1911.09107 [hep-ph].
- [4] A. Andreassen et al. “Scaffolding Simulations with Deep Learning for High-dimensional Deconvolution”. In: *ICLR simDL workshop* (2021). arXiv: 2105.04448 [stat.ML].
- [5] H1 Collaboration. “Measurement of Lepton-Jet Correlation in Deep-Inelastic Scattering with the H1 Detector Using Machine Learning for Unfolding”. In: *Phys. Rev. Lett.* 128.13 (2022), p. 132002. DOI: 10.1103/PhysRevLett.128.132002. arXiv: 2108.12376 [hep-ex].
- [6] H1 Collaboration. “Unbinned Deep Learning Jet Substructure Measurement in High Q^2 ep collisions at HERA”. In: *Phys. Lett. B* (2024). arXiv: 2303.13620 [hep-ex].
- [7] K. Datta, D. Kar, and D. Roy. “Unfolding with Generative Adversarial Networks”. In: (2018). arXiv: 1806.00433 [physics.data-an].
- [8] M. Bellagente et al. “How to GAN away Detector Effects”. In: *SciPost Phys.* 8 (2020), p. 070. DOI: 10.21468/SciPostPhys.8.4.070. arXiv: 1912.00477 [hep-ph].
- [9] M. Bellagente et al. “Invertible Networks or Partons to Detector and Back Again”. In: *SciPost Phys.* 9 (2020), p. 074. DOI: 10.21468/SciPostPhys.9.5.074. arXiv: 2006.06685 [hep-ph].
- [10] M. Backes et al. “An unfolding method based on conditional Invertible Neural Networks (cINN) using iterative training”. In: (2022). arXiv: 2212.08674 [hep-ph].
- [11] J. Chan and B. Nachman. “Unbinned Profiled Unfolding”. In: *Phys. Rev. D* 108 (2023), p. 016002. DOI: 10.1103/PhysRevD.108.016002. arXiv: 2302.05390 [hep-ph].
- [12] A. Höcker and V. Kartvelishvili. “SVD approach to data unfolding”. In: *Nucl. Instrum. Meth. A* 372 (1996), p. 469. DOI: 10.1016/0168-9002(95)01478-0. arXiv: hep-ph/9509307.
- [13] S. Schmitt. “TUnfold: an algorithm for correcting migration effects in high energy physics”. In: *JINST* 7 (2012), T10003. DOI: 10.1088/1748-0221/7/10/T10003. arXiv: 1205.6201 [physics.data-an].
- [14] T. Adye. “Unfolding algorithms and tests using RooUnfold”. In: *PHYSTAT 2011 Workshop on Statistical Issues*. 2011, p. 313. arXiv: 1105.1160 [physics.data-an].

- [15] L. Aliaga et al. “Design, Calibration, and Performance of the MINERvA Detector”. In: *Nucl. Instrum. Meth. A* 743 (2014), p. 130. DOI: 10.1016/j.nima.2013.12.103. arXiv: 1305.5199 [physics.ins-det].
- [16] L. Aliaga et al. “Neutrino flux predictions for the NuMI beam”. In: *Phys. Rev. D* 94.9 (2016), p. 092005. DOI: 10.1103/PhysRevD.94.092005. arXiv: 1607.00704 [hep-ex].
- [17] C. Andreopoulos et al. “The GENIE Neutrino Monte Carlo Generator”. In: *Nucl. Instrum. Meth. A* 614 (2010), p. 87. DOI: 10.1016/j.nima.2009.12.009. arXiv: 0905.2517 [hep-ph].
- [18] T. Golan, J. T. Sobczyk, and J. Żmuda. “NuWro: the Wrocław Monte Carlo Generator of Neutrino Interactions”. In: *Nucl. Phys. B Proc. Suppl.* 229-232 (2012), p. 499. DOI: 10.1016/j.nuclphysbps.2012.09.136.
- [19] O. Buss et al. “Transport-theoretical Description of Nuclear Reactions”. In: *Phys. Rept.* 512 (2012), p. 1. DOI: 10.1016/j.physrep.2011.12.001. arXiv: 1106.1344 [hep-ph].
- [20] Y. Hayato. “A neutrino interaction simulation program library NEUT”. In: *Acta Phys. Polon. B* 40 (2009), p. 2477.
- [21] A. Bashyal et al. “Use of neutrino scattering events with low hadronic recoil to inform neutrino flux and detector energy scale”. In: *JINST* 16.08 (2021), P08068. DOI: 10.1088/1748-0221/16/08/P08068. arXiv: 2104.05769 [hep-ex].
- [22] F. Canelli et al. “A Practical Guide to Unbinned Unfolding”. In: (2025). arXiv: 2507.09582 [hep-ph].
- [23] V. Mikuni and B. Nachman. “OmniLearn: A Method to Simultaneously Facilitate All Jet Physics Tasks”. In: (2024). arXiv: 2404.16091 [hep-ph].
- [24] S. Henry, H. Su, et al. “Measurement of Electron Neutrino and Antineutrino Cross Sections at Low Momentum Transfer”. In: *Phys. Rev. D* 109.9 (2024), p. 092008. DOI: 10.1103/PhysRevD.109.092008. arXiv: 2312.16631 [hep-ex].
- [25] P. A. Rodrigues et al. “Identification of Nuclear Effects in Neutrino-Carbon Interactions at Low Three-Momentum Transfer”. In: *Phys. Rev. Lett.* 116.7 (2016), p. 071802. DOI: 10.1103/PhysRevLett.116.071802. arXiv: 1511.05944 [hep-ex].
- [26] “Inclusive charged-current ν_μ cross section at low momentum transfer”. In: (2022). arXiv: 2110.13372 [hep-ex].
- [27] “Measurement of the ν_μ charged-current quasielastic cross section on hydrocarbon”. In: (2013). arXiv: 1305.2243 [hep-ex].
- [28] “Measurement of the $\bar{\nu}_\mu$ charged-current quasielastic cross section on hydrocarbon”. In: (2013). arXiv: 1305.2234 [hep-ex].
- [29] “Ratios of ν_μ charged-current cross sections on C, Fe, and Pb to CH”. In: (2014). arXiv: 1403.2103 [hep-ex].
- [30] “Coherent $\pi^\pm$ production in $\nu_\mu/\bar{\nu}_\mu$ interactions on carbon (1.5–20 GeV)”. In: (2014). arXiv: 1409.3835 [hep-ex].
- [31] B. Eberly et al. “Charged Pion Production in ν_μ Interactions on Hydrocarbon at $\langle E_\nu \rangle = 4.0$ GeV”. In: *Phys. Rev. D* 92.9 (2015), p. 092008. DOI: 10.1103/PhysRevD.92.092008. arXiv: 1406.6415 [hep-ex].
- [32] “Muon-plus-proton final states in ν_μ interactions on hydrocarbon”. In: (2015). arXiv: 1409.4497 [hep-ex].

- [33] “Single neutral-pion production by $\bar{\nu}_\mu$ on hydrocarbon”. In: (2015). arXiv: 1503.02107 [hep-ex].
- [34] “Measurement of electron-neutrino quasielastic-like scattering on hydrocarbon”. In: (2016). arXiv: 1509.05729 [hep-ex].
- [35] “Cross sections for ν_μ and $\bar{\nu}_\mu$ induced pion production on hydrocarbon”. In: (2016). arXiv: 1606.07127 [hep-ex].
- [36] “Measurement of partonic nuclear effects in deep-inelastic ν_μ scattering”. In: (2016). arXiv: 1601.06313 [hep-ex].
- [37] “Charged-current $\nu_\mu K^+$ production on hydrocarbon”. In: (2016). arXiv: 1604.03920 [hep-ex].
- [38] “Measurement of ν_μ charged-current single π^0 production on hydrocarbon”. In: (2017). arXiv: 1708.03723 [hep-ex].
- [39] “Direct measurement of the nuclear dependence of charged-current quasielastic-like ν_μ interactions”. In: (2017). arXiv: 1705.03791 [hep-ex].
- [40] C. E. Patrick et al. “Measurement of the muon antineutrino double-differential cross section for quasielastic-like scattering on hydrocarbon at $E_\nu \sim 3.5$ GeV”. In: *Phys. Rev. D* 97.5 (2018), p. 052002. DOI: 10.1103/PhysRevD.97.052002. arXiv: 1801.01197 [hep-ex].
- [41] X.-G. Lu et al. “Measurement of final-state correlations in neutrino muon-proton mesonless production on hydrocarbon at $\langle E_\nu \rangle = 3$ GeV”. In: *Phys. Rev. Lett.* 121.2 (2018), p. 022504. DOI: 10.1103/PhysRevLett.121.022504. arXiv: 1805.05486 [hep-ex].
- [42] “Coherent $\pi^\pm$ production in $\nu_\mu/\bar{\nu}_\mu$ charged-current interactions on carbon”. In: (2018). arXiv: 1711.01178 [hep-ex].
- [43] “Measurement of quasielastic-like ν_μ scattering on hydrocarbon”. In: (2019). arXiv: 1811.02774 [hep-ex].
- [44] “Measurement of antineutrino single π^- production on hydrocarbon”. In: (2019). arXiv: 1906.08300 [hep-ex].
- [45] MINERvA Collaboration. “Double-differential inclusive charged-current ν_μ cross sections on hydrocarbon in MINERvA at $\langle E_\nu \rangle \sim 3.5$ GeV”. In: *Phys. Rev. D* 101.11 (2020), p. 112007. DOI: 10.1103/PhysRevD.101.112007. arXiv: 2002.12496 [hep-ex].
- [46] T. Cai et al. “Nucleon binding energy and transverse momentum imbalance in neutrino-nucleus reactions”. In: *Phys. Rev. D* 101.9 (2020), p. 092001. DOI: 10.1103/PhysRevD.101.092001. arXiv: 1910.08658 [hep-ex].
- [47] “High-statistics measurement of ν_μ charged-current quasielastic-like cross sections”. In: (2020). arXiv: 1912.09890 [hep-ex].
- [48] “ ν_μ charged-current neutral-pion production and nuclear effects on hydrocarbon”. In: (2020). arXiv: 2002.05812 [hep-ex].
- [49] “Measurement of ν_μ charged-current quasielastic-like cross sections in muon and proton kinematics on hydrocarbon”. In: (2022). arXiv: 2203.08022 [hep-ex].
- [50] MINERvA Collaboration. “Measurement of ν_μ charged-current single π^+ production on hydrocarbon”. In: (2023). arXiv: 2209.07852 [hep-ex].
- [51] “Measurement of coherent π^+ production in ν_μ charged-current interactions on nuclear targets”. In: (2023). arXiv: 2210.01285 [hep-ex].

- [52] MINERvA Collaboration. “Measurement of the Multi-Neutron $\bar{\nu}_\mu$ Charged Current Differential Cross Section at Low Available Energy on Hydrocarbon”. In: (2023). arXiv: 2310.17014 [hep-ex].
- [53] J. Kleykamp et al. “Measurement of the A dependence of the ν_μ charged-current quasielastic-like cross section as a function of muon and proton kinematics at $\langle E_\nu \rangle \sim 6$ GeV”. In: *Phys. Rev. D* (2025). arXiv: 2503.15047 [hep-ex].